

Disentangling Autocorrelated Intraday Returns

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Abstract

We propose a semiparametric framework to disentangle the autocovariance structure of high-frequency equity returns. The observed price is modeled as the sum of an efficient component, evolving as a nonparametric continuous-time Itô-semimartingale, and a market microstructure component governed by a discrete-time moving-average process. Our estimation strategy is based on a quasi-likelihood approach that employs a misspecified moving-average model selected via information criteria. We establish a pointwise central limit theorem for the resulting autocovariance estimator, as well as uniform consistency over a broad class of models allowing for arbitrary noise magnitude and flexible dependence structures. Simulation results demonstrate that our estimator outperforms leading nonparametric alternatives, particularly in regimes with small noise magnitude, and highlight that the performance of all estimators is critically driven by the level of noise. Applying our methodology, alongside existing nonparametric approaches, to S&P 1500 constituents from 1996 to 2024, we find that noise magnitude is persistently small throughout the sample. This empirical finding has important implications for the measurement of noise magnitude, noise persistence, and volatility.

KEYWORDS: Moving-Average Models, Noise Autocorrelations, QMLE, Small Noise

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1 Introduction

Autocorrelations in stock returns are ubiquitous. The earlier literature regards such autocorrelations as evidence against market efficiency. Nonetheless, as market efficiency has improved over past decades, autocorrelations have remained a salient feature of intraday stock returns sampled at sufficiently high frequencies. The modern view of such autocorrelations is that they arise from market microstructure frictions, such as bid-ask bounces, nonsynchronous trading, price discreteness, etc., which coalesce into efficient equilibrium prices and lead to the convoluted dynamics of returns.

To disentangle the observed autocorrelations in intraday returns, we model the transaction price as a discretized continuous-time semimartingale process plus a discrete-time moving-average process. The former represents the efficient price process that features return heteroscedasticity in the form of stochastic volatility and jumps, but does not contribute to any autocovariance; the latter serves as a reduced-form description of microstructure noise that generates the observed return autocovariances. Motivated by empirical evidence that microstructure noise can be small, as documented by [Hansen and Lunde \(2006\)](#), we explicitly allow for potentially shrinking noise magnitude, aiming for more refined quantitative measurement of noise autocovariances under the empirically relevant scenarios.

To estimate the autocovariances and autocorrelations of the noise process, we construct a tractable quasi-maximum likelihood estimator (QMLE), pretending that the transaction price arrives regularly and comprises a Brownian motion with constant volatility and an $\text{MA}(q)$ noise component. We adopt the moving-average specification rather than ARMA because ARMA likelihoods are prone to near-cancellation of common autoregressive and moving-average roots—the so-called “pile-up” problem—when noise is weak, an empirically prominent regime in our data. We select q based on the Akaike/Bayesian information criteria (AIC/BIC). Although our estimator shares the same likelihood as that from an $\text{MA}(q + 1)$ model, our asymptotic design is in-fill, i.e., the number of observations increases within a fixed window—say, a trading day—which renders our analysis rather different from the usual long-span asymptotics in the classic time-series setting. [Da and Xiu \(2021\)](#) also employ a QMLE framework but focus on inference for volatility, treating noise autocovariances as nuisance parameters. In contrast, the present paper studies the noise autocovariances themselves as primary objects of interest.

When noise is large and follows a finite-order moving-average structure, we provide a pointwise central limit theorem for our autocovariance estimator at the usual $n^{1/2}$ rate, together with a quadratic representation of the estimator. This central limit theorem also provides a semiparametric efficiency benchmark for noise autocovariances, extending the line of inquiry

on semiparametric efficiency in high-frequency volatility inference (Jacod (1994); Jacod and Rosenbaum (2013)) from volatility to noise dynamics. We show that uniformly valid distributional results for our estimator are not attainable due to data-driven order selection. We establish uniform consistency of our estimator over a broad class of $\text{MA}(\infty)$ models that allow for asymptotically vanishing noise with flexible dependence structures. Methodologically, we leave the QMLE unconstrained so that an *inadmissible* output—one in which the implied moving-average representation is undefined—serves as an informative diagnostic of weak noise signal, rather than being masked by a constrained estimator.

Jacod et al. (2017) (JLZ) and Li and Linton (2021) (ReMeDI) develop important nonparametric estimators of the autocovariances and higher-order moments of microstructure noise based on local averaging and differencing strategies, respectively. They derive central limit theorems under general dependence structures and focus on settings with large noise. Relative to these nonparametric approaches, our likelihood-based method is by construction limited to second-moment estimation, and is computationally more demanding as it requires optimization over model parameters and order. On the other hand, our BIC-based order selection provides a clear in-sample objective for tuning—an advantage the bandwidth parameters of nonparametric methods lack—which enables a fully data-adaptive empirical analysis across the full S&P 1500 universe over 1996–2024 (roughly 10 million stock-days) without per-stock or per-period tuning.

We examine the finite-sample performance of JLZ, ReMeDI, and our estimator through Monte Carlo simulations, with an emphasis on better informing our empirical analysis. Because whether noise is statistically small depends on the sample size, we vary both noise magnitude and sampling frequency in our simulations. The results show that performance deteriorates substantially in low-signal regimes with small noise magnitude and limited observations, with large RMSEs for autocorrelation estimates across all three methods. In sufficiently weak-signal regimes, they frequently fail to produce meaningful autocorrelation estimates. Our simulations further identify the BIC-selected moving-average order as the most informative observable diagnostic of signal strength: an autocorrelation at any given lag is reliably estimated on the subset of stock-days where the QMLE produces admissible output and the selected order is at least as large as that lag, motivating the empirical estimation pipeline we adopt.

We apply our estimator, together with JLZ and ReMeDI approaches, to intraday returns of S&P 1500 index constituents from 1996 to 2024 and obtain four main findings. First, microstructure noise is statistically detectable on the typical stock-day throughout the sample, but its dependence structure is often weakly identified: for a substantial fraction of stock-days,

QMLE fails to produce noise autocovariances consistent with any moving-average representation, especially in the early years. Second, the estimated noise variance has declined by roughly two orders of magnitude over the sample period across all three methods. Third, noise autocorrelations can be reliably estimated on a non-trivial share of stock-days. On these subsamples, tick-time autocorrelations decline only modestly, while the implied calendar-time persistence has shortened by roughly two orders of magnitude—from about 9–15 minutes in 1996 to 5–25 seconds in 2024—driven primarily by the rise in trading frequency. Finally, even when noise is pervasively small, accounting for it remains important for volatility estimation. We find that sparsely sampled realized volatility estimators exhibit substantial bias in the early sample, with bias shrinking but standard-error inefficiency growing over time.

Empirical evidence of autocorrelations in the returns of transaction prices goes back to as early as [Niederhoffer and Osborne \(1966\)](#), [Simmons \(1971\)](#), and [Garbade and Lieber \(1977\)](#). Among others, [Hasbrouck and Ho \(1987\)](#) document positive autocorrelations in intraday stock returns, in returns of quote midpoints, and in the arrival of buy and sell orders. They thus propose a model of the return-generating process, which is observationally equivalent to an ARMA(2,2) model. While classical time-series models such as ARMA are convenient for dependent data, they are not appropriate for intraday returns because of the heteroscedasticity in returns.

Why do higher-order autocorrelations of returns exist? There are many economic hypotheses, such as strategic order splitting ([Garbade and Lieber \(1977\)](#)); optimal control of execution cost ([Bertsimas and Lo \(1998\)](#)); price impact and inventory control ([Kyle \(1985\)](#), [Amihud and Mendelson \(1980\)](#)); the crowd effect or herding ([Tóth et al. \(2015\)](#)); and high-frequency trading [Brogaard et al. \(2014\)](#). Our objective here is modest. We aim to estimate parameters in a general class of reduced-form models, since many structural economic models yield specific reduced-form models—see, for example, [Hasbrouck \(2007\)](#)—with differences only in how the reduced-form parameters relate to structural parameters.

There is an enormous literature on the estimation of quadratic variation or its components using noisy high-frequency data; e.g., the two-scale or multi-scale estimators by [Zhang et al. \(2005\)](#) and [Zhang \(2006\)](#); the realized kernel estimator and its extensions by [Barndorff-Nielsen et al. \(2008\)](#) and [Barndorff-Nielsen et al. \(2011\)](#); the pre-averaging estimator by [Jacod et al. \(2009\)](#) and [Jacod et al. \(2010\)](#); the quasi-maximum likelihood estimator (QMLE) by [Xiu \(2010\)](#); and the local method of moments estimator by [Reiß \(2011\)](#).¹ An “essentially” white noise assumption is most common in this strand of the literature, with the exception of [Jacod et al. \(2019\)](#), [Varneskov \(2016\)](#), [Da and Xiu \(2021\)](#), [Li et al. \(2020\)](#), and [Li and Linton \(2026\)](#),

¹On the other hand, [Laurent and Shi \(2020\)](#) and [Andersen et al. \(2023\)](#) emphasize the impact of a large drift component on volatility estimation and propose corresponding solutions.

who tackle general colored-noise processes for the purpose of volatility estimation. Related work also include Aït-Sahalia et al. (2005), Aït-Sahalia et al. (2011), Kalnina and Linton (2008), and Bibinger et al. (2019). Unlike the above papers, which treat noise as nuisance parameters in the estimation of quadratic variation, our target here is mainly the temporal dependence of intraday returns beyond the first-order autocorrelations. Chang et al. (2018) also focus on analyzing the statistical properties of the noise process and propose an estimator of noise density and noise moments in an i.i.d. noise setting.

The paper is organized as follows. Section 2 sets up the model. Section 3 introduces the estimator and provides the main asymptotic results. Section 4 reports Monte Carlo simulations. In Section 5, we apply our approach, JLZ, and ReMeDI to study the noise component in returns of S&P Composite 1,500 index constituents. Section 6 concludes. The online supplemental appendix contains all mathematical proofs.

2 Model Assumptions

We assume that transaction prices $\widetilde{X}$ are observed at t_i , for $i = 1, 2, \dots, n_T$, within a fixed window $[0, T]$. They comprise two components: $\widetilde{X}_{t_i} = X_{t_i} + U_i$, where X_{t_i} is (the logarithm of) the efficient equilibrium price and U_i is the microstructure noise associated with the i th observation. Furthermore, the efficient price satisfies:

Assumption 1. *The logarithm of the efficient price process X_t is an Itô-semimartingale defined on some filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ and satisfies*

$$X_t = X_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s + (\delta \mathbb{1}_{\{|\delta| \leq 1\}}) \star (\underline{\mu} - \underline{\nu})_t + (\delta \mathbb{1}_{\{|\delta| > 1\}}) \star \underline{\mu}_t, \quad (2.1)$$

where μ_t and σ_t are adapted and locally bounded, W is a standard Brownian motion, and $\underline{\mu}$ is a Poisson random measure on $\mathbb{R}_+ \times E$, where E is a Polish space. The compensator $\underline{\nu}$ satisfies $\underline{\nu}(dt, du) = dt \otimes \lambda(du)$ for some σ -finite measure λ on E . Moreover, $|\delta(\omega, t, u)| \wedge 1 \leq \Gamma_m(u)$ for all (ω, t, u) with $t \leq \tau_m(\omega)$, where $\{\tau_m\}$ is a localizing sequence of stopping times and $\{\Gamma_m\}$ a sequence of deterministic functions satisfying $\int \Gamma_m^2(u) \lambda(du) < \infty$.

In addition, the process $Z_t = (\mu_t, \sigma_t^2)$ is also an Itô semimartingale on the space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ with the form

$$Z_t = Z_0 + \int_0^t \tilde{\mu}_s ds + \int_0^t \tilde{\sigma}_s d\widetilde{W}_s + (\tilde{\delta} \mathbb{1}_{\{|\tilde{\delta}| \leq 1\}}) \star (\underline{\mu} - \underline{\nu})_t + (\tilde{\delta} \mathbb{1}_{\{|\tilde{\delta}| > 1\}}) \star \underline{\mu}_t, \quad (2.2)$$

where $\tilde{\mu}_t$ and $\tilde{\sigma}_t$ are locally bounded adapted processes, $\widetilde{W}$ is a multivariate Brownian motion, potentially correlated with W , and $\tilde{\delta}$ is a predictable function such that for some deterministic function $\tilde{\Gamma}_m(u)$, $\|\tilde{\delta}(\omega, t, u)\| \wedge 1 \leq \tilde{\Gamma}_m(u)$ for all $\omega \in \Omega$, $t \leq \tau_m(\omega)$, and $\int \tilde{\Gamma}_m^2(u) \lambda(du) < \infty$.

While the efficient prices are defined in continuous time, we only observe their noisy versions at discrete time points. We now describe the assumption of the arrival times of transactions:

Assumption 2. *The sequence of observation times $\{t_i : i \geq 0\}$ satisfies $t_0 = 0$ and $t_i = t_{i-1} + \frac{T}{n} \xi_{t_{i-1}} \chi_i$, where the sequence $\{\chi_i : i \geq 1\}$ is i.i.d., $(0, \infty)$ -valued, defined on $(\Omega, \mathcal{F}, \mathbb{P})$, and independent of the σ -field $\mathcal{F}_\infty = \bigvee_{t \geq 0} \mathcal{F}_t$, with $m_j = \mathbb{E}((\chi_i)^j) < \infty$ and $m_1 = 1$, for all $j > 0$. In addition, the process $\xi = (\xi_t)_{t \geq 0}$ is a nonnegative Itô-semimartingale defined on $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ in the form of (2.2), such that neither ξ_t nor ξ_{t-} vanishes.*

The intervals between adjacent transactions are determined by a continuous-time process, ξ_t , and a discrete-time process, χ_i , jointly. This assumption allows for dependence between trading times and the underlying driving forces of efficient prices, and thereby accommodates a large class of sampling schemes; see Jacod et al. (2017) for detailed discussions.

Next, we impose a discrete-time moving-average process for the microstructure noise to capture the potential temporal dependence in the transaction prices:²

Assumption 3. *The noise sequence $\{U_i : i \geq 0\}$ consists of random variables defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that $\{U_i : i \geq 0\}$ has an $MA(\infty)$ representation with mean 0:*

$$U_i = \eta_{t_i} \iota^{(n)} \theta^{(n)}(B) \varepsilon_i, \quad \text{with} \quad \theta^{(n)}(x) = 1 + \sum_{j=1}^{\infty} \theta_j^{(n)} x^j, \quad (2.3)$$

where B is the lag operator; $\varepsilon_i \stackrel{i.i.d.}{\sim} (0, 1)$, defined on $(\Omega, \mathcal{F}, \mathbb{P})$, is independent of $\mathcal{F}_\infty$ and $\{\chi_i : i \geq 1\}$, and has finite moments of all orders; $(\eta_t)_{t \geq 0}$ is an $(\mathcal{F}_t)$ -adapted nonnegative Itô-semimartingale that satisfies the same form of (2.2); and $\iota^{(n)}$ is a deterministic nonnegative number that characterizes the noise magnitude and satisfies $\iota^{(n)} \leq K$.

The noise again depends on a continuous-time process η_t and a discrete-time moving-average process U . The former introduces dependence between noise and the underlying efficient price, whereas the latter dictates the temporal dependence of the noise. Combining the two allows for heteroscedastic, temporally dependent, and endogenous noise. Importantly, Assumption 3 imposes zero correlation between X and U , without which identification would require a fully specified structural microstructure model along with additional observables (e.g., bid-ask prices). As Jacod et al. (2017) note, a colored noise model is not “consistent” with

²We use a superscript (n) on noise parameters to facilitate discussion of uniformity over different sequences of data-generating processes (DGPs) of noise indexed by n . n is a nonobservable mathematical abstraction. All limits are taken as $n \rightarrow \infty$. K is a generic n -independent positive constant that may vary from line to line.

the in-fill asymptotics. We adopt the same interpretation as theirs: the observation frequency is fixed by the available data and does not really go to infinity, so the in-fill asymptotics should be understood as the frequency being high enough for the asymptotic approximation to be adequate.³

Finally, we need some regularity assumption on the behavior of the spectral density of the noise process so that it is uniformly invertible and its long range dependence cannot be overly strong.

Assumption 4. *For each $n \geq 1$, the spectral density function of $\theta^{(n)}(\mathbf{B})\varepsilon$ satisfies for some fixed $\alpha > 3$,*

$$\inf_{\lambda} g(\lambda; \theta^{(n)}) \geq \frac{1}{K} \quad \text{and} \quad \left| \int_{-\pi}^{\pi} g(\lambda; \theta^{(n)}) e^{i\lambda j} d\lambda \right| \leq K j^{-\alpha}, \quad \forall j \geq 0.$$

Technically, our assumptions on the efficient price, observation times, and noise process are somewhat less general than those adopted in the leading nonparametric approaches of [Jacod et al. \(2017\)](#) and [Li and Linton \(2021\)](#). In particular, our observation time and noise specifications capture prominent empirical features such as random arrival rates and noise serial dependence by adopting explicit structures, i.e., a predictably modulated random walk sampling scheme and an $\text{MA}(\infty)$ noise process with i.i.d. innovation, and imposing restrictions on their elements such as trade arrival intensity and moving-average coefficients. In contrast, JLZ and ReMeDI impose restrictions directly through general moment conditions, which nests our assumptions. However, our framework also accommodates noise whose magnitude shrinks with sample size, which is empirically relevant and substantially increases challenges for noise estimation.

The parameters of interest in our study are autocovariances $\{\gamma_j^{(n)} : j \geq 0\}$ and autocorrelations $\{\rho_j^{(n)} : j \geq 1\}$ of the noise process, defined as

$$\gamma_j^{(n)} = (\iota^{(n)})^2 \frac{\int_0^T \eta_s^2 \xi_s^{-1} ds}{\int_0^T \xi_s^{-1} ds} \times \kappa_j^{(n)}, \quad j \geq 0, \quad \text{and} \quad \rho_j^{(n)} = \kappa_j^{(n)} / \kappa_0^{(n)}, \quad j \geq 1, \quad (2.4)$$

where $\kappa_j^{(n)}$ is given by

$$\kappa_j^{(n)} = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(\lambda; \theta^{(n)}) e^{i\lambda j} d\lambda, \quad j \geq 0, \quad (2.5)$$

³Given that our asymptotic analysis explicitly models the moving-average parameters as n -dependent, this interpretation shares the spirit of [Bekker \(1994\)](#): “the parameter sequence is designed to make the asymptotic distribution fit the finite sample distribution better. It is completely irrelevant whether or not further sampling will lead to samples conforming to this sequence or not.”

and $g(\lambda; \theta^{(n)}) = |\theta^{(n)}(e^{i\lambda})|^2$ is the spectral density of $\theta^{(n)}(B)\varepsilon$. While the autocovariances depend on the underlying processes η_t and ξ_t that drive the sampling times and noise magnitudes, respectively, the autocorrelations are entirely determined by the set of parameters $\{\theta_j^{(n)} : j = 1, 2, \dots, \infty\}$ in the MA process.

3 Main Results

In what follows, we will discuss the constructed estimators and their asymptotic properties.

3.1 Quasi-likelihood Estimation

To estimate noise autocovariances, we adopt a quasi-likelihood approach based on a misspecified model for observed returns. Specifically, we pretend that the efficient price X (in logarithm) is a Brownian motion with constant volatility but no drift, and that the noise U follows a Gaussian MA(q) model with the order q to be determined:

$$dX_t = \sigma dW_t; \quad U_i = \iota \theta(B)\varepsilon_i, \quad \text{with} \quad \theta(x) = 1 + \sum_{j=1}^q \theta_j x^j, \quad \text{and} \quad \varepsilon_i \sim \mathcal{N}(0, 1).$$

Under this model, the observed log-return vector $Y_n = (Y_{n,1}, Y_{n,2}, \dots, Y_{n,n_T})^\top$,

$$Y_{n,i} = X_{t_i} - X_{t_{i-1}} + U_i - U_{i-1}, \quad 1 \leq i \leq n_T. \quad (3.6)$$

follows a reduced-form Gaussian MA($q + 1$) model, whose $n_T \times n_T$ covariance matrix Σ_n is given by

$$\Sigma_n(\sigma^2, \iota^2, \theta) = \sigma^2 \Delta_n \mathbb{I}_n + \sum_{h=0}^{n_T-1} (2\gamma_h - \gamma_{h+1} - \gamma_{h-1}) \mathbb{G}_n^h, \quad (3.7)$$

where $(\mathbb{I}_n)_{ij} = \delta_{i,j}$, $(\mathbb{G}_n^h)_{ij} = \delta_{h,|i-j|}$, and γ_h is the h -th order autocovariance of U :

$$\gamma_h = \frac{\iota^2}{2\pi} \int_{-\pi}^{\pi} g(\lambda; \theta) e^{i\lambda h} d\lambda, \quad \text{where} \quad g(\lambda; \theta) = \left| \theta(e^{i\lambda}) \right|^2. \quad (3.8)$$

Since we are interested in the noise autocovariances, we reparameterize the likelihood function in terms of (σ^2, γ) :

$$L_n(\sigma^2, \gamma) = -\frac{1}{2} \log \det(\Sigma_n(\sigma^2, \gamma)) - \frac{1}{2} \text{tr}(\Sigma_n(\sigma^2, \gamma)^{-1} Y_n Y_n^\top), \quad (3.9)$$

where $\Sigma_n(\sigma^2, \gamma) := \Sigma_n(\sigma^2, \iota^2, \theta)$ and γ is the $(q + 1)$ -dimensional vector of the noise autocovariances. Notably, the parameterization in terms of (σ^2, γ) is made possible by the choice of MA(q) noise specification—an autoregressive or ARMA specification would always lead to an

autocovariance vector with infinite nonzero components. Moreover, the $\text{MA}(q)$ noise leads to that the observed returns follow an $\text{MA}(q+1)$ model, which greatly facilitates the estimator implementation, explained in Section 3.2.^{4,5}

Beyond the parameter-constraint observation above, ARMA likelihood is prone to *near-cancellation of common AR and MA roots*—a phenomenon often called the “pile-up” problem in the time-series literature. Because the autoregressive and moving-average polynomials of the reduced-form ARMA share a common structure inherited from the noise dynamics, as the noise magnitude shrinks toward zero their roots converge, leading to exact cancellation in the limit. Away from the limit, near-cancellation renders the individual AR and MA coefficients poorly identified even though the autocovariances themselves remain well defined. In our high-frequency setting, the issue is particularly acute because the noise-to-signal ratio varies across stocks and sampling frequencies, so that near-cancellation regimes are empirically relevant rather than merely a theoretical curiosity. By adopting the MA specification, we sidestep this difficulty: the quasi-likelihood is parameterized directly through the autocovariances γ , which remain identifiable regardless of the noise magnitude.

We define $(\hat{\sigma}_n^2(q), \hat{\gamma}_n(q))$ as the maximizer of $L_n(\sigma^2, \gamma)$:

$$(\hat{\sigma}_n^2(q), \hat{\gamma}_n(q)) = \arg \max_{(\sigma^2, \gamma) \in \Pi_n(q)} L_n(\sigma^2, \gamma), \quad (3.10)$$

where, following Da and Xiu (2021), the parameter space $\Pi_n(q)$ is defined as

$$\Pi_n(q) = \left\{ (\sigma^2, \gamma) \in \mathbb{R}^{q+2} : \inf_{\lambda} f(\lambda; \sigma^2, \gamma, \Delta_n) \geq \frac{\Delta_n}{K}, \quad \sigma^2 + |\gamma_0| + \frac{\sum_{j=1}^{\infty} j^2 |\gamma_j|}{\inf_{\lambda} |\sigma^2 \Delta_n + f(\lambda; \gamma)|} \leq K \right\}. \quad (3.11)$$

Here $f(\lambda; \sigma^2, \gamma, \Delta_n)$ stands for the spectral density of the observed log-return Y_n under the quasi-model: $f(\lambda; \sigma^2, \gamma, \Delta_n) = \sigma^2 \Delta_n + (2 - 2 \cos \lambda) f(\lambda; \gamma)$, with $f(\lambda; \gamma) = \sum_{j=-\infty}^{\infty} \gamma_{|j|} e^{ij\lambda}$.

To determine an appropriate order q , we use information criteria, such as BIC, which in our setting can be written as

$$\text{BIC}_n(q) = q \log n_T - 2 \max_{(\sigma^2, \gamma) \in \Pi_n(q)} L_n(\sigma^2, \gamma).$$

⁴In contrast, an $\text{AR}(p)$ noise would imply that the observed returns follow $\text{ARMA}(p, p)$, in which the AR and MA coefficients are intertwined and jointly satisfy $p-1$ nonlinear constraints.

⁵Although the microstructure literature often includes an autoregressive component with economic interpretations (Ho and Stoll, 1981, Hasbrouck and Ho, 1987, Hendershott and Menkveld, 2014), our method with the MA specification, as we show later, can consistently estimate the noise autocovariances under $\text{MA}(\infty)$ noise, which nests $\text{ARMA}(p, q)$ as special cases. Hence, the resulting autocovariance estimates can serve as inputs for structural modeling, as in Li et al. (2020).

Our choice of order q will be based on⁶

$$\hat{q}_n = \arg \min_{q \leq n_T^{1/3}} \text{BIC}_n(q). \quad (3.12)$$

We can define a similar criterion based on AIC, by replacing $q \log n_T$ above by $2q$. Hannan (1980) shows that using BIC results in consistent order selection for ARMA models. We demonstrate that a similar result with BIC also holds in our setting. We will therefore focus on BIC in the following discussion.

Our approach shares the same QMLE framework as Da and Xiu (2021), which employs the AIC-based version to establish uniformly valid inference for integrated volatility. In other words, maximizing a single quasi-likelihood simultaneously yields both volatility and noise autocovariance estimates.⁷ Notably, unlike the nonparametric approaches by Jacod et al. (2017) and Li and Linton (2021), which estimate both second-moment and higher-order moments in a unified way, the likelihood-based framework is, by construction, suited for second-moment estimation, but intrinsically more difficult to extend to higher-order noise moments. As our approach requires optimization over both model parameters and order, it is computationally more demanding to implement, which we now turn to.

3.2 Implementation

We implement the exact likelihood via an auxiliary reduced-form MA($q + 1$) model of the observed noisy returns:

$$Y_{n,i} = \phi(B)\epsilon_i, \quad \text{with} \quad \phi(x) = 1 + \sum_{j=1}^{q+1} \phi_j x^j, \quad 1 \leq i \leq n, \quad \epsilon \sim \mathcal{N}(0, \chi^2). \quad (3.13)$$

Algorithm 1. *Our algorithm starts as follows:*

1. *Select the optimal order, $\hat{q}_n$, of the MA process (3.13) for Y_n using BIC, defined by (3.12) but rewritten equivalently in terms of χ^2 and ϕ .*
2. *Obtain exact quasi-likelihood estimates of $\hat{\chi}^2$ and $\hat{\phi}_j$ for $1 \leq j \leq \hat{q}_n + 1$, using the*

⁶The theoretical restriction $q \leq n_T^{1/3}$ is imposed solely for technical reasons and is not enforced in practice; moreover, $n_T^{1/3}$ can be replaced by $c n_T^{1/3}$ for any constant $c > 0$ without affecting the theoretical results, so the restriction is considerably less stringent than it may appear at first glance. In both simulations and empirical analysis, we select the order over a fixed and common candidate set $q \in \{0, 1, \dots, 12\}$ across all stock-days and all years, and selected orders rarely approach the upper bound, alleviating the concern that less liquid stocks might be mechanically assigned lower maximal orders.

⁷Our noise autocovariance estimator, as a function of the return data, necessarily differs from the volatility estimator of Da and Xiu (2021), just as nonparametric estimators for noise autocovariances (e.g., JLZ and ReMeDI) and those for volatility (e.g., pre-averaging and realized kernel methods) are constructed differently.

state-space representation of (3.13) and Kalman filtering,

3. Construct volatility and noise autocovariance estimators using the above estimates:

$$\begin{aligned}\hat{\gamma}_n(\hat{q}_n)_j &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\hat{\chi}^2 e^{ij\lambda}}{|1 - e^{i\lambda}|^2} \left(\left| 1 + \sum_{l=1}^{\hat{q}_n+1} \hat{\phi}_l e^{il\lambda} \right|^2 - \left(1 + \sum_{l=1}^{\hat{q}_n+1} \hat{\phi}_l \right)^2 \right) d\lambda, \quad 0 \leq j \leq \hat{q}_n, \\ \hat{\sigma}_n^2(\hat{q}_n) &= \Delta_n^{-1} \hat{\chi}^2 \left(1 + \sum_{j=1}^{\hat{q}_n+1} \hat{\phi}_j \right)^2,\end{aligned}$$

which are obtained by comparing different parameterizations of the return autocovariances.

4. Solve $\hat{q}_n + 1$ nonlinear equations for $\hat{q}_n + 1$ model parameters $(\hat{\iota}_n^2(\hat{q}_n), \hat{\theta}_n(\hat{q}_n))$ from $\hat{\gamma}_n(\hat{q}_n)$ obtained in Step 3:

$$\hat{\gamma}_n(\hat{q}_n)_j = \hat{\iota}_n^2(\hat{q}_n) \sum_{l=0}^{\hat{q}_n-j} \hat{\theta}_n(\hat{q}_n)_l \hat{\theta}_n(\hat{q}_n)_{l+j}, \quad 0 \leq j \leq \hat{q}_n. \quad (3.14)$$

A Newton-Raphson algorithm that converges quadratically is available from [Wilson \(1969\)](#).

Effectively, Step 4 is to find $\hat{q}_n + 1$ model parameters of the $\text{MA}(\hat{q}_n)$ noise process from up-to- $\hat{q}_n$ th-order autocovariances $\hat{\gamma}_n(\hat{q}_n)_j$, $0 \leq j \leq \hat{q}_n$. This practice is common in the classic time-series analysis. For instance, [Box et al. \(2007\)](#) recommend using this algorithm to find initial values based on autocovariances for the maximum likelihood estimation of an MA model.

Step 3 is sufficient for volatility and noise autocovariance estimation, and it is rather simple to implement. If one is further interested in (ι^2, θ) , a unique solution $(\hat{\iota}_n^2(q), \hat{\theta}_n(q))$ exists from Step 4, with probability approaching 1 when noise is sufficiently large relative to the sample size.

When noise is small, however, (3.14) may have no solution such that $\hat{\iota}_n^2(q)$ is positive and $\hat{\theta}_n(q)$ is real. The mechanism is the standard one: as $(\iota^{(n)})^2$ approaches zero, the structural decomposition $\gamma \mapsto (\iota^2, \theta)$ degenerates, and the structural pair is weakly identified in the drifting-strength sense of [Andrews and Cheng \(2012\)](#). Here, ι^2 plays the role of the identification-strength parameter for θ : $(\iota^{(n)})^2$ bounded away from zero corresponds to strong identification, $(\iota^{(n)})^2$ converging to zero sufficiently fast to weak identification, and $(\iota^{(n)})^2 = 0$ to lack of identification. Under weak identification, the distribution of $(\hat{\iota}_n^2(q), \hat{\theta}_n(q))$ becomes dispersed and puts non-trivial mass outside the admissible set. The DGP class we entertain

spans this entire continuum, which is why $\iota^{(n)}$ -driven asymptotics dominate the results in Sections 3.4–3.5. The inadmissibility frequencies in Section 5.2 are the empirical manifestation of this weak identification phenomenon.

A close econometric parallel is the weak-instrument literature initiated by Stock and Wright (2000), where the strength of the first-stage relationship plays the role our $(\iota^{(n)})^2$ plays here and where standard CLT-based inference loses its uniform validity once that strength is allowed to shrink with the sample size. More generally, our $\theta^{(n)}$ —recovered from $\gamma^{(n)}$ as a ratio of jointly vanishing components in the limit $(\iota^{(n)})^2 \rightarrow 0$ —is a weakly regular parameter in the semiparametric framework of Kaji (2021). Unlike the focus on a constrained estimator in Andrews (1999), we deliberately adopt the unconstrained QMLE rather than imposing a constraint on parameters that (3.14) must have an admissible solution. An inadmissible estimate preserves the identification signal: it indicates that the noise magnitude is too small to support a meaningful moving-average structure.

3.3 Model Selection Consistency

We now discuss the asymptotic properties of the proposed estimators. The asymptotic analysis here is more involved than the classic time-series analysis, because the DGP of observed returns is misspecified. Moreover, the asymptotic design is in-fill, so that not only the dimensions, but also the entries of the covariance matrix Σ_n in the quasi-likelihood, depend on the sample size n_T ; see (3.7). Consequently, prior results from classic time-series studies are not applicable. Even worse, the quasi-likelihood estimator does not have an explicit form.

We start with a model selection consistency result based on BIC, which allows us to conduct pointwise inference on autocovariance parameters. We thereby impose a finite-order moving-average model for the DGP of noise.

Theorem 1. *Suppose Assumptions 1 - 4 hold. We further assume a non-vanishing noise process with an exact $MA(q^*)$ structure, i.e., $\iota^{(n)} \geq K^{-1}$ and $\theta^{(n)} \in \mathbb{R}^{q^*}$ for all $n \geq 1$ and $\sqrt{n}(\log n)^{-1}|\theta_{q^*}^{(n)}| \rightarrow \infty$, for some fixed $q^* \geq 0$. Then it holds that*

$$\lim_{n \rightarrow \infty} \mathbb{P}(\hat{q}_n = q^*) = 1.$$

As the sample size increases, the likelihood is asymptotically dominated by that of the noise component. Therefore, the same intuition from the classic time-series result applies here. The likelihood estimator effectively minimizes the Kullback-Leibler divergence, but only when the selected order is no smaller than the truth. Moreover, the BIC imposes a penalty just large enough to rule out orders that are greater than the truth asymptotically. The combination of these two results leads to the desired consistency in model selection.

3.4 Asymptotic Distribution of Noise Autocovariances and Autocorrelations

Recall that in (3.10) and Step 3 of Algorithm 1, we defined and implemented estimators of noise autocovariances. We now propose estimators of autocorrelations, denoted by $\hat{\rho}_n(\hat{q}_n)$, which are defined as follows.

If (3.14) has a solution such that $\hat{\iota}_n^2(\hat{q}_n)$ is positive and $\hat{\theta}_n(\hat{q}_n)$ is real, we set⁸

$$\hat{\rho}_n(\hat{q}_n)_j = \frac{\hat{\gamma}_n(\hat{q}_n)_j}{\hat{\gamma}_n(\hat{q}_n)_0}, \quad j \geq 1.$$

Otherwise, we set

$$\hat{\rho}_n(\hat{q}_n) = 0.$$

In light of their definitions, we can regard these estimators as “hard-thresholding” estimators, in that higher-order autocovariance and autocorrelation estimates are truncated to zero beyond the selected order $\hat{q}_n$. Not requiring (3.14) have an admissible solution can reveal when the noise magnitude is too small. In those cases, our choice for the autocorrelation estimates is to set all of them to zero, in the absence of a better *a priori* alternative.

Next, we prove the pointwise central limit theorem for estimators of noise autocovariances in the finite-order moving average model. The corresponding result for autocorrelations follows straightforwardly.

Theorem 2. *Suppose Assumptions 1 - 4 hold. We further assume $\iota^{(n)} \geq K^{-1}$ and $\theta^{(n)} \in \mathbb{R}^{q^*}$ for all $n \geq 1$ and some fixed $q^* \geq 0$. Let $\gamma^{(n)}$ be the $(q^* + 1)$ -dimensional vector of up-to- q^* th-order autocovariances of U , whose components are defined in equation (2.4).⁹ Assume there exists a $(q^* + 1)$ -dimensional vector γ^* such that $\gamma^{(n)} - \gamma^* = o_P(1)$. Then it holds that¹⁰*

$$n^{1/2}(\hat{\gamma}_n(q^*) - \gamma^{(n)}) \xrightarrow{\mathcal{L}_s - \mathcal{F}_\infty} \mathcal{MN}(0_{q^*+1}, \text{AVAR}_1),$$

where

$$\text{AVAR}_1 = \left(2W(\gamma^*)^{-1} + \gamma^* \gamma^{*\top} (\text{cum}_4(\varepsilon) + m_2 - 1) \right) \frac{T \int_0^T \eta_s^4 \xi_s^{-1} ds}{\left(\int_0^T \eta_s^2 \xi_s^{-1} ds \right)^2} - \frac{T(m_2 - 1) \gamma^* \gamma^{*\top}}{\int_0^T \xi_s^{-1} ds},$$

⁸Estimates of autocovariances and autocorrelations are, of course, zero beyond the $\hat{q}_n$ -th lag.

⁹Recall that the vectors $\gamma^{(n)}$ and γ^* are indexed from 0. We refer to $\gamma^{(n)}$ here as a $(q^* + 1)$ -dimensional vector simply because $\gamma_j^{(n)} = 0$ for all $j > q^*$, since $\theta^{(n)} \in \mathbb{R}^{q^*}$. For this reason, in most of our discussions, we do not distinguish it from an ∞ -dimensional vector. The same applies to other ∞ -dimensional vectors.

¹⁰Here and throughout the appendix, $\xrightarrow{\mathcal{L}_s - \mathcal{F}_\infty}$ stands for stable convergence in law with respect to $\mathcal{F}_\infty$.

$\text{cum}_4(\varepsilon)$ denotes the fourth cumulant of ε ,

$$W(\gamma) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{\partial \log f(\lambda; \gamma)}{\partial \gamma} \right)^\top \frac{\partial \log f(\lambda; \gamma)}{\partial \gamma} d\lambda.$$

This result shows that our estimator achieves the best convergence rate possible— $n^{1/2}$. In addition, the nonparametric estimation of volatility, which serves as a nuisance parameter here, does not influence the asymptotic variance of noise parameters. In fact, the asymptotic variance has the same form as in the classic time-series analysis—e.g., [Brockwell and Davis \(1991\)](#)—barring η and ξ terms, which are irrelevant in discrete time settings, as if the observed prices were purely made of noise. This further suggests that when ε indeed follows a Gaussian distribution and η and ξ are constants, our estimator achieves the optimal efficiency. Semiparametric efficiency results have played a central role in the high-frequency volatility inference literature (e.g., [Jacod \(1994\)](#); [Jacod and Rosenbaum \(2013\)](#)); Theorem 2 extends this line of inquiry from volatility to noise autocovariances.

The CLT in Theorem 2 remains valid for the first $q^* + 1$ entries of the plug-in estimator $\hat{\gamma}_n(\hat{q}_n)$ conditional on correct model selection, which Theorem 1 guarantees with probability approaching 1. However, Theorem 1 requires the strong separation condition $\sqrt{n}(\log n)^{-1}|\theta_{q^*}^{(n)}| \rightarrow \infty$ that does not hold uniformly over a broader DGP class; we will discuss this issue further in the next subsection.

The next corollary presents the central limit result for autocorrelations:

Corollary 1. *Suppose the same assumptions as those in Theorem 2 hold. Let $\rho^{(n)}$ be the q^* vector of up-to- q^* -th-order autocorrelations of U whose components are defined in equation (2.4). Then it holds that*

$$n^{1/2}(\hat{\rho}_n(q^*) - \rho^{(n)}) \xrightarrow{\mathcal{L}_s - \mathcal{F}_\infty} \mathcal{MN}(0_{q^*}, \text{AVAR}_2),$$

where the ij th entry of the $q^* \times q^*$ matrix AVAR_2 is given by

$$(\text{AVAR}_2)_{ij} = \frac{\gamma_i^* \gamma_j^*}{\gamma_0^{4^*}} (\text{AVAR}_1)_{11} + \frac{1}{\gamma_0^{2^*}} (\text{AVAR}_1)_{i+1, j+1} - \frac{\gamma_i^*}{\gamma_0^{3^*}} (\text{AVAR}_1)_{1, j+1} - \frac{\gamma_j^*}{\gamma_0^{3^*}} (\text{AVAR}_1)_{1, i+1}.$$

Next, we construct an estimator of the asymptotic variance, AVAR_1 , in Theorem 2, which naturally leads to an estimator for AVAR_2 in Corollary 1.

Proposition 1. *Suppose the same assumptions as those in Theorem 2 hold. Define*

$$\widehat{\text{AVAR}}_1 = \left(2W(\hat{\gamma}_n(\hat{q}_n))^{-1} + \hat{\gamma}_n(\hat{q}_n)\hat{\gamma}_n(\hat{q}_n)^\top \left(\widehat{\text{cum}}_4(\varepsilon) + \widehat{m} \right) \right) \hat{B}_n - \widehat{m} \hat{\gamma}_n(\hat{q}_n) \hat{\gamma}_n(\hat{q}_n)^\top,$$

where, with $k_n \sim \log n$,

$$\widehat{cum}_4(\varepsilon) = k_n \widehat{B}_n'^{-1} \widehat{B}_n'' - 2k_n - \left(\widehat{\gamma}_n(\widehat{q}_n)_0 - \widehat{\gamma}_n(\widehat{q}_n)_1 \right)^{-2} \frac{1}{\pi} \int_{-\pi}^{\pi} f(\lambda; \widehat{\gamma}_n(\widehat{q}_n))^2 (1 - \cos \lambda)^2 d\lambda,$$

$$\widehat{m} = \frac{1}{n_T} \sum_{i=k_n}^{n_T-2-k_n} \frac{\left(k_n(t_{i+1} - t_i) - t_{i+2+k_n} + t_{i+2} \right)^2}{(t_i - t_{i-k_n})^2 \vee (n^{-2} k_n^{7/4})}, \quad \widehat{B}_n = \left(\widehat{\gamma}_n(\widehat{q}_n)_0 - \widehat{\gamma}_n(\widehat{q}_n)_1 \right)^{-2} \widehat{B}_n',$$

$$\widehat{B}_n' = \frac{1}{4n_T k_n} \sum_{i=1}^{n_T-2k_n} Y_{n,i}^2 \sum_{j=k_n+1}^{2k_n} Y_{n,i+j}^2, \quad \text{and} \quad \widehat{B}_n'' = \frac{1}{4n_T k_n} \sum_{i=1+k_n}^{n_T-k_n} Y_{n,i}^2 \sum_{j=-k_n}^{k_n} Y_{n,i+j}^2.$$

Then, we have

$$\left\| \widehat{\text{AVAR}}_1 - n_T n^{-1} \text{AVAR}_1 \right\| = o_P(1).$$

With this proposition in place, we can build confidence intervals for noise autocovariances and autocorrelations using $n_T^{-1} \widehat{\text{AVAR}}_1$, which does not involve the unobservable scalar n in the CLT.

3.5 Uniform Consistency of Noise Autocovariances and Autocorrelations

The asymptotic distribution established here is pointwise, in the sense that it does not allow for model-selection mistakes. As pointed out by [Leeb and Pötscher \(2005\)](#), [Leeb and Pötscher \(2008\)](#), model selection errors matter in finite samples, to the extent that the prescribed asymptotic distribution could be seriously distorted. Moreover, in standard settings uniformly valid inference is generally not available.

To demonstrate this point in our context, we consider a simple setting in which the noise process has no autocorrelation beyond the first lag (so that we use BIC to select q from $\{0, 1\}$), and the noise magnitude $(\iota^{(n)})^2$ is constant (so that the optimal rate of the noise autocovariance estimator is $n^{1/2}$). The next proposition shows that no uniformly consistent estimator exists for the cumulative distribution function $G_n(x)$, where

$$G_n(x) = \mathbb{P}\left(n^{1/2} \left(\widehat{\gamma}_{n,1}(\widehat{q}_n \wedge 1) - \gamma_1^{(n)} \right) \leq x\right).$$

Proposition 2. *For each $x \in \mathbb{R}$, there exists a DGP sequence satisfying Assumptions 1–4 with $\iota^{(n)} \geq K$, and a single parameter $\theta^{(n)} \in \mathbb{R}$, such that*

$$\liminf_{n \rightarrow \infty} \inf_{\widehat{G}_n(x)} \mathbb{P}\left(\left| \widehat{G}_n(x) - G_n(x) \right| > \frac{1}{K}\right) > 0,$$

where the infimum extends over all estimators $\widehat{G}_n(x)$ of $G_n(x)$.

Given the challenge in establishing uniformly valid inference that Proposition 2 presents,

we set a less ambitious target and provide a uniform consistency result for $\hat{\gamma}_n(\hat{q}_n)$ and $\hat{\rho}_n(\hat{q}_n)$ with respect to $\gamma^{(n)}$ and $\rho^{(n)}$ under $\mathbb{L}^2$ -norm, where all vectors are regarded as ∞ -dimensional. Importantly, this result does not require uniqueness of the selected model or the resulting estimate—what matters is that any estimate produced by the selection-based QMLE procedure approaches the true parameter with bounded error. This result sheds light on the asymptotic behavior of these estimators when noise DGPs are allowed to vary within a larger class beyond $\text{MA}(q)$ with fixed q , allowing for a vanishing magnitude and a more flexible dependence structure. We characterize the class of noise DGPs we consider in the next assumption.

Assumption 5. Define $q_n^*(k) := \min q$, subject to $n\psi_n^4 \sum_{j=q}^{2q} |\tilde{\kappa}_j^{(n)}|^2 \leq kq \log n$, where $\psi_n := (1 + n^{-1/2}/\iota^{(n)})^{-1}$ and $\tilde{\kappa}_j^{(n)} := \sum_{i=0}^{\infty} (i+1)\psi_n^i (2\kappa_{j+1+i}^{(n)} - \kappa_{j+i+2}^{(n)} - \kappa_{j+i}^{(n)})$. We assume for any $0 < k < K$,

$$q_n^*(k) = o(n^{1/3}(\iota^{(n)} \vee n^{-1/2})^{4/9}), \quad \text{and} \quad n\psi_n^4 \sum_{j=q_n^*(k)}^{\infty} |\kappa_j^{(n)}|^2 = O(q_n^*(k) \log n).$$

Intuitively, $q_n^*(k)$ mimics the “oracle” order that BIC selects. Effectively, Assumption 5 requires that this order cannot be too large and imposes an upper bound on the approximation error induced by a selected MA model. Nevertheless, these conditions in Assumption 5 are not restrictive. They accommodate common processes such as $\text{MA}(\infty)$, with $|\kappa_j^{(n)}| \sim j^{-\alpha}$ for some $\alpha > 3 \vee \frac{3}{2+4\log \iota^{(n)}/\log n}$, as well as any finite order $\text{ARMA}(p, q)$ with an arbitrarily shrinking noise magnitude $\iota^{(n)} \lesssim 1$.

We are now ready to present the uniform consistency result for autocovariances and autocorrelations:

Theorem 3. For any sequence of DGPs that satisfies Assumptions 1 - 5, we have

$$\|\hat{\gamma}^{(n)}(\hat{q}_n) - \gamma^{(n)}\|^2 = O_P\left(n^{-1}(\iota^{(n)})^4(\hat{q}_n + 1)^2 \log n + n^{-3}(n^{1/2}\iota^{(n)} + 1)(\hat{q}_n + 1)^4 \log n\right).$$

If, in addition, we assume $\iota^{(n)} \geq Kn^{-2/3}(\log n)^{1/4}$, it holds that

$$\|\hat{\rho}_n(\hat{q}_n) - \rho^{(n)}\|^2 = O_P\left((\iota^{(n)})^{-4}\|\hat{\gamma}^{(n)}(\hat{q}_n) - \gamma^{(n)}\|^2\right).$$

In general, the autocorrelation $\rho^{(n)}$ is weakly identified in the presence of small noise. The last part of Theorem 3 rules out this scenario, restricting the class of DGPs such that the noise variance cannot be too small.

Whereas consistent estimation of autocorrelations requires a more restrictive class of DGPs,

Theorem 3 allows for arbitrarily small and vanishing noise for autocovariances. The case of small noise is highly relevant in practice, as shown from our empirical study below. Our result is complementary to the asymptotic theory developed by Jacod et al. (2017) and Li and Linton (2021), who focus on the case of non-vanishing noise.

We offer some remarks on the interpretation of the selected model and the resulting estimate. Under the broader DGP class considered here, the order selection consistency established in Theorem 1 no longer holds—indeed, the true order is not even unequivocally defined under $\text{MA}(\infty)$. Similar to the classic time series literature, e.g., Shibata (1980), the selected order balances approximation error against model complexity. The order chosen by BIC should not be interpreted as a literal statement about the persistence structure of the noise unless one is willing to invoke model-selection consistency, which requires strong separation conditions. Instead, we interpret the selected order as a truncation that balances approximation error against model complexity—a higher selected order does not automatically imply significant higher-lag dependence, as the magnitudes of the estimated coefficients are equally important. The resulting estimate $\hat{\gamma}^{(n)}(\hat{q}_n)$ corresponds to an infinite-dimensional parameter vector in which all entries beyond the selected order are set to zero. Because Theorem 3 provides an explicit bound on $\|\hat{\gamma}^{(n)}(\hat{q}_n) - \gamma^{(n)}\|$, this truncated representation remains informative: it closely reflects the true noise autocovariances up to the selected order and omits only components whose contribution is provably small.

3.6 Quadratic Representation

The QMLE estimator appears to have a rather different structure compared with alternative nonparametric estimators in the literature, e.g., realized kernels, which can be regarded as quadratic estimators. In this section, we propose an alternative but equivalent quadratic form of the QMLE, which sheds light on its connection with and distinction from these quadratic estimators. We do so for both volatility and noise autocovariance estimators.

Theorem 4. *Suppose the same assumptions as those in Theorem 2 hold and that $\gamma^{(n)} = \gamma^*$. The QMLE $(\hat{\sigma}_n^2(q^*), \hat{\gamma}_n(q^*))$ satisfies that for $0 \leq j \leq q^*$,*

$$\hat{\sigma}_n^2(q^*) = Y_n^\top \mathcal{W}_n(\hat{\sigma}_n^2(q^*), \hat{\gamma}_n(q^*); 1) Y_n, \quad \hat{\gamma}_n(q^*)_j = Y_n^\top \mathcal{W}_n(\hat{\sigma}_n^2(q^*), \hat{\gamma}_n(q^*); j+2) Y_n, \quad (3.15)$$

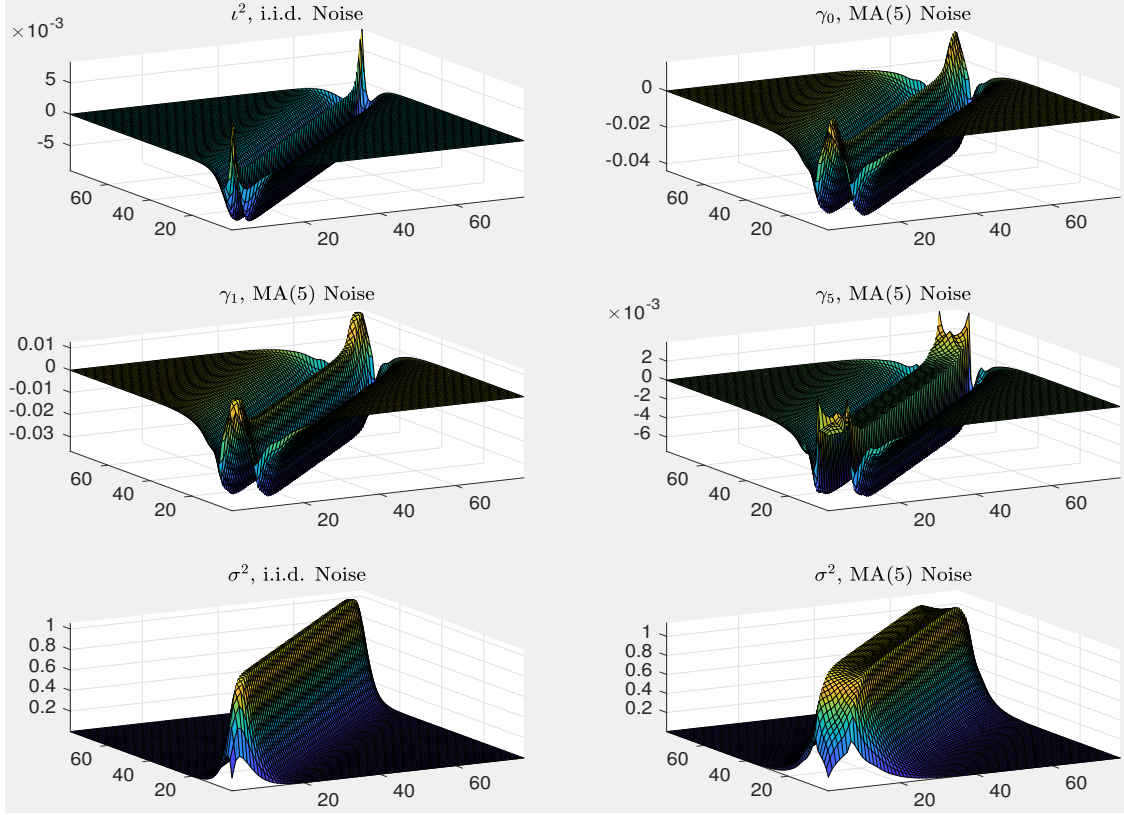
where the set of $n_T \times n_T$ weighting matrices $\mathcal{W}_n(\sigma^2, \gamma; l)$, $l = 1, 2, \dots, q^* + 2$, is defined by¹¹

$$\text{vec}(\mathcal{W}_n(\sigma^2, \gamma; l)) = \Sigma_n^{-1}(\sigma^2, \gamma) \frac{\partial \Sigma_n(\sigma^2, \gamma)}{\partial (\sigma^2, \gamma)} \Sigma_n^{-1}(\sigma^2, \gamma) \widetilde{W}_n^{-1}(\sigma^2, \gamma) (0_{l-1}, 1, 0_{q^*+2-l}),$$

¹¹ 0_d is the d -dimensional vector of 0s. All vectors are column vectors. We write (a, b, c) in place of $(a^\top, b^\top, c^\top)$ for simplicity.

with $\Sigma_n(\sigma^2, \gamma)$ given by (3.7), and the $(q^* + 2) \times (q^* + 2)$ matrix $\widetilde{W}_n(\sigma^2, \gamma)$ given by

$$\widetilde{W}_n(\sigma^2, \gamma)_{i,j} = \text{tr} \left(\Sigma_n^{-1}(\sigma^2, \gamma) \frac{\partial \Sigma_n(\sigma^2, \gamma)}{\partial (\sigma^2, \gamma)_i} \Sigma_n^{-1}(\sigma^2, \gamma) \frac{\partial \Sigma_n(\sigma^2, \gamma)}{\partial (\sigma^2, \gamma)_j} \right).$$



Note: This figure compares weighting matrices W s in the quadratic representations of the QMLE for σ^2 and ι^2 in the case of i.i.d. noise, as well as those matrices for σ^2 , γ_0 , γ_1 , and γ_5 in the case of MA(5) noise. We scale the volatility weighting matrices by T . In both cases, we fix $\sigma^* = 0.3$, $\iota^* = 0.005$, $\Delta = 5$ minutes, $T = 1$ day. The moving-average parameters of the MA(5) process are given by $\theta^* = (0.25, 0.2, 0.15, 0.1, 0.05)$.

Figure 1: Quadratic Representations of the Estimators

Theorem 4 shows that the QMLE can be written as an iterative quadratic estimator. It also suggests an alternative algorithm for estimation. With some initial values given, we can iteratively update parameters via equations given by (3.15) until convergence. Figure 1 plots these weighting matrices for both volatility and noise parameters, and compares them in the case of i.i.d. and MA(5) noises. The noise weighting matrices feature a “W” shape along the diagonal, and the magnitude of weighting matrices for autocovariance decays as their order increases. With respect to the volatility estimator, the bottom panel shows notable “flatness” at

the top of the volatility weighting matrix for the MA(5) model, which helps cancel out the impact of dependent noise. This pattern motivates us to investigate the connection between the QMLE and the flat-top realized kernel introduced by Varneskov (2016) to the high-frequency environment in the context of volatility estimation. We also provide an equivalent kernel for autocovariances.

Theorem 5. *Suppose the same assumptions as those in Theorem 2 hold. In addition, suppose $q \geq 0$ is fixed and $(\sigma^2, \gamma) \in \Pi_n(q)$ such that $K^{-1} \leq \inf_{\lambda} f(\lambda; \gamma) \leq \sup_{\lambda} f(\lambda; \gamma) \leq K$. Then for all $n^{1/2+\alpha} \leq i, j \leq n - n^{1/2+\alpha}$ with $0 < \alpha < \frac{1}{2}$, the weighting matrix $\mathcal{W}_n(\sigma^2, \gamma; l)$ satisfies for $l \geq 1$,*

$$\begin{aligned} (i) \quad & \mathcal{W}_n(\sigma^2, \gamma; 1)_{i,j} = T^{-1}k(H_n^{-1}|i-j|)(1+o(1)), \quad \mathcal{W}_n(\sigma^2, \gamma; l)_{i,j} = \lambda_l \tilde{k}(H_n^{-1}|i-j|) + O(1); \\ (ii) \quad & \sup_{|i-j| \leq q+1} \left| \mathcal{W}_n(\sigma^2, \gamma; 1)_{i,j} - \mathcal{W}_n(\sigma^2, \gamma; 1)_{i,i} \right| = O(\Delta_n^{3/2}); \\ (iii) \quad & \sup_{|i-j| \leq q+1} \left| \mathcal{W}_n(\sigma^2, \gamma; l)_{i,j} + \mathbb{1}_{\{l \leq |i-j|+1\}} \frac{|i-j|+2-l}{2n_T} \right| = O(\Delta_n^{3/2}), \end{aligned}$$

where the implied equivalent kernels are $k(x) = (1+x)e^{-x}$ and $\tilde{k}(x) = xe^{-x}$, the implied bandwidth is $H_n = \zeta \sigma^{-1} \Delta_n^{-1/2} + O(1)$ with $\zeta^2 = \sum_{|j| \leq q} \gamma_{|j|}$, and $\lambda_l = (2\sigma \zeta^3 \Delta_n^{1/2} n_T)^{-1} \sum_{r=1}^{q+1} (2 - \delta_{r,1}) W(\gamma)_{r,l-1}^{-1}$, with $W(\gamma)$ defined in Theorem 2.

Theorem 5 suggests that the bulk of the QMLE weighting matrices can be approximately written as that of a nonparametric kernel estimator with an implicit bandwidth. Despite this equivalence, it is more convenient to implement the QMLE using Algorithm 1 in Section 3.2, which does not require tuning parameters barring order selection, or any special adjustment to the border effect. Also note that this equivalence result is only established under the assumption that the spectral density of the noise (and hence its magnitude) is bounded from below, which rules out the case of small noise.

4 Monte Carlo Simulations

This section examines the finite-sample performance of the estimators across a range of simulation settings. Throughout we fix $T = 1$ day and have 1,000 Monte Carlo trials. We simulate X_t and σ_t^2 according to the same log-volatility model as in Li and Xiu (2016):

$$\begin{cases} dX_t &= (0.05 + 0.5\sigma_t^2)dt + \sigma_t dW_t + J^X dN_t, \\ \sigma_t^2 &= D_t \exp(-2.8 + 6F_t), \quad dF_t = -4F_t dt + 0.8d\tilde{W}_t + J^F dN_t - 0.02\lambda_N dt, \end{cases} \quad (4.16)$$

where $\mathbb{E}[dW_t d\widetilde{W}_t] = -0.8dt$, $J^X \sim \mathcal{N}(0, 0.02^2)$, $J^F \sim \mathcal{N}(0.02, 0.02^2)$, N_t is a Poisson process with intensity $\lambda_N = 25$, and D_t captures the diurnal effect:

$$D_t = 0.75 \exp(-10t/T) + 0.25 \exp(-10(1 - t/T)) + 0.8.$$

The arrival of trades follows an inhomogeneous Poisson process with rate $nT^{-1}\xi_t^{-1} = nT^{-1}(1 + \cos(2\pi t/T)/2)$, so that fewer trades arrive in the middle of the day. The average sampling interval will be set as either 5 seconds or 60 seconds. With respect to the noise, the innovation ε_i follows a Student's t -distribution with 7 degrees of freedom, and the heteroskedasticity process η_t follows

$$d\eta_t = 10 \times \left(\left(1 + 10^{-1} \cos(2\pi t/T) \right) - \eta_t \right) dt + 0.1 dW_t,$$

where W_t is the same Brownian motion that drives X . The noise magnitude and autocorrelation structure, governed by $\iota^{(n)}$ and $\theta^{(n)}$, will be specified in each subsection below. We also round the observed prices to the nearest cent: $\widetilde{X}_t = \log([100 \times \exp(X_t)]) - \log 100$, where $[\cdot]$ means rounding to the nearest integer.¹²

Throughout this section we work with a single fixed noise DGP per scenario, so the superscript (n) on noise parameters introduced in Section 2 adds no information; for brevity we write ι , θ , and γ in place of $\iota^{(n)}$, $\theta^{(n)}$, and $\gamma^{(n)}$ in the discussion below.

4.1 Verification of the Asymptotic Results

We first verify the asymptotic results developed in Section 3.4. We fix the average sampling frequency at every 5 seconds, and set the noise to follow an MA(5) model with $\iota = 2.5 \times 10^{-3}$ and $\theta^* = (0.25, 0.2, 0.15, 0.1, \theta_5)$. According to Theorem 1, the separation condition $\sqrt{n}(\log n)^{-1}|\theta_5| \rightarrow \infty$ is required for selection consistency and hence the validity of the asymptotic distribution of the plug-in estimator $\widehat{\gamma}(\widehat{q})$. We therefore consider two values of θ_5 : 0.05 and 0.20.¹³

Figure 2 provides the histograms of the standardized plug-in autocovariance estimates $\widehat{\gamma}_k(\widehat{q})$ for $k \in \{0, 2, 3, 4, 5\}$, using the corresponding asymptotic variances, along with the histograms of the BIC-selected order, for both values of θ_5 . When $\theta_5 = 0.20$, the histograms of the standardized estimates closely match the standard normal density, and BIC predominantly selects the correct order of 5. When $\theta_5 = 0.05$, the finite-sample distribution is substantially distorted from the standard normal for all orders of autocovariances considered, and BIC more

¹²Our theory does not allow for this type of rounding errors. We simulate this model to demonstrate that the rounding effect appears negligible.

¹³Under 5-second sampling, $n_T \simeq 4,680$ and $n_T^{-1/2} \log n_T \simeq 0.12$.

frequently underselects the order, consistent with the weaker separation.

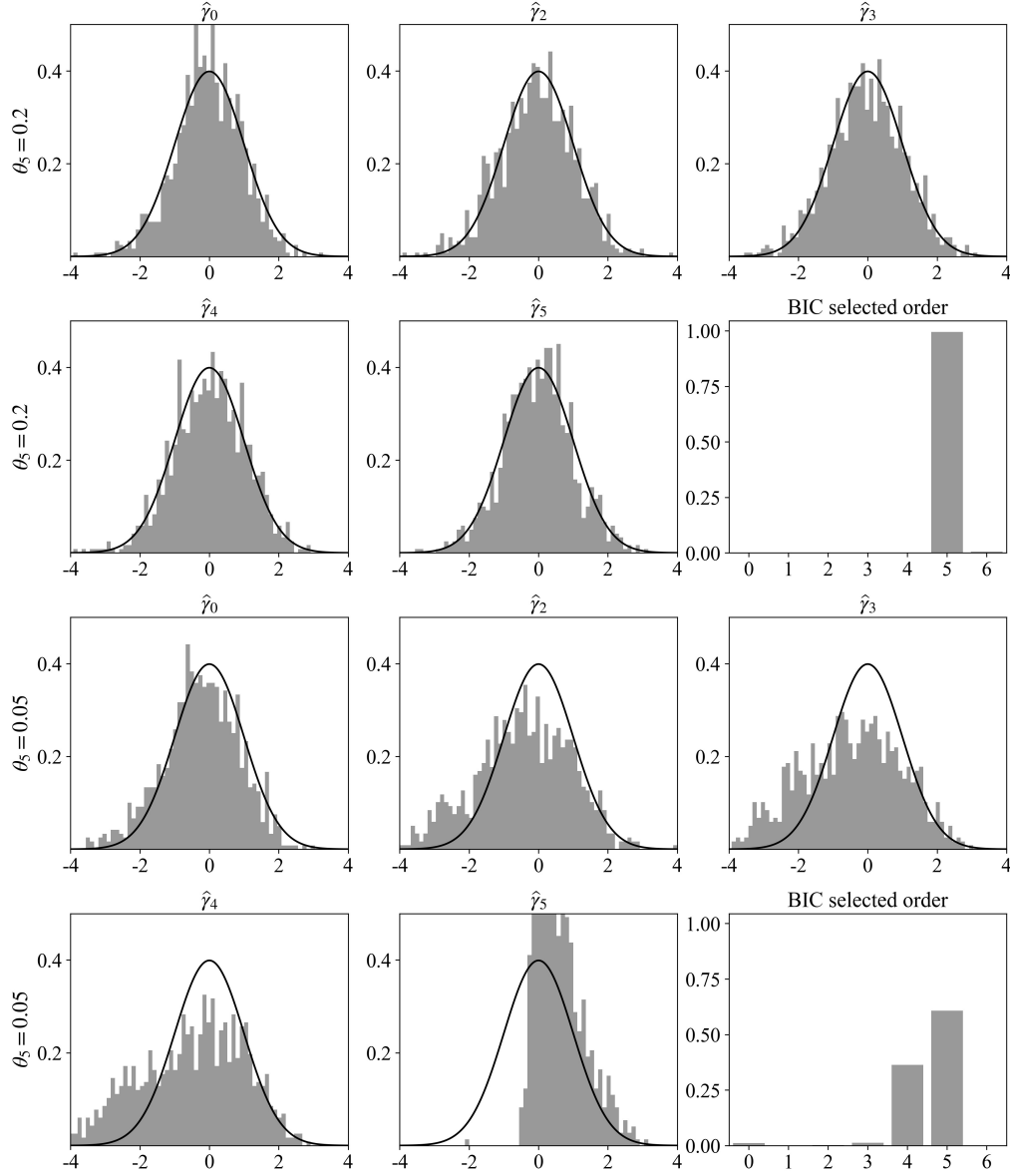


Figure 2: Histograms of Autocovariance Estimates and Selected Order

Note: The first two rows plot the histograms of the standardized plug-in autocovariance estimates $\hat{\gamma}_k(\hat{q})$ for $k \in \{0, 2, 3, 4, 5\}$, along with the density of the standard normal distribution, and the histogram of the BIC-selected order. The last two rows repeat the same layout. The first two rows use $\theta^* = (0.25, 0.2, 0.15, 0.1, 0.20)$ and the last two rows use $\theta^* = (0.25, 0.2, 0.15, 0.1, 0.05)$. In both cases, $\iota = 2.5 \times 10^{-3}$.

4.2 Performance of Noise Variance and Autocorrelation Estimation

We study the finite-sample performance of noise autocovariance estimation using our approach and the nonparametric methods of [Jacod et al. \(2017\)](#) (JLZ) and [Li and Linton \(2021\)](#) (ReMeDI). We examine the autocovariances γ_j at different lags through their two natural components: the variance γ_0 and the autocorrelations $\rho_j = \gamma_j/\gamma_0$ for $j \geq 1$, the latter being unit-free in $[-1, 1]$ and thus directly comparable across noise magnitudes. The noise follows an ARMA(1,1) process with MA coefficient 0.2 and AR coefficient $\psi \in \{-0.4, -0.6\}$, corresponding to moderate and high levels of serial dependence. To examine how noise signal strength and sampling frequency affect the estimation of these components, we consider six combinations of noise magnitude and sampling interval. The noise magnitude ι takes values 10^{-4} (small noise), 5×10^{-4} (medium noise), and 2.5×10^{-3} (large noise), and the average sampling interval is either 5 seconds or 60 seconds. Our estimator uses either BIC or AIC for order selection, whereas JLZ and ReMeDI each involve a tuning parameter.

JLZ propose to estimate autocovariances, γ , by approximating efficient prices using their local averages:

$$\hat{\gamma}_j^{\text{JLZ}} = \frac{1}{n_T} \sum_{i=0}^{n_T+1-j-4h_n} \left(\tilde{X}_{t_i} - \frac{1}{h_n} \sum_{l=0}^{h_n-1} \tilde{X}_{t_{i+j+l+h_n}} \right) \left(\tilde{X}_{t_{i+j}} - \frac{1}{h_n} \sum_{l=0}^{h_n-1} \tilde{X}_{t_{i+j+l+3h_n}} \right).$$

Here h_n is a sequence of integers satisfying $h_n \sim n^{-\eta}$ with $\frac{1}{2v+1} < \eta < \frac{1}{2}$, where v is the ρ -mixing exponent of ε . It determines the local window size used to estimate realization of the noise. Their paper selects $h_n = 6$ in simulations with 1-second data. According to their criterion, when data are sampled at 5-second or 1-minute frequency, h_n must be an even smaller integer in a finite sample, so we report the autocorrelation estimates for $h_n = 2, 4$, and 6.

ReMeDI proposes an alternative construction that takes the differences of log prices over longer horizons to dampen the impact of efficient prices:

$$\hat{\gamma}_j^{\text{ReMeDI}} = -\frac{1}{n_T} \sum_{i=1}^{n_T-2k_n-j} (\tilde{X}_{i+k_n} - \tilde{X}_i)(\tilde{X}_{i+j+2k_n} - \tilde{X}_{i+j+k_n}),$$

where k_n is a tuning parameter that satisfies: $k_n \rightarrow \infty$, $k_n n^{-\eta} \rightarrow 0$, for $\frac{1}{2v} < \eta < \frac{1}{3}$. We select $k_n = k'_n \log n$, where $k'_n = 0.25, 0.5$, and 1 in simulations.

Given the estimated autocovariances, autocorrelations are estimated as $\hat{\rho}_j^{\text{JLZ}} = \hat{\gamma}_j^{\text{JLZ}}/\hat{\gamma}_0^{\text{JLZ}}$ and $\hat{\rho}_j^{\text{ReMeDI}} = \hat{\gamma}_j^{\text{ReMeDI}}/\hat{\gamma}_0^{\text{ReMeDI}}$. We find it necessary to winsorize the estimated autocorrelations for both nonparametric estimators, when the noise magnitude is small, to ensure that

their estimates are within the natural bound $[-1, 1]$.¹⁴

4.2.1 Performance Across Noise Magnitudes and Sampling Frequencies

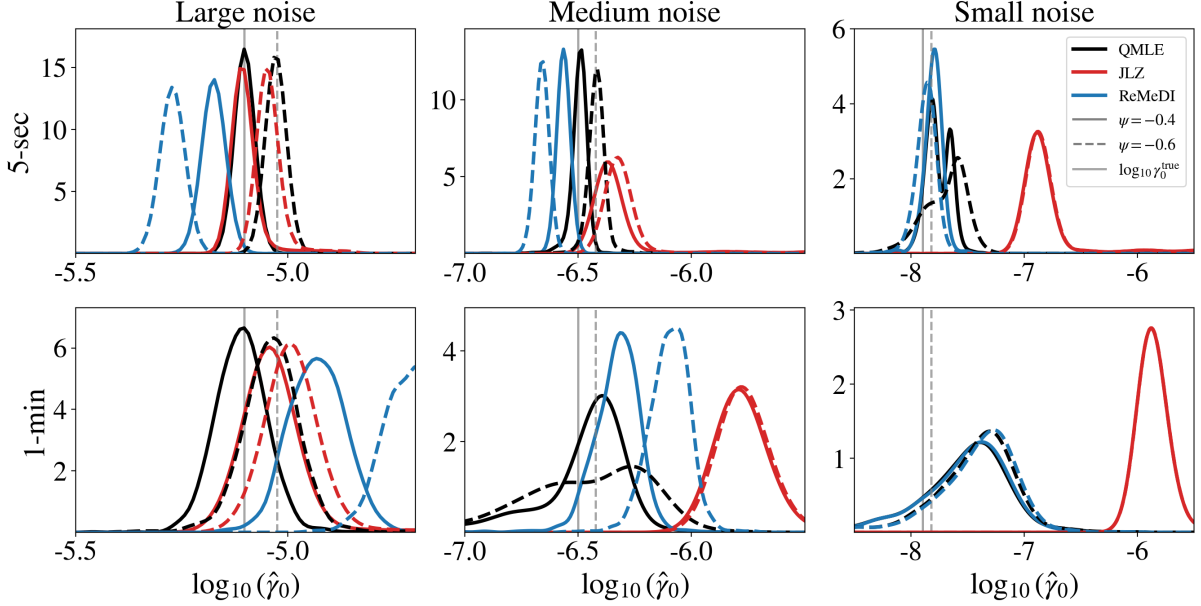


Figure 3: Distributions of Noise Variance Estimates

Note: Kernel densities of $\log_{10}(\hat{\gamma}_0)$ across 1,000 Monte Carlo sample paths. Columns correspond to the three noise levels (Large, Medium, Small); rows correspond to the two sampling frequencies (5-sec and 1-min). Within each panel, densities are shown for QMLE (black), JLZ (red), and ReMeDI (blue), with solid lines for $\psi = -0.4$ and dashed lines for $\psi = -0.6$. The two vertical gray lines (solid for $\psi = -0.4$, dashed for $\psi = -0.6$) mark the path-mean of the true noise variance $\gamma_0^{(n)}$ in equation (2.4), which equals $\iota^2 \mathbb{E} \left[\eta^2 \xi^{-1}_T / \xi^{-1}_T \right] \kappa_0(\psi)$ and depends on ψ through $\kappa_0(\psi) = (1 + 2\psi\theta_{\text{MA}} + \theta_{\text{MA}}^2)/(1 - \psi^2)$. JLZ uses $h_n = 2$; ReMeDI uses $k'_n = 0.25$.

We begin with the noise variance γ_0 , which underpins the conversion of estimated autocovariances into autocorrelations. Figure 3 reports kernel densities of $\log_{10}(\hat{\gamma}_0)$ for BIC-QMLE, JLZ, and ReMeDI across the six combinations of three noise magnitudes (large, medium, small) and two sampling frequencies (5-sec, 1-min), separately for $\psi = -0.4$ (solid) and $\psi = -0.6$ (dashed); within each panel the two vertical gray lines mark the path-mean estimand under the two values of ψ . Only positive estimates are plotted; the non-positive rate is below 0.5% in every cell except the 1-min small-noise cell, where it is about 21% for QMLE and 19% for ReMeDI (JLZ produces no non-positive estimates). For JLZ and ReMeDI, we adopt the tuning values that minimize the average RMSE of $\log_{10} \hat{\gamma}_0$ across the twelve combinations of noise magnitude, sampling frequency, and serial dependence— $h_n = 2$ for JLZ and

¹⁴If a correlation estimate exceeds 1 (resp. -1), we reset it to be 1 (resp. -1).

$k'_n = 0.25$ for ReMeDI—and report all bias and RMSE numbers below at these values.

QMLE is essentially unbiased relative to the estimand whenever the signal is non-trivial: mean deviations stay within 2% in all 5-sec cells with at least medium noise, within 3% in the 1-min large-noise cell, and within 13% at 1-min medium noise. They become substantial only in the small-noise cells (about +42% at 5-sec and over +230% at 1-min), where the signal is too weak to identify $\gamma_0^{(n)}$ accurately. ReMeDI’s bias depends sharply on ψ —e.g., −15% ($\psi = -0.4$) vs. −43% ($\psi = -0.6$) in 5-sec large noise, with a sign flip at 1-min sampling. JLZ’s bias varies little with ψ but grows rapidly as noise weakens, from +1% in 5-sec large noise to over 24× in 5-sec small noise.

In RMSE terms ($\log_{10} \hat{\gamma}_0$, averaged across ψ), QMLE attains the lowest RMSE in five of the six cells. Its advantage over JLZ is roughly threefold even at the strongest signal (0.014 vs. 0.039 in 5-sec large noise) and widens to an order of magnitude as the signal weakens (0.288 vs. 0.819 at 1-min medium noise). Its advantage over ReMeDI is largest at strong signal—an order of magnitude in 5-sec large noise (0.014 vs. 0.161), about 0.11 in 5-sec medium noise, and 0.18 in 1-min large noise—and the two methods are tied in the remaining 1-min cells. ReMeDI attains a lower RMSE only in the 5-sec small-noise cell (0.171 vs. 0.204).

We next examine performance of the autocorrelation estimation. Table 1 reports the bias and RMSE of the autocorrelation estimates $\hat{\rho}_1, \hat{\rho}_3, \hat{\rho}_5$ across the six combinations of noise magnitude (large, medium, small) and average sampling interval (5-sec, 1-min), with the two levels of serial dependence ($\psi = -0.4$ and $\psi = -0.6$) reported side by side. For JLZ and ReMeDI, the reported bias and RMSE are those of the estimates using the tuning value that minimizes RMSE for each combination of noise magnitude, sampling frequency, and order of autocorrelation. The smallest tuning dominates ex post for both JLZ and ReMeDI: across the 36 combinations, JLZ’s best tuning is always $h_n = 2$, and ReMeDI’s is $k'_n = 0.25$ in 28 combinations, $k'_n = 0.5$ in 6, and $k'_n = 1$ in 2—the eight exceptions all occurring at 5-sec sampling under medium or large noise.

In the large-noise, 5-sec scenario, all three estimators perform reasonably well, with small bias and RMSE. QMLE typically exhibits the smallest RMSE, with AIC-QMLE slightly outperforming BIC-QMLE. ReMeDI is the stronger of the two nonparametric estimators and is broadly competitive with QMLE in this scenario, with bias and RMSE noticeably smaller than JLZ’s at every lag.

As signal strength weakens—through smaller noise magnitude, lower sampling frequency, or both—finite-sample estimation error naturally rises across methods, with BIC-QMLE delivering the most favorable performance overall. An important reason behind the advantageous performance of BIC-QMLE is the following: with weaker signal, it more frequently produces

Table 1: Finite-Sample Performance of Noise Autocorrelation Estimators

		Moderate serial dependence				High serial dependence			
		QMLE		JLZ	ReMeDI	QMLE		JLZ	ReMeDI
		BIC	AIC			BIC	AIC		
Large noise, 5-sec sampling									
ρ_1	BIAS	0.007	0.001	0.004	0.005	0.008	0.002	−0.047	0.005
	RMSE	0.019	0.019	0.083	0.037	0.025	0.019	0.104	0.035
ρ_3	BIAS	0.025	0.001	0.003	0.000	0.023	0.004	−0.038	0.002
	RMSE	0.032	0.019	0.072	0.038	0.033	0.020	0.085	0.045
ρ_5	BIAS	0.005	0.001	0.003	−0.001	0.039	0.006	−0.035	0.003
	RMSE	0.005	0.016	0.070	0.038	0.046	0.020	0.078	0.046
Medium noise, 5-sec sampling									
ρ_1	BIAS	0.010	−0.001	0.363	0.036	0.061	0.005	0.363	0.037
	RMSE	0.060	0.086	0.414	0.059	0.142	0.101	0.439	0.072
ρ_3	BIAS	0.027	0.001	0.309	0.007	0.043	0.004	0.290	0.013
	RMSE	0.035	0.055	0.354	0.049	0.066	0.069	0.353	0.061
ρ_5	BIAS	0.005	0.001	0.300	−0.001	0.045	0.006	0.263	0.014
	RMSE	0.005	0.031	0.344	0.051	0.052	0.039	0.321	0.060
Small noise, 5-sec sampling									
ρ_1	BIAS	0.044	0.093	1.064	0.075	0.390	0.172	1.254	0.225
	RMSE	0.172	0.480	1.066	0.245	0.403	0.553	1.255	0.349
ρ_3	BIAS	0.034	0.027	0.904	0.017	0.155	0.041	1.001	0.098
	RMSE	0.034	0.375	0.906	0.215	0.159	0.419	1.003	0.258
ρ_5	BIAS	0.005	0.021	0.877	−0.007	0.057	0.018	0.909	0.019
	RMSE	0.005	0.287	0.880	0.237	0.057	0.310	0.912	0.255

(to be continued)

Table 1—*Continued*

		Moderate serial dependence				High serial dependence			
		QMLE		JLZ	ReMeDI	QMLE		JLZ	ReMeDI
		BIC	AIC			BIC	AIC		
Large noise, 1-min sampling									
ρ_1	BIAS	0.027	−0.006	0.202	−0.058	0.118	0.009	0.176	−0.080
	RMSE	0.116	0.154	0.282	0.113	0.191	0.167	0.292	0.104
ρ_3	BIAS	0.032	−0.002	0.170	−0.007	0.123	0.015	0.140	−0.024
	RMSE	0.035	0.105	0.242	0.110	0.142	0.119	0.238	0.101
ρ_5	BIAS	0.005	0.000	0.167	0.001	0.057	0.024	0.128	−0.006
	RMSE	0.005	0.067	0.242	0.114	0.057	0.089	0.223	0.108
Medium noise, 1-min sampling									
ρ_1	BIAS	0.205	0.151	0.982	−0.052	0.413	0.289	1.132	−0.074
	RMSE	0.214	0.599	0.987	0.207	0.429	0.694	1.138	0.159
ρ_3	BIAS	0.034	0.079	0.826	−0.006	0.159	0.123	0.899	−0.020
	RMSE	0.034	0.412	0.838	0.206	0.160	0.518	0.912	0.165
ρ_5	BIAS	0.005	0.029	0.786	0.001	0.057	0.062	0.805	−0.005
	RMSE	0.005	0.254	0.806	0.218	0.057	0.323	0.826	0.175
Small noise, 1-min sampling									
ρ_1	BIAS	0.211	0.669	1.175	0.092	0.442	0.939	1.403	0.184
	RMSE	0.213	0.794	1.176	0.748	0.444	1.027	1.404	0.704
ρ_3	BIAS	0.033	0.219	0.959	0.011	0.158	0.355	1.083	0.079
	RMSE	0.033	0.400	0.965	0.747	0.158	0.491	1.088	0.684
ρ_5	BIAS	0.005	0.069	0.897	0.032	0.057	0.118	0.947	0.029
	RMSE	0.005	0.212	0.908	0.748	0.057	0.235	0.958	0.685

Note: Bias and RMSE of the 1st-, 3rd-, and 5th-order noise autocorrelation estimates across 1,000 Monte Carlo sample paths per scenario. The noise follows an ARMA(1,1) with MA coefficient 0.2 and AR coefficient ψ ; moderate serial dependence corresponds to $\psi = -0.4$ and high serial dependence to $\psi = -0.6$. The true values of (ρ_1, ρ_3, ρ_5) are $(-0.209, -0.033, -0.005)$ and $(-0.440, -0.158, -0.057)$ for the two cases respectively. Each of the six panels combines one noise level (large / medium / small) with one sampling frequency (5-sec / 1-min). The first part reports the three 5-sec panels; the continuation reports the three 1-min panels. QMLE columns are indexed by the order-selection criterion (BIC or AIC). For JLZ and ReMeDI, the reported bias and RMSE are those of the estimates using the tuning value that minimizes RMSE for each combination of noise magnitude, sampling frequency, and lag, with $h_n \in \{2, 4, 6\}$ for JLZ and $k'_n \in \{0.25, 0.5, 1\}$ for ReMeDI. JLZ and ReMeDI estimates are clipped to $[-1, 1]$.

inadmissible outputs, in which case all autocorrelations are set to zero, or selects an MA model with a smaller order, thereby setting a larger set of higher-order autocorrelations to zero. Because higher-order autocorrelations are typically small, zero can yield a lower RMSE than noisy direct estimates. This pattern is particularly evident in the weakest-signal scenarios, where BIC-QMLE almost always yields zero estimates, so that the RMSE is almost entirely driven by bias, while AIC-QMLE produces many nontrivial estimates and correspondingly higher RMSE. The estimation error of the nonparametric estimators is most pronounced at 1-min sampling, where JLZ’s RMSE exceeds 0.8 for all three lags under medium and small noise, and ReMeDI’s RMSE rises to 0.6–0.8 in the small-noise scenarios.

Holding noise magnitude and sampling frequency fixed, RMSE rises with serial dependence for QMLE and JLZ in nearly all cells; for ReMeDI the comparison is mixed, with RMSE often falling at 1-min sampling. For QMLE, the rise reflects the need to fit higher-order moving-average models when noise persistence is stronger. Against JLZ, QMLE-BIC’s edge widens along both dimensions of Table 1, both as noise weakens and as serial dependence strengthens from $\psi = -0.4$ to $\psi = -0.6$. The comparison with ReMeDI is more mixed. At higher lags (ρ_3, ρ_5), QMLE-BIC dominates across cells. At ρ_1 under $\psi = -0.6$, ReMeDI matches or outperforms QMLE-BIC in four moderate-signal cells (5-sec medium: RMSE 0.072 vs. 0.142; 5-sec small: 0.349 vs. 0.403; 1-min large: 0.104 vs. 0.191; 1-min medium: 0.159 vs. 0.429), while QMLE-BIC retains the lead in 1-min small noise (0.444 vs. 0.704) and across all $\psi = -0.4$ cells. The structural feature of QMLE-BIC—the data-adaptive choice of $\hat{q}_n$, which the nonparametric estimators (with their pre-specified h_n and k'_n) cannot reproduce—underlies its dominance in most cells. Beyond finite-sample performance, BIC provides a clear in-sample objective for selecting $\hat{q}_n$, whereas h_n and k'_n lack a comparable in-sample objective in practice; moreover, the selected $\hat{q}_n$ is itself a meaningful signal of estimation quality, indicating up to which lag autocorrelations can be reliably estimated.

4.2.2 Noise Strength Indicators and Estimation Error

Table 1 shows that the error of autocorrelation estimation varies substantially across noise magnitude and sampling frequency. The underlying driver is the noise signal strength of the data-generating process, but signal strength is unobservable in practice. We therefore ask whether observable statistics can serve as proxies for signal strength and inform us about estimation error. Figure 4 examines three candidates across the six Monte Carlo scenarios: the frequency of QMLE “inadmissibility” (left column), the unconditional distribution of the BIC-selected moving-average order $\hat{q}_n$ (middle column), and the kernel density of $\log_{10}(\hat{T}_n)$ (right column). Here “inadmissibility” refers to the failure of (3.14) to admit a solution with $\hat{\tau}_n^2(\hat{q}_n)$ positive and $\hat{\theta}_n(\hat{q}_n)$ real. $\hat{T}_n$ is the noise-presence test statistic of Aït-Sahalia and Xiu

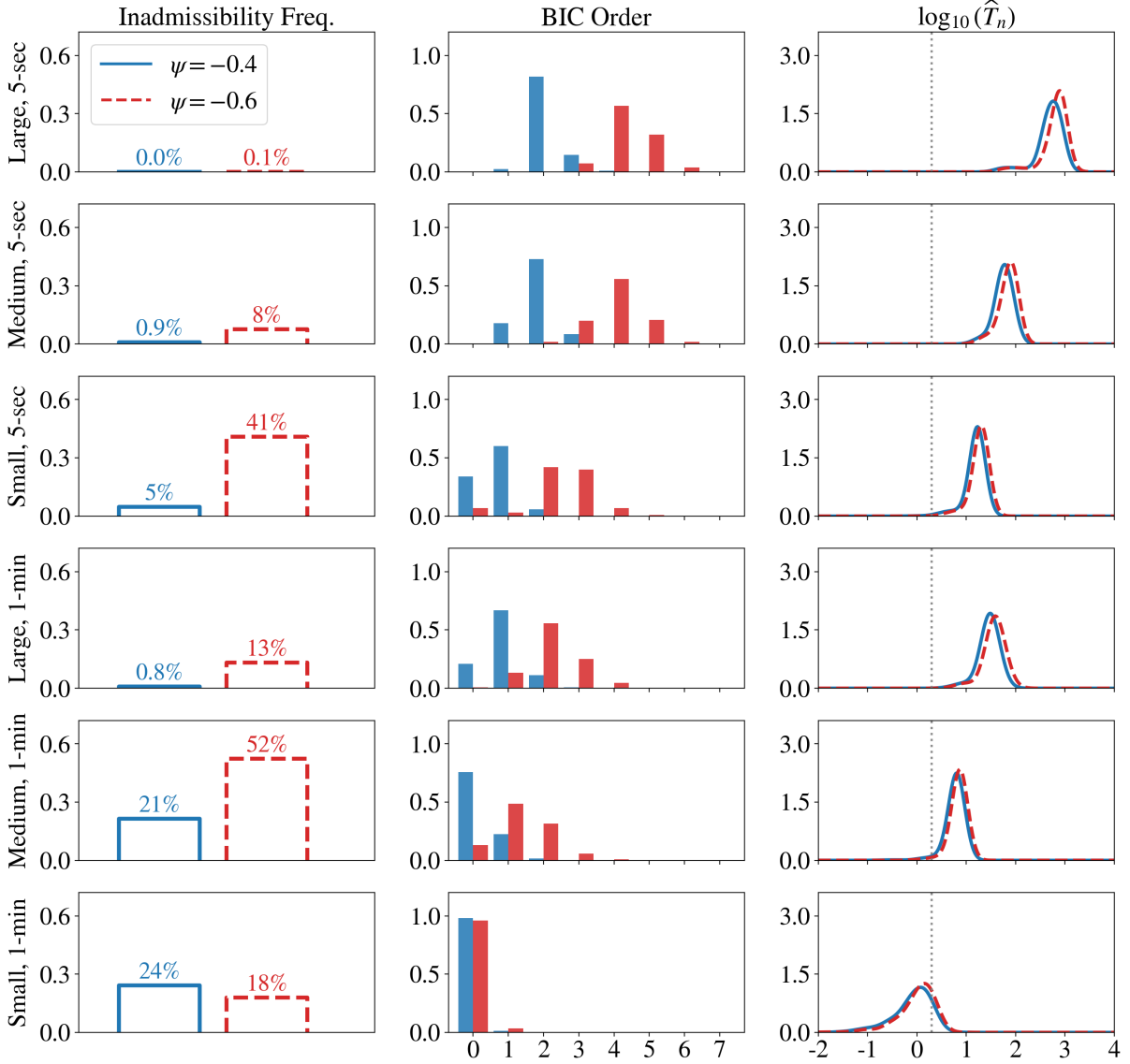


Figure 4: Diagnostic Statistics across Monte Carlo Scenarios

Note: Diagnostic statistics across the six Monte Carlo scenarios (rows: Large / Medium / Small noise $\times$ 5-sec / 1-min sampling). Left column: frequency of QMLE inadmissibility. Middle column: unconditional histogram of the BIC-selected MA order $\hat{q}_n$ across all 1,000 sample paths (not conditional on admissibility). Right column: kernel density of $\log_{10}(\hat{T}_n)$, where $\hat{T}_n$ is the noise-presence test statistic of Aït-Sahalia and Xiu (2019). Solid blue bars/lines correspond to $\psi = -0.4$; dashed red bars/lines to $\psi = -0.6$.

(2019), constructed explicitly as,

$$\hat{T}_n = \frac{\hat{\gamma}_0(0)}{\widehat{\text{SE}}(\hat{\gamma}_0(0))},$$

where $\hat{\gamma}_0(0)$ denotes the QMLE noise-variance estimate obtained when the noise order is fixed at $q = 0$ (the i.i.d.-noise specification; under this null, the auxiliary observed-returns model in eq. (3.13) is MA(1)), and $\widehat{\text{SE}}(\cdot)$ denotes its asymptotic standard error from in-fill asymptotics. The MA(1) specification is used because, under the no-noise null $H_0: \gamma_0 = 0$, $\hat{T}_n$ converges to a standard normal distribution. Importantly, fixing the order at $q = 0$ a priori avoids the data-driven model-selection step that BIC would otherwise introduce, so the asymptotic distribution of $\hat{T}_n$ is not contaminated by the discontinuity associated with order selection—a clean limiting distribution that is essential for the test to have the nominal 5% size. Because the alternative is one-sided ($H_a: \gamma_0 > 0$), we reject H_0 at the 5% level when $\hat{T}_n > z_{0.95} \approx 1.645$. The histograms of $\log_{10}(\hat{T}_n)$ are conditional on $\hat{T}_n > 0$; the proportion of non-positive $\hat{T}_n$ is negligible except in the 1-min, small-noise case, where it is about 20%. The vertical dotted line in the right column marks $\hat{T}_n = 1.645$, the one-sided 5% critical value of the no-noise test.

All three statistics shift systematically as noise magnitude shrinks or sampling frequency falls. The inadmissibility frequency rises from near zero to substantial portions, as it becomes more difficult for noisy autocovariance estimates to be consistent with a moving-average model. The distribution of $\hat{q}_n$ moves toward zero, since the selection criterion prefers simpler models under more limited data. The density of $\log_{10}(\hat{T}_n)$ shifts leftward, as expected from its definition. Serial dependence affects the three differently. Stronger dependence tends to raise inadmissibility, as it is more challenging to reconcile a higher-dimensional noisy autocovariance vector with a moving-average model. It also raises the selected order, as simpler models now face larger approximation errors. The lone exception is the small-noise, 1-min cell: there the signal is so weak that BIC selects $\hat{q}_n = 0$ on more than 95% of paths for both values of ψ , so admissibility reduces to $\hat{\gamma}_0 > 0$. On the other hand, $\log_{10}(\hat{T}_n)$ is largely unaffected by serial dependence: its distributions overlap substantially across $\psi = -0.4$ and $\psi = -0.6$, and the small rightward shift visible at $\psi = -0.6$ reflects that the true γ_0 is larger at $\psi = -0.6$ than at $\psi = -0.4$, as Figure 3 shows.

The BIC-order distributions in the middle column display two features. First, in the high-signal cells (large or medium noise at 5-sec sampling), the selected order rises with persistence—from $\hat{q}_n = 2$ at $\psi = -0.4$ to $\hat{q}_n = 4$ at $\psi = -0.6$ —reflecting the longer MA approximation needed when noise autocorrelations decay more slowly. Second, and more notably, as we move down the rows the distribution collapses toward $\hat{q}_n = 0$. This reflects that, as signal weakens, BIC’s complexity penalty dominates and the selected order shrinks: the data no longer support resolving any noise-driven autocovariance structure beyond zero. The selected order $\hat{q}_n$ therefore encodes both the signal strength and the persistence structure, which is precisely why it turns out to be the most informative diagnostic of the three.

Among the three statistics, the selected order $\hat{q}_n$ is the most useful for guiding autocorrelation estimation. Inadmissibility flags weak signal, but admissibility is not a sufficient indicator of strong signal: even in the weakest-signal scenarios, a substantial fraction of sample paths still produce admissible QMLE outputs. Rejection of the no-noise hypothesis under $\hat{T}_n$ is also not a guarantee of low-error estimation of noise autocorrelations: in all but the weakest scenarios, rejection at the 5% level happens almost certainly, yet the estimation error in many of those scenarios are sizable. The magnitude of $\hat{T}_n$ can instead provide more direct guidance on signal strength; that said, $\hat{T}_n$ does not distinguish between different levels of serial dependence, nor does it indicate how many orders of autocorrelations can be estimated with small error. In this regard, the selected order $\hat{q}_n$ provides a complete picture: it responds to both signal strength and serial dependence, and indicates up to which specific order autocorrelations should be estimated rather than simply set to zero. For the binary noise-presence question, by contrast, $\hat{T}_n$ remains the natural statistic, and Section 5.2 uses it that way.

Table 2 quantifies to what extent the selected order $\hat{q}_n$, the statistic $\hat{T}_n$, or the noise variance estimate $\hat{\gamma}_0$ inform about the error of autocorrelation estimation. We focus on the estimation of ρ_j , with $j \in \{1, 2, 3, 4, 5\}$. We pool sample paths across all 12 scenarios (3 noise levels $\times$ 2 sampling frequencies $\times$ 2 values of ψ , totaling 12,000 paths). We then partition the pooled sample into three disjoint groups for each lag j , using a criterion based on $\hat{q}_n$, $\hat{T}_n$, or $\hat{\gamma}_0$. The three groups are: group (i), where the criterion holds and the path is admissible; group (ii), where the criterion holds but the path is inadmissible; and group (iii), where the criterion does not hold. The criterion at lag j is $\{j \leq \hat{q}_n\}$ when using the selected-order criterion, $\{\hat{T}_n \geq c^T\}$ when using the $\hat{T}_n$ criterion, and $\{\hat{\gamma}_0 \geq c^\gamma\}$ when using the $\hat{\gamma}_0$ criterion; the cutoffs c^T and c^γ are stratified by admissibility, so that for each lag, the selection fraction matches that when using the selected-order criterion.

Table 2: Relative RMSE Conditional on Selected Order, $\hat{T}_n$, or $\hat{\gamma}_0$

Fraction		$\hat{q}_n \geq \text{order of autocorrelation}$				$\hat{T}_n \geq c^T$				$\hat{\gamma}_0 \geq c^\gamma \times 10^{-6}$			
		QMLE	QMLE*	JLZ	ReMeDI	c^T	QMLE	JLZ	ReMeDI	c^γ	QMLE	JLZ	ReMeDI
Group (i): admissible and criterion holds													
ρ_1	59.0%	0.44		1.97	0.45	12.0	0.43	2.10	0.46	0.11	0.50	1.98	0.47
ρ_2	45.9%	0.33		1.83	0.61	23.5	0.47	1.50	0.65	0.42	0.58	2.48	0.92
ρ_3	20.6%	0.54		2.27	0.82	84.6	0.62	0.86	0.79	8.38	0.75	3.80	1.63
ρ_4	15.1%	0.56		2.63	1.02	332	0.79	1.58	2.02	9.29	0.75	8.29	2.84
ρ_5	4.6%	0.26		2.33	0.79	800	0.81	1.13	0.88	10.3	0.98	21.3	6.16
Group (ii): inadmissible and criterion holds													
ρ_1	12.2%	1.00	1.57	3.16	0.76	-0.02	1.00	3.16	0.74	-0.00	1.00	3.16	0.74
ρ_2	5.2%	1.00	1.42	2.30	1.16	14.0	1.00	3.41	1.08	0.47	1.00	2.79	0.58
ρ_3	5.1%	1.00	2.34	5.94	0.95	15.1	1.00	8.99	1.47	0.47	1.00	7.45	1.20
ρ_4	0.4%	1.00	2.50	4.67	2.12	45.9	1.00	2.12	0.77	11.3	1.00	5.19	1.23
ρ_5	0.4%	1.00	3.80	8.69	2.48	46.5	1.00	3.97	1.26	11.4	1.00	10.0	3.69
Group (iii): criterion does not hold													
ρ_1	28.8%	1.00		4.50	2.21		1.01	4.39	2.21		0.94	4.50	2.20
ρ_2	48.9%	1.00		7.10	4.16		0.94	7.13	4.15		0.88	6.89	4.12
ρ_3	74.3%	1.00		16.6	8.29		0.99	16.5	8.29		0.96	16.5	8.25
ρ_4	84.5%	1.00		36.4	19.4		0.97	36.4	19.4		0.98	36.2	19.4
ρ_5	95.0%	1.00		86.7	45.8		0.99	86.7	45.8		0.98	86.5	45.8

Note: Relative RMSE = $\sqrt{\mathbb{E}[(\hat{\rho}_k - \rho_k^{\text{true}})/\rho_k^{\text{true}}]^2}$, for $k = 1, \dots, 5$, pooled across all 12 scenarios (12,000 paths). The three column groups select paths by $\{k \leq \hat{q}_n\}$, $\hat{T}_n \geq c^T$, or $\hat{\gamma}_0 \geq c^\gamma \times 10^{-6}$, with c^T and c^γ stratified by admissibility so all three share identical panel fractions per k . The c^T and c^γ columns report per-stratum cutoffs (blank in the not-selected panel; $\hat{\gamma}_0$ in units of 10^{-6}). QMLE sets $\hat{\rho}_k = 0$ on every inadmissible path; QMLE* disables this safeguard and instead reports the raw plug-in ratio $\hat{\rho}_k = \hat{\gamma}_k/\hat{\gamma}_0$ throughout, so QMLE* and QMLE differ only in group (ii). JLZ uses $h_n = 2$; ReMeDI uses $k'_n = 0.25$. All estimates clipped to $[-1, 1]$.

We evaluate estimation performance via the relative RMSE, defined as

$$\text{Rel. RMSE} = \sqrt{\widehat{\mathbb{E}} \left[\left(\frac{\widehat{\rho}_j - \rho_j^{\text{true}}}{\rho_j^{\text{true}}} \right)^2 \right]},$$

where the expectation is taken over all 12,000 paths. For each group and criterion choice, we report the relative RMSE of BIC-QMLE, JLZ, and ReMeDI. For the selected-order criterion, we additionally report the relative RMSE of QMLE*, which keeps the raw autocorrelation ratio $\widehat{\rho}_k = \widehat{\gamma}_k / \widehat{\gamma}_0$ on inadmissible paths instead of zeroing them. Hence QMLE* estimates differ from QMLE estimates only in group (ii). For JLZ and ReMeDI, we use $h_n = 2$ and $k'_n = 0.25$, which minimize the average relative RMSE over lags and groups. The second column provides the proportion of paths falling into each group at each lag, identical across the three criteria by construction. The first columns in the middle ($\widehat{T}_n$) and right ($\widehat{\gamma}_0$) panels report the cutoff values defining each group's selected set.

Two design choices warrant brief comment. First, we use relative rather than absolute RMSE because the true autocorrelation values ρ_j^{true} span more than an order of magnitude across lags (see the note to Table 1); normalizing by $|\rho_j^{\text{true}}|$ yields a unit-free metric directly comparable across lags. Second, the stratification of c^T and c^γ by admissibility ensures that all three criteria select the same fraction of paths at each lag, so RMSE differences across criteria reflect genuine differences in informativeness rather than differences in selection thresholds. Without this stratification, a more selective criterion would mechanically achieve lower RMSE simply by retaining the easiest paths.

Three patterns emerge. First, all three criteria are informative about estimation error. In 44 of 45 method-lag-criterion combinations, the relative RMSE in group (i) is significantly lower than in group (iii). Under JLZ and ReMeDI, the same comparison holds between groups (ii) and (iii). For QMLE, however, the RMSEs in both groups (ii) and (iii) are close to one—since QMLE zeros out estimates on all inadmissible paths and most criterion-failing paths. In contrast, the relative RMSEs of the nonparametric estimators in group (iii) are quite large: for JLZ, they range from about 4.4 at ρ_1 to 87 at ρ_5 ; for ReMeDI, from 2.2 to 46.

Second, the selected-order criterion is most useful in practice. In group (i), it delivers the lowest RMSE for QMLE and ReMeDI at most lags; for JLZ within group (i), the $\widehat{T}_n$ criterion produces somewhat smaller RMSE. Importantly, even though the $\widehat{T}_n$ and $\widehat{\gamma}_0$ criteria help identify paths on which estimation has small error, their cutoffs c^T and c^γ vary substantially across lags and are unknown in practice, whereas the selected-order criterion uses the data-driven $\widehat{q}_n$ directly.

Third, imposing admissibility is in general beneficial. The relative RMSE in group (i) is

better than in group (ii) for the majority of lags across all 9 criterion-method combinations except $\hat{\gamma}_0$ -ReMeDI. Under the selected-order criterion specifically, group (i) is strictly better than group (ii) for all 15 method-lag combinations. Relatedly, within group (ii), QMLE* yields worse outcomes than QMLE, confirming the success of zeroing out raw QMLE estimates under inadmissibility.

Taken together, these patterns suggest that, for each lag j , the autocorrelation ρ_j can be estimated with the smallest error on paths where both the selected-order criterion (i.e., $j \leq \hat{q}_n$) and admissibility hold. Within this set, QMLE attains the smallest relative RMSE, with errors that remain moderate across all five lags; ReMeDI ranks second; JLZ exhibits higher RMSE. This motivates an empirical estimation pipeline: for each lag j , retain stock-day pairs that are admissible and satisfy the selected-order criterion (i.e., $j \leq \hat{q}_n$), and report the resulting QMLE autocorrelation estimates on this set.

5 Empirical Analysis of U.S. Equity

5.1 Sample, Data Cleaning, and Method Tuning

To demonstrate the empirical relevance of the proposed approach, we conduct a large-scale study of noise autocovariances for S&P 1500 index constituents from January 1, 1996, to December 31, 2024. There are approximately 1,500 tickers every day, and about 3,500 tickers in total due to changes in index constituents. To illustrate, we summarize cross-sectional findings here though all estimates are available upon request. We use BIC-QMLE for noise-related parameters because of the model selection consistency result discussed earlier and the supporting finite-sample evidence from the simulations in Section 4. Although the empirical analysis is centered on noise, we also report volatility estimates—with an emphasis on the implications of microstructure noise for volatility inference—using AIC*-QMLE, as suggested by Da and Xiu (2021). For JLZ, we set the tuning parameter $h_n = 2$; for ReMeDI, we set $k'_n = 0.25$. These choices are guided by our simulation results in Section 4.

We download the trades and quotes of all equities at their highest frequency available (up to a millisecond after January 1, 2007, and a microsecond from July 27, 2015) from the TAQ database.¹⁵ Next, we remove trades and quotes with special condition codes or suffix codes, as well as those that occur outside regular trading hours.¹⁶ We then construct national

¹⁵Because companies change their tickers from time to time for mergers, acquisitions, or other reasons, the same ticker in the TAQ database may correspond to different stocks. We therefore keep track of these changes and use CRSP PERMINOs to index all stocks that do not change over time.

¹⁶We remove trades and quotes with condition codes Z, B, U, T, L, G, W, K, J and corresponding odd-lot trades, which have an additional letter I, as well as those with non-empty suffix codes (preferred shares). We identify opening trades as those with condition codes O, Q, OI, or QI; closing trades with 6, M, 6I, or MI; and remove all trades beyond the window of opening and closing time points. We only keep trades with correction

best bid and offer (NBBO) data using quotes from all exchanges at a 1-second frequency.¹⁷ We then match trades with NBBOs by their recorded time points and remove those trades that are outside the range of the corresponding NBBOs.¹⁸ Our approach is less aggressive than that of Barndorff-Nielsen et al. (2009), in that we maintain trades and quotes from all exchanges, whereas they retain only entries originating from a single exchange. Next, we remove redundant trades, retaining only nonzero returns.¹⁹ This step helps alleviate model misspecification due, for example, to the effect of rounding, latency or delay across exchanges, and so on. Finally, we remove any stock days that have fewer than 12 observations after cleaning. The remaining cleaned panel covers 7,231 trading days and roughly 10.34 million stock-day records, on which all subsequent figures and tables are computed. We organize the empirical evidence into four subsections, beginning with a diagnostic figure that motivates the analysis (Section 5.2), followed by the core findings on noise magnitude (Section 5.3), persistence (Section 5.4), and implications for volatility inference (Section 5.5).

5.2 Noise Presence and Dependence Structure: Diagnostics

Figure 5 asks whether the data are informative about two distinct features of microstructure noise: its *presence* and its *dependence structure*. Informativeness about the first is captured by the no-noise test statistic $\hat{T}_n$; informativeness about the second is captured by the inadmissibility frequencies and the BIC-selected order $\hat{q}_n$. Inadmissibility can arise not only from weak presence, when noise is not even statistically detectable, but also from weak structural identification as discussed in Section 3.2. Because the inversion recovers multiple parameters jointly, weak identification can arise even with detectable noise, particularly at higher $\hat{q}_n$. $\hat{T}_n$ separates the two cases: large $\hat{T}_n$ with high inadmissibility points to the latter; small $\hat{T}_n$ with high inadmissibility points to the former.

The QMLE inadmissibility fraction is moderate and reasonably stable, fluctuating within 25%–65% throughout the sample. By contrast, JLZ and ReMeDI inadmissibility tracks QMLE in the early years but declines markedly after 2010 and is close to zero by the late 2010s. The cross-sectional ordering is consistent across all three methods: S&P 500 (large caps) display

indicator 00 or 01.

¹⁷We construct NBBOs from the millisecond dataset by adapting the SAS codes from <https://wrds-web.wharton.upenn.edu/wrds/research/applications/microstructure/NBBO%20derivation/>. Although this database has more precise timestamps, we do not construct NBBOs at any frequency higher than every second.

¹⁸For trades that are observed at millisecond or microsecond intervals, we match them with the NBBOs of the previous second. Our SAS codes for cleaning the data are available upon request.

¹⁹This step is called “tick-time sampling” by Griffin and Oomen (2008). Our assumptions on the noise process and sampling scheme, when the impact of rounding is negligible, can accommodate both transaction-time and tick-time sampling. Resorting to tick-time sampling allows us to extract richer information about the persistence of microstructure noise. In transaction time, the serial dependence is naturally dampened quite substantially by the prevalence of zero returns.

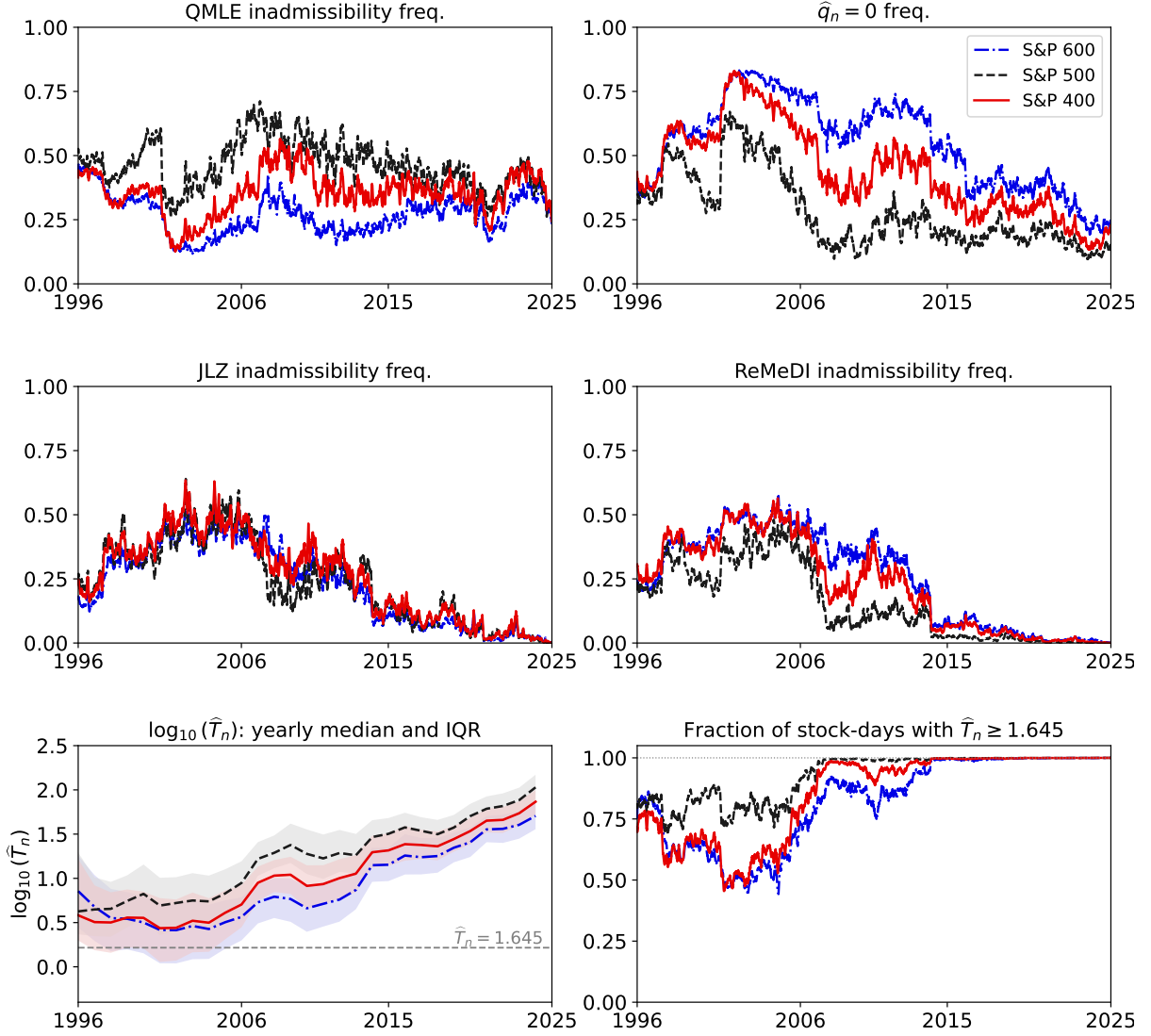


Figure 5: Diagnostics of Noise Presence and Dependence Structure

Note: The figure consolidates six diagnostic statistics of microstructure-noise signal strength, all on the S&P 1500 universe with cross-sectional medians (lines) and inter-quartile ranges (shaded bands when shown), separately for S&P 400, S&P 500, and S&P 600 constituents. *Top row:* smoothed daily fraction of stock-days yielding inadmissible QMLE output (left) and the daily fraction with BIC-selected order $\hat{q}_n = 0$ (right). *Middle row:* smoothed daily fraction of inadmissible JLZ output (left) and inadmissible ReMeDI output (right); a JLZ (resp. ReMeDI) estimate is classified as inadmissible if at least one of $\hat{\rho}_1$, $\hat{\rho}_3$, or $\hat{\rho}_5$ lies outside $[-1, 1]$. *Bottom-left:* yearly median and IQR of $\log_{10}(\hat{T}_n)$, where $\hat{T}_n$ is the no-noise test statistic of [Aït-Sahalia and Xiu \(2019\)](#), defined in Section 4.2.2; the horizontal dashed line marks the approximate one-sided 5% rejection threshold $\hat{T}_n = 1.645$. *Bottom-right:* smoothed daily fraction of stock-days for which $\hat{T}_n \geq 1.645$ (i.e., the one-sided no-noise null is rejected at the 5% level). All time-series panels use a 21-day equal-weight trailing moving window for smoothing; the bottom-left panel uses yearly bins.

the highest inadmissibility rates and S&P 600 (small caps) the lowest. The $\hat{q}_n = 0$ frequency mirrors this ordering: BIC selects $q = 0$ most frequently for S&P 600 and least frequently for S&P 500. The two patterns are intuitively connected—when fewer autocovariance lags need to be reconciled with a moving-average representation, admissibility is easier to achieve. Taken together, these four panels show that “raw” admissibility rates conflate two distinct phenomena: the size of the parameter vector being fit (captured by $\hat{q}_n$) and the quality of the underlying noise signal. Reading inadmissibility in isolation can be misleading; a high rate need not mean that noise is absent or undetectable.

The bottom-left panel addresses the noise-detectability question directly. The yearly median $\log_{10}(\hat{T}_n)$ rises by roughly 1.5 orders of magnitude over the sample, from around 0.6–0.9 in the late 1990s to about 1.7–2.0 by 2024. Even at the start of the sample, the median sits well above the $\log_{10}(1.645) \approx 0.22$ rejection threshold for every S&P group. By the mid-2010s the entire IQR sits above the $\hat{T}_n = 1.645$ line for all three groups. The cross-sectional ordering is broadly stable: S&P 500 tends to deliver the largest $\hat{T}_n$, followed by S&P 400 and then S&P 600. The fraction of stock-days with $\hat{T}_n \leq 0$ (excluded from the log scale) is small and shrinks from about 3.5%–4.5% in 1996 and 2006 to under 0.3% by 2016 and effectively zero by 2024. The bottom-right panel translates the same evidence into a binary rejection rate: the fraction of stock-days at which the no-noise null is rejected at the 5% level rises from roughly 60%–85% in the late 1990s to essentially 100% by 2016, and stays near 100% thereafter for all three S&P groups.

The six-panel figure delivers the central diagnostic message of this section: noise is *pervasive*—it is statistically detectable on the typical stock-day throughout the entire 1996–2024 sample—but its identifiability as an MA process varies meaningfully across stock-days and over time. As trading frequency grows, $\hat{T}_n$ shifts rightward and the JLZ and ReMeDI inadmissibility rates fall sharply. By contrast, QMLE inadmissibility remains in the 25%–65% range throughout the sample, indicating that the dependence structure of the noise can be weakly identified even when its presence is unambiguous. Together with our simulation results, this evidence motivates selecting subsets of stock-days carefully to estimate different noise parameters to ensure the estimation quality.

5.3 Volatility, Trading Frequency, and the Decline of Noise Variance

We next examine the time-series behavior of volatility, trading frequency, and noise variance. Figure 6 compresses these three quantities into four panels: a cross-sectional view of volatility (top-left) and trading activity (top-right), a cross-method view of the trajectory of noise variance (bottom-left), and a cross-sectional view of the noise variance itself (bottom-right).

Median annualized volatility shares a common pattern across the three S&P groups, with

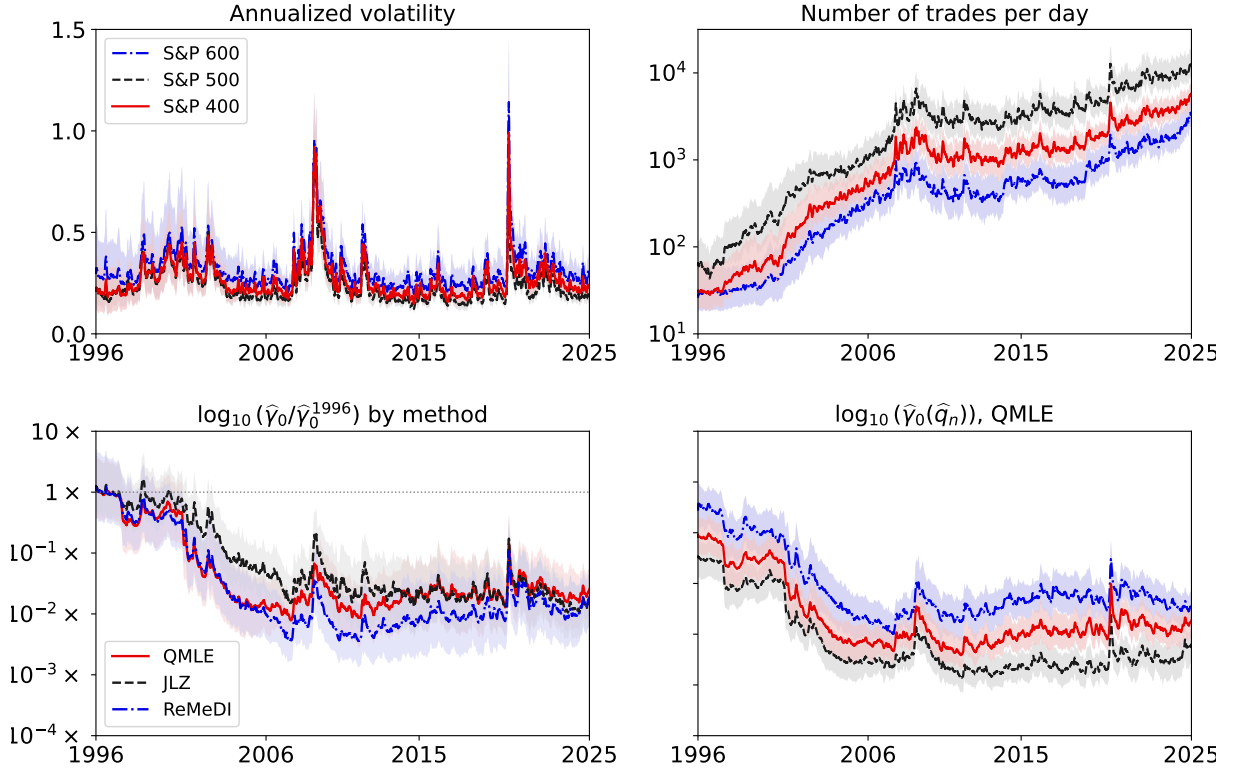


Figure 6: Time Series of Volatility, Trading Frequency, and Noise Variance

Note: The figure plots smoothed time series with cross-sectional median (lines) and inter-quartile range (shaded bands). *Top-left:* annualized volatility, separately for S&P 400, S&P 500, and S&P 600 constituents. *Top-right:* same statistics for the daily number of trades per stock-day, on a $\log_{10}$ y-axis. *Bottom-left:* pooled cross-sectional median and IQR of $\log_{10}(\hat{\gamma}_0/\hat{\gamma}_0^{1996})$ for QMLE, JLZ, and ReMeDI on each method's admissible subset; the y-axis is normalized so each method starts at $1 \times$ in 1996, isolating the relative decline in noise magnitude over time. *Bottom-right:* cross-sectional median and IQR of $\log_{10}(\hat{\gamma}_0(\hat{q}_n))$ from QMLE by S&P group, on the absolute scale. Series are smoothed with equal weights over a 21-day moving window.

conspicuous spikes during the 2008 financial crisis and the 2020 COVID episode; S&P 600 (small caps) are on average the most volatile and S&P 500 (large caps) the least, with S&P 400 in between. The number of trades per stock-day rises by roughly two orders of magnitude over the sample, with S&P 500 trading most frequently throughout. These two panels also reconcile the evidence from $\hat{T}_n$ in Figure 5: the rightward shift in $\hat{T}_n$ over the sample is driven primarily by the growth in trading frequency—which raises the effective sample size n and thereby the power of the no-noise test.

The bottom-left panel plots $\log_{10}(\hat{\gamma}_0/\hat{\gamma}_0^{1996})$, the pooled cross-sectional median of each method’s noise variance estimate normalized to its own 1996 level. All three estimators trace nearly parallel trajectories: a sharp drop of roughly an order of magnitude between 2001 and 2007, a brief reversal around the 2008 crisis, and a continued slow decline thereafter. By the late 2010s, all three methods agree that the median noise variance has fallen to between 1% and 5% of its 1996 level—about two orders of magnitude—and the trajectories of QMLE, JLZ, and ReMeDI track each other tightly to within ~ 0.3 log units. Across the six simulation scenarios in Figure 4, the QMLE RMSE of $\log_{10}(\hat{\gamma}_0)$ is roughly 0.30 in the next-worst (1-min medium noise) scenario and smaller elsewhere, so the two-order-of-magnitude movement documented here is well outside estimation error.

Within QMLE, the absolute level of $\log_{10}(\hat{\gamma}_0(\hat{q}_n))$ shows a stable cross-sectional ordering throughout the sample: S&P 600 (small caps) carries the highest noise variance, S&P 500 (large caps) the lowest, and S&P 400 in between. The pairwise time-series correlations of yearly log-median changes are 0.96 between S&P 500 and S&P 400, 0.86 between S&P 500 and S&P 600, and 0.92 between S&P 400 and S&P 600—so the three groups share a strong common factor in noise dynamics. Conditioning on the noise-presence rejection $\hat{T}_n \geq 1.645$ would raise the estimated noise variance slightly in the early years and would leave the post-2005 series essentially unchanged, since by then $\hat{T}_n \geq 1.645$ holds on virtually every stock-day (Figure 5). We therefore omit a separate panel and report the unconditional series.

Both the time-series decline and the cross-sectional patterns are consistent with classic information-based microstructure models. Grossman and Stiglitz (1980) suggest that technological and institutional developments lower the costs of acquiring and processing information, enabling more participants to operate with timely information and thereby reducing the equilibrium price discrepancy. Kyle (1989) adds that increased competition among informed traders further suppresses microstructure noise. These mechanisms extend naturally to the cross-section: large-cap stocks attract more informed participation, leading to smaller noise, while small-cap stocks exhibit larger noise.

5.4 Persistence of Microstructure Noise: Durations and Strength

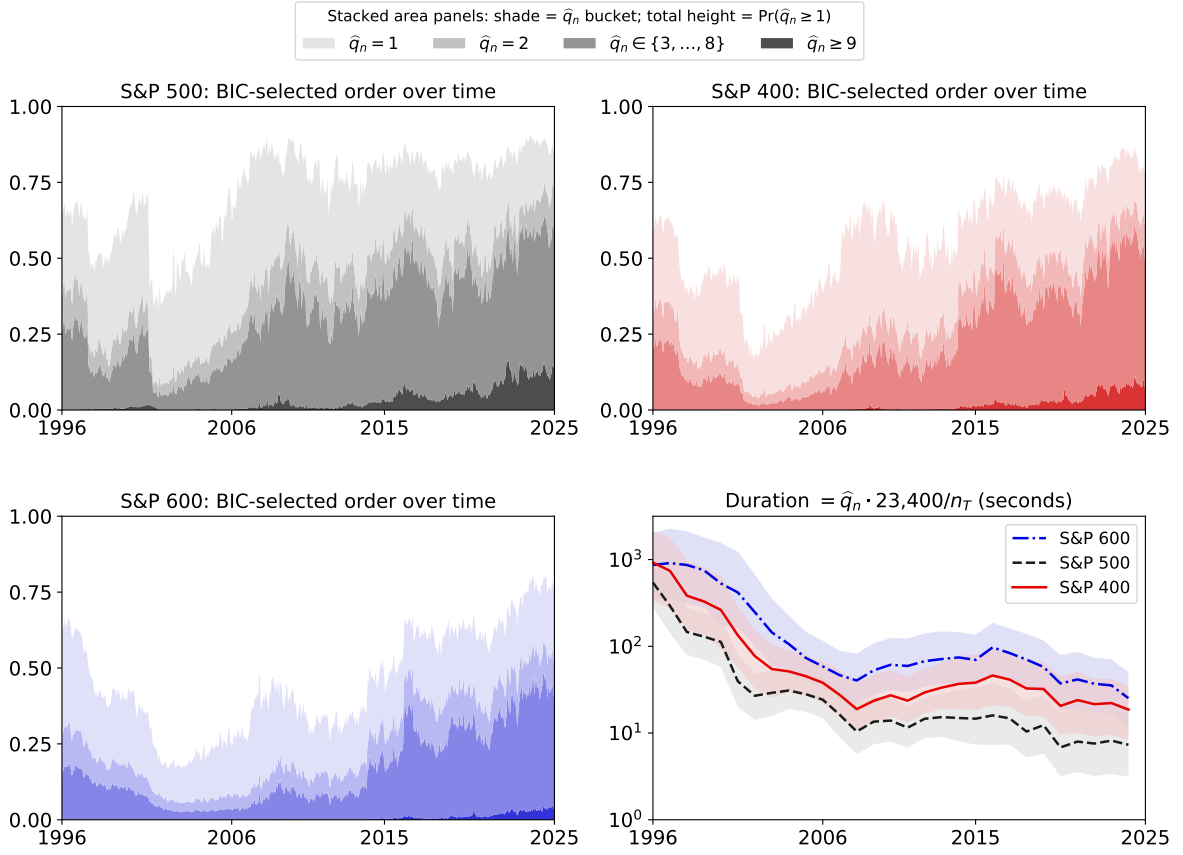


Figure 7: Selected Orders and Durations of Noise Autocorrelations

Note: Top-left, top-right, and bottom-left panels: smoothed daily fraction of stock-days falling into each $\hat{q}_n$ bucket, $\{q = 1, q = 2, q \in [3, \dots, 8], q \geq 9\}$, stacked from bottom (darkest, $q \geq 9$) to top (lightest, $q = 1$), separately for S&P 500, S&P 400, and S&P 600 constituents. Each panel uses its own hue (gray, red, blue); shade encodes the bucket. The total stack height at any date equals $\Pr(\hat{q}_n \geq 1)$ on that date for that S&P group. Daily fractions are smoothed with a 21-day trailing moving average. Bottom-right panel: yearly cross-sectional median (lines) and inter-quartile range (shaded bands) of $\log_{10}(\text{duration})$ in seconds, where duration is $\hat{q}_n \times (23,400/n_T)$, restricted to stock-days with $\hat{q}_n \geq 1$, separately for the three S&P groups.

We now turn to the dependence structure of the noise. We start by examining the BIC-selected order $\hat{q}_n$, which proxies the depth of the autocorrelated component, and the implied calendar-time duration of that component. In Figure 7, the top-left, top-right, and bottom-left panels stack the daily fraction of stock-days in each $\hat{q}_n$ bucket, separately for S&P 500, S&P 400, and S&P 600 constituents; the bottom-right panel reports the yearly cross-sectional distribution of duration.

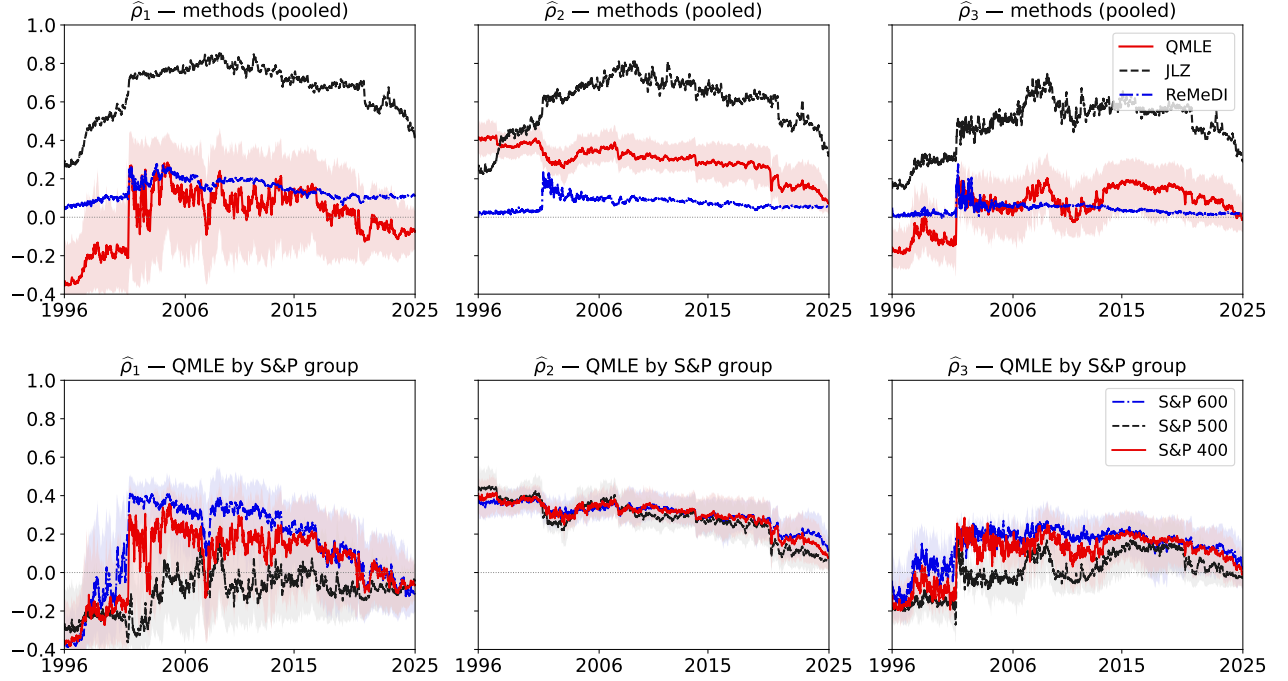


Figure 8: Time Series of Noise Autocorrelations

Note: One panel per lag $j \in \{1, 2, 3\}$, arranged in two rows. *Top row:* smoothed cross-sectional median of $\hat{\rho}_j$ for the three estimators QMLE (with BIC), JLZ ($h_n = 2$), and ReMeDI ($k'_n = 0.25$), pooled across S&P 1500 constituents. The lighter shaded band is the QMLE inter-quartile range. *Bottom row:* QMLE only, broken out by S&P group (S&P 400 / 500 / 600), each with its own median line and IQR band. *Per-lag conditioning matches Table 2:* panel j is restricted to stock-days with $\hat{q}_n \geq j$ from BIC-QMLE and QMLE admissibility (panel-1 sample ≈ 2.23 million stock-days; panel-2 ≈ 1.57 million; panel-3 ≈ 0.91 million). Within a panel the three methods are evaluated on identical (date, permno, ticker) triples; JLZ and ReMeDI $\hat{\rho}_j$ are clipped to $[-1, 1]$ to bound the small share of estimates outside the unit interval (0.3% for JLZ, 5.8% for ReMeDI on the $\hat{q}_n \geq 3$ subset). All series are smoothed with a 21-day trailing moving window.

The total stack height in each panel equals $\Pr(\hat{q}_n \geq 1)$ on that date, and each shade contributes one bucket. Three patterns are visible at a glance. First, the total stack height rises gradually from roughly 0.5–0.7 in 1996 to 0.8–0.9 by 2024 in all three S&P groups, dips to a low in 2002–2005 (the period of weakest signal strength documented in Figure 5), and recovers thereafter. Second, the composition of the stack shifts upward over time: the lightest layer ($\hat{q}_n = 1$) dominates in the early years; from the mid-2000s onward the $\hat{q}_n \in \{3, \dots, 8\}$ band thickens substantially, and a non-negligible $\hat{q}_n \geq 9$ band emerges by 2016. Third, the cross-sectional ordering is consistent: the S&P 500 panel has the largest total stack throughout, S&P 600 the smallest, with S&P 400 in between, paralleling the signal-strength ordering in Figure 5.

The lower selected orders in 1996 and 2006 partly reflect weaker signal strength rather than genuinely lower noise persistence: the simulation evidence in Figure 4 shows that, holding the true autocorrelation structure fixed, the distribution of $\hat{q}_n$ shifts leftward as signal strength weakens. Translating selected order into calendar time, duration $\equiv \hat{q}_n \times (23,400/n_T)$ seconds shifts dramatically downward over the sample. The yearly cross-sectional median falls from about 9–15 minutes in 1996 to roughly 5–25 seconds in 2024, with the IQR shrinking commensurately. In every year S&P 500 has the shortest duration (about 7 seconds in 2024), followed by S&P 400 (about 19 seconds) and S&P 600 (about 25 seconds). Read together with the stacked-area panels, the central message is: the rise in selected orders in tick time, visible in the upper composition of the stacks, should not be read as an increase in noise persistence in calendar time, since the average trading interval has decreased substantially over the same period (Figure 6). On the contrary, calendar-time persistence has shortened by roughly two orders of magnitude.

Next, we examine how the autocorrelation estimates themselves change across lags, methods, and time. Table 2 from our simulation shows that the j th-order autocorrelation ρ_j can be accurately estimated by QMLE on the subset of stock-days with $\hat{q}_n \geq j$ and admissible QMLE output. We honor that conditioning panel by panel: each panel j in Figure 8 restricts to the corresponding $\hat{q}_n \geq j$ subset, and within each panel JLZ and ReMeDI are evaluated on the *same* stock-day population as QMLE (about 2.23, 1.57, and 0.91 million stock-days for $j = 1, 2, 3$, respectively). On the most-restrictive $\hat{q}_n \geq 3$ subset, the share of stock-days for which JLZ (resp. ReMeDI) produces at least one of $\hat{\rho}_1, \hat{\rho}_2, \hat{\rho}_3$ outside $[-1, 1]$ is small—0.3% (resp. 5.8%)—versus 18.3% (resp. 20.4%) unconditionally; we clip these out-of-bounds estimates to $[-1, 1]$ so the displayed medians are not pulled by a few wild values.

JLZ estimates sit substantially higher than QMLE—hovering around 0.4–0.8 across all three lags—while ReMeDI estimates are uniformly smaller than QMLE but preserve the lag-

by-lag ordering $|\hat{\rho}_1| > |\hat{\rho}_2| > |\hat{\rho}_3|$. The vertical gap of 0.3–0.5 between JLZ and QMLE is a real method-level discrepancy, not an artifact of the panel sample. We do not attempt to verify the precise mechanism behind these gaps; the most plausible attributions are differences in estimator construction—QMLE fits a parametric $\text{MA}(\hat{q}_n)$ model by quasi-likelihood, JLZ uses local averaging of high-frequency returns, and ReMeDI uses moments of disjoint return increments—and the choice of bandwidth-type tuning parameters for the nonparametric estimators ($h_n = 2$ for JLZ and $k'_n = 0.25$ for ReMeDI), which we fix a priori following simulation evidence. A data-adaptive selection of these tuning parameters would in principle deliver more reliable estimates, and this is precisely the advantage of QMLE: the BIC-selected order $\hat{q}_n$ adapts to the data. Reading the panels left to right, the time-series patterns for the three methods nonetheless rhyme with each other—all peak in the early 2000s, decline through the 2010s, and stabilize thereafter—though the absolute levels differ in a method-specific way.

Restricting to QMLE, the median $\hat{\rho}_1$ falls from roughly 0.4 in the early 2000s to under 0.05 by 2024; $\hat{\rho}_2$ from about 0.5 to about 0.1; $|\hat{\rho}_3|$ remains smallest throughout and moves least. Two transient episodes around 2001–2002 and 2008 elevate $\hat{\rho}_1$ above the surrounding trend, both coinciding with documented liquidity stress. Cross-sectionally, the bottom row shows that S&P 600 (small caps) carries the largest $\hat{\rho}_1$ and $\hat{\rho}_3$, followed by S&P 400, with S&P 500 (large caps) the smallest—reading off the gap between the blue, red, and black bands. $\hat{\rho}_2$ is essentially indistinguishable across the three S&P groups. Yearly-change pairwise correlations between S&P groups average 0.43 for $\hat{\rho}_1$, 0.58 for $\hat{\rho}_2$, and 0.59 for $\hat{\rho}_3$, indicating that the three index groups share a substantial common component for each lag.

The gradual decline of $|\hat{\rho}_1|$ and $|\hat{\rho}_2|$ together with the relatively stable $|\hat{\rho}_3|$ in tick time implies that calendar-time persistence has declined considerably, since the number of trades per day rose by roughly two orders of magnitude (Figure 6). This strengthens the duration analysis in Figure 7: the same data viewed through both lenses (tick-time autocorrelation magnitude and calendar-time duration) tells the same story of vanishing serial dependence.

5.5 Implications for Realized Volatility Inference

Finally, we discuss the importance of modeling the microstructure noise through the lens of volatility inference. Given the pervasiveness of small noise documented in Figure 5, we might choose to use realized volatility estimators on stock-days for which the data suggest small noise—whether based on informal volatility signature plots, formal tests of noise (Aït-Sahalia and Xiu (2019), Li and Yang (2026)), or that QMLE generates inadmissible outputs for noise. But such a practice amounts to a pre-testing-based estimation strategy and does not deliver valid volatility inference due to uniformity concerns when noise exists but is too small to be detected. We compare the biases and standard errors of the popular realized volatility

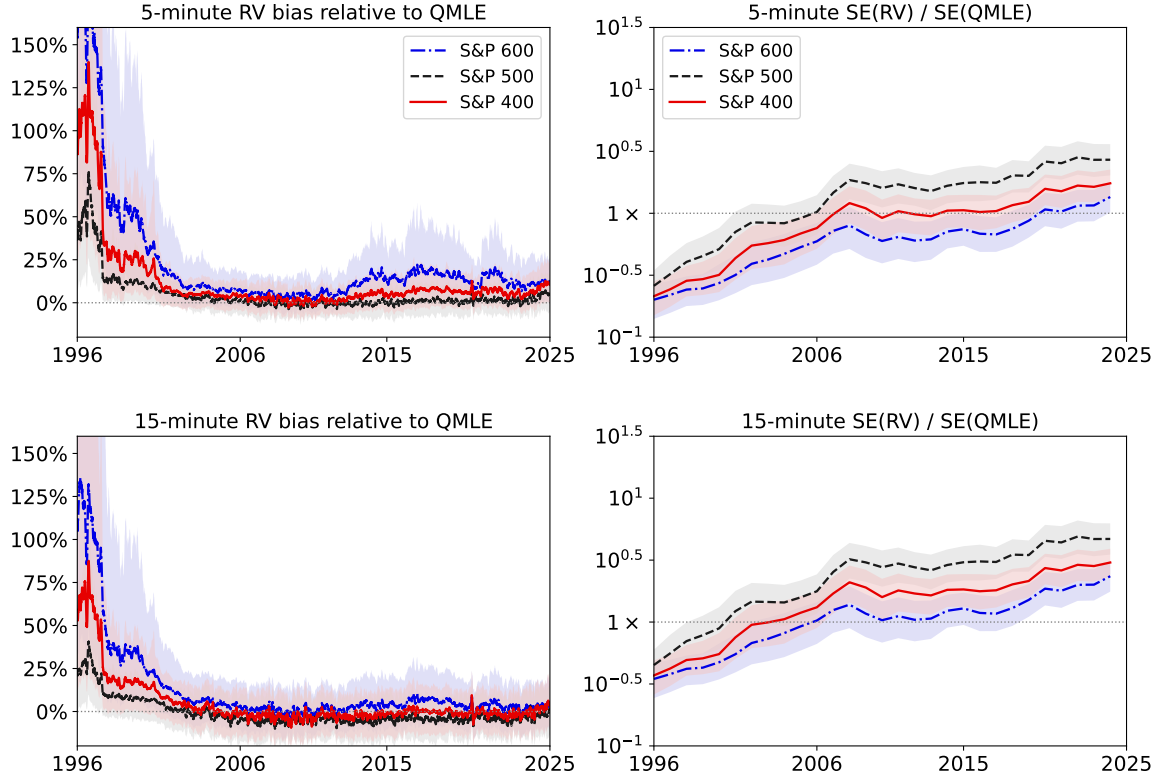


Figure 9: Relative Biases and Standard Errors of the Realized Volatility

Note: All four panels plot smoothed time series, with cross-sectional median (lines) and inter-quartile range (shaded bands), separately for S&P 400, S&P 500, and S&P 600 constituents. *Left column:* percentage bias of the realized-volatility estimator relative to QMLE for 5-minute (top) and 15-minute (bottom) sampling, $(\hat{\sigma}_{RV}/\hat{\sigma}_{QMLE}) - 1$, computed daily and smoothed with a 21-day moving window. *Right column:* yearly cross-sectional median and IQR of $\log_{10}(\text{SE}(\hat{\sigma}_{RV})/\text{SE}(\hat{\sigma}_{QMLE}))$ for the same two sampling frequencies. Values larger than $1 \times$ on the right-column y -axis indicate that the realized-volatility estimator is less efficient than QMLE.

estimator and the QMLE, to indirectly shed light on the influence of noise. The former estimator, based on data sampled at a prespecified frequency—say, every 5 or 15 minutes—is most commonly adopted in practice. Figure 9 reports both quantities (bias and standard-error ratio) over the full sample.

The left panels of Figure 9 compare the cross-sectional medians (lines) and inter-quartile ranges (bands) of realized-volatility estimates relative to QMLE, at 5-minute (top) and 15-minute (bottom) sampling frequencies, separately for S&P 500, S&P 400, and S&P 600 constituents. The 5-minute realized-volatility estimator exhibits a large upward bias on average—over 120% for small caps in 1996—likely due to the presence of microstructure noise. The bias has decreased steadily over the past two decades, with a slight increase post-2008, and stabilizes in the 5–25% range post-2010, with small caps still slightly above large caps. Large caps trade more frequently than every five minutes, so their cross-sectional median bias is essentially indistinguishable from zero post-2002; this does not imply that 5-minute sampling is safe for each individual constituent, since the IQR band remains wide. At 15-minute frequency, the median bias is much smaller—near zero for all three groups by the late 2000s—though the bands remain non-trivial.

The right panels of Figure 9 report the yearly cross-sectional median (lines) and IQR (bands) of $\log_{10}(\text{SE}(\hat{\sigma}_{\text{RV}})/\text{SE}(\hat{\sigma}_{\text{QMLE}}))$ over the full sample, with the horizontal $1\times$ reference indicating equal efficiency. The relative efficiency loss of the realized-volatility estimator grows continuously over the sample. At 5-minute frequency, the median ratio rises from roughly $1\times$ in 1996 to about $3\times$ – $5\times$ by 2024 across S&P groups—i.e., the realized-volatility estimator is now several times less efficient than QMLE on the typical stock-day. At 15-minute frequency the loss is even more dramatic: by 2024 the median ratio sits between $3\times$ and $10\times$, with the upper quartile reaching $10\times$ and beyond for S&P 500. The growth of these ratios over time reflects the sharp rise in trading frequency documented in Figure 6: as more high-frequency data become available, fixed-frequency subsampling discards an increasingly large fraction of the information. To sum up, without accounting for noise, the realized volatility estimator faces a bias and variance dilemma. Estimates using 5-minute data are subject to severe biases, whereas 15-minute estimates suffer from considerable efficiency losses. Additionally, the standard errors could still be understated because the noise might not be sufficiently small to be safely ignored. Figure 9 further shows that bias has shrunk over time while efficiency loss has grown, so the relative cost of ignoring noise has shifted from a bias-dominated problem in the early sample to a variance-dominated one today.

6 Conclusion

We propose a semiparametric approach to disentangling autocovariances and autocorrelations due to the microstructure frictions associated with observed prices. Our approach resembles a threshold estimator, which gives zero autocovariance estimates beyond the lag selected by the information criteria. This feature delivers appealing performance in the finite sample, particularly when noise is relatively small, compared with alternative nonparametric estimators. Our empirical study of S&P 1500 stocks finds that microstructure noise is statistically detectable on the typical stock-day throughout the sample, yet its dependence structure is often weakly identified through a finite-order moving-average representation; that the noise variance has shrunk by roughly two orders of magnitude over the sample period; and that, when restricted to stock-days on which the noise autocorrelations can be accurately estimated, calendar-time persistence has shortened substantially in recent years. These findings indicate that market efficiency has improved substantially, potentially due to the popularity of electronic and algorithmic trading. In a cross-sectional comparison, large-cap (S&P 500) stocks consistently exhibit smaller noise variance and shorter calendar-time autocorrelation duration than small-cap (S&P 600) stocks, with mid-cap (S&P 400) stocks in between, consistent with the hypothesis that more informed participation suppresses noise. Finally, the empirical evidence on realized-volatility bias and efficiency—bias dominant in the early sample, efficiency loss dominant in the recent years—demonstrates that noise-robust inference remains essential even as raw noise magnitude declines.

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Web Supplement to

Disentangling Autocorrelated Intraday Returns

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Abstract

This supplement contains mathematical proofs.

Appendix A Proofs of Technical Lemmas

A.1 Notation

In this section, we prepare the notation to be used throughout the proofs. Below we will introduce additional notation that applies only to the corresponding proofs unless otherwise indicated.

Part 1. The probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ can be constructed more explicitly. Specifically, we define X , Z , ξ , and η (which satisfy the relevant assumptions) on a space $(\Omega_{(0)}, \mathcal{F}_{\infty}, (\mathcal{F}_t), \mathbb{P}_{(0)})$, and define $\{\chi_i\}$ and $\{\varepsilon_i\}$ on a different space $(\Omega_{(1)}, \mathcal{F}_{(1)}, \mathbb{P}_{(1)})$. We then set $\Omega = \Omega_{(0)} \times \Omega_{(1)}$, $\mathcal{F} = \mathcal{F}_{\infty} \otimes \mathcal{F}_{(1)}$, and $\mathbb{P}(d\omega_{(0)}, d\omega_{(1)}) = \mathbb{P}_{(0)}(d\omega_{(0)})\mathbb{P}_{(1)}(d\omega_{(1)})$.

For any $x = (x_1, x_2, \dots, x_q) \in \mathbb{R}^q$, we set $x_j = 0$ for any $j \geq q+1$ and denote $\|x\|_{(q')}^2 = \sum_{j=q'+1}^{\infty} x_j^2$, $\|x\|_{1,(q')} = \sum_{j=q'+1}^{\infty} |x_j|$, $\|x\|_1 = \|x\|_{1,(0)}$, and $1\mathbb{T} \cdot x = \sum_{j=1}^{\infty} x_j$. For an integer i and a random variable x , we denote the i -th cumulant of x by $\text{Cum}_i(x)$. Let $\mathcal{M}_d$ denote the set of all $d \times d$ matrices. For any m and h , let $O_m, \mathbb{D}_m^h, \mathbb{F}_m^h, \mathbb{I}_m \in \mathcal{M}_m$ be defined by $(\mathbb{I}_m)_{i,j} = \delta_{i,j}$, $(O_m)_{ij} = \sqrt{\frac{2}{m+1}} \sin \frac{ij\pi}{m+1}$, $(\mathbb{D}_m^h)_{ij} = \delta_{i,j}(2 - \delta_{h,0}) \cos \frac{hj\pi}{m+1}$, and $(\mathbb{F}_m^h)_{ij} = \mathbb{1}_{\{h=|i-j|\}} - \mathbb{1}_{\{h=i+j\}} - \mathbb{1}_{\{h=2m+2-(i+j)\}}$. We also introduce $\mathbb{I}_n, O_n, \mathbb{D}_n^h, \mathbb{F}_n^h \in \mathcal{M}_{n_T}$ (instead of $\mathcal{M}_n$) with similar entries. We let $n_d = \lfloor n^{7/8} \rfloor$, $J_d = \lfloor n_T/n_d \rfloor - 1$ and $n'_d = n_T - n_d J_d$. For any m , we define

$$D_m = \sum_{h=0}^{\infty} \gamma_h \mathbb{D}_m^h, \quad V_m = \sigma^2 \Delta_n \mathbb{I}_m + (2\mathbb{I}_m - \mathbb{D}_m^1) D_m, \quad \Omega_m = O_m V_m O_m, \quad \Omega_{D,n} = (\mathbb{I}_{J_d} \otimes \Omega_{n_d}) \oplus \Omega_{n'_d}. \quad (\text{A.1})$$

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Here the dependence of $(D_m, V_m, \Omega_m, \Omega_{D,n})$ on $(\sigma^2, \gamma, \Delta_n)$ is omitted.

Part 2. We use $\Delta_i^n A$ to denote $A_{t_i} - A_{t_{i-1}}$ when A is a continuous-time stochastic process and to denote $A_i - A_{i-1}$ when A is a discrete-time stochastic process. Further, for $j \geq 1$, we introduce $t(j)_i = t_{(j-1)n_d+i}$. When A is a continuous-time process, we let $A_C(j) = A_{t_{(j-1)n_d}}$, $A_{C,t} := \sum_{j=1}^{\infty} A_C(j) \mathbb{1}_{\{t_{(j-1)n_d} \leq t < t_{jn_d}\}}$, and $A(j)_i = A_{t_{(j-1)n_d+i}}$. When A is a discrete-time process, we let $A(j)_i = A_{(j-1)n_d+i}$. In both cases, $A(j)$ can be regarded as a discrete-time process. We further let $\varepsilon_C(j)_i := \varepsilon_{(j-1)n_d+i}$ for $i \geq 1$ and $\varepsilon_C(j)_i := \tilde{\varepsilon}(j)_i$ for $i < 1$, where $\{\tilde{\varepsilon}(j)_i : i \leq 0, j \geq 1\}$ is a set of standard normal random variables that are independent across (i, j) and with everything else. We define for all $i \geq 1$,

$$U^C(j)_i = \eta_C(j) \iota^{(n)} \theta^{(n)}(B) \varepsilon_C(j)_i,$$

and write $U^C(j) := (U^C(j)_1, \dots, U^C(j)_{n_d})^\top$.

Part 3. For $(m = n_d, 1 \leq j \leq J_d)$ or $(m = n'_d, j = J_d + 1)$, we define $\Omega_m^U(j) \in \mathcal{M}_m$ by

$$\begin{aligned} \Omega_m^U(j)_{ik} &= (\iota^{(n)})^2 (\eta(j)_i \eta(j)_k \kappa_{|i-k|}^{(n)} + \eta(j)_{i-1} \eta(j)_{k-1} \kappa_{|i-k|}^{(n)} \\ &\quad - \eta(j)_i \eta(j)_{k-1} \kappa_{|i-k+1|}^{(n)} - \eta(j)_{i-1} \eta(j)_k \kappa_{|i-k-1|}^{(n)}). \end{aligned}$$

Using $\Omega_m^U(j)$, we define $\Omega_n^U = \left(\bigoplus_{j=1}^{J_d} \Omega_{n_d}^U(j) \right) \oplus \Omega_{n'_d}^U(J_d + 1)$. For any n , let $\Omega_n^B, \Omega_n^J, \Omega_n^Y, \Omega_n^{Y,B} \in \mathcal{M}_{n_T}$ be defined by

$$(\Omega_n^B)_{ij} = \delta_{i,j} \int_{t_{i-1}}^{t_i} \sigma_s^2 ds, \quad (\Omega_n^J)_{ij} = \delta_{i,j} \sum_{t_{i-1} < s \leq t_i} (\Delta X_s)^2, \quad \Omega_n^{Y,B} = \Omega_n^U + \Omega_n^B, \quad \Omega_n^Y = \Omega_n^{Y,B} + \Omega_n^J.$$

Then, for $j \geq 1$ we introduce an ∞ -dimensional vector $\gamma_C(j) := (\gamma_C(j)_k)_{k \geq 0}$ with $\gamma_C(j)_k = (\iota^{(n)})^2 \eta_C^2(j) \kappa_k^{(n)}$, and a scalar $\zeta_C(j) := \sum_{k=-\infty}^{\infty} \gamma_C(j)_{|k|}$. Finally, we introduce

$$\Omega_n^{U,C}(j) = O_{n_d} (2\mathbb{I}_{n_d} - \mathbb{D}_{n_d}^1) D_{n_d}(\gamma_C(j)) O_{n_d},$$

where D_{n_d} is defined in (A.1), whose dependence on $\gamma_C(j)$ is made explicit.

Part 4. We introduce shorthand notation $\mathcal{L}(A) = -\frac{1}{2} \log \det \Omega_{D,n} - \frac{1}{2} \text{tr}(\Omega_{D,n}^{-1} A)$ and let

$$L_{A,n} = -\frac{1}{2} \log \det \Omega_n - \frac{1}{2} \text{tr}(\Omega_n^{-1} Y_n Y_n^\top), \quad L_{D,n} = \mathcal{L}(Y_n Y_n^\top), \quad \bar{L}_n = \mathcal{L}(\Omega_n^Y),$$

where we omit the argument (σ^2, γ) of $(\Omega_n, \Omega_{D,n})$ and $(L_{A,n}, L_{D,n}, \bar{L}_n)$. Finally, we define

$$\begin{aligned} \bar{L}_n^*(\sigma^2, \gamma) &= -\frac{n_T}{4\pi} \int_{-\pi}^{\pi} \left(\log f(\lambda; \sigma^2, \gamma, \Delta_n) + \frac{f(\lambda; C_T, \gamma^{(n)}, \Delta_n)}{f(\lambda; \sigma^2, \gamma, \Delta_n)} \right) d\lambda, \\ \chi^2(\sigma^2, \gamma, \Delta_n) &= \exp \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} \log f(\lambda; \sigma^2, \gamma, \Delta_n) d\lambda \right). \end{aligned}$$

Part 5. With any given n , (σ, γ) , and q , we define

$$R_n(\sigma^2, \gamma) = |\sigma^2 - C_T| + \sup_{\lambda} |f(\lambda; \gamma) - f(\lambda; \gamma^{(n)})|,$$

$$\widehat{\mathcal{R}}_n(q) = R_n(\widehat{\sigma}_n^2(q), \widehat{\gamma}_n(q)), \quad \mathcal{R}^{(n)}(q) = R_n(\sigma^{(n)}(q)^2, \gamma^{(n)}(q)).$$

Part 6. We introduce a framework to conduct reparameterization. To avoid ambiguity, throughout the proof we use $\Pi_n^{(\sigma^2, \gamma)}(q)$ to refer to the parameter space $\Pi_n(q)$ defined in (3.11). We let

$$(\sigma^{(n)}(q)^2, \gamma^{(n)}(q)) = \arg \min_{(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q)} \bar{L}_n^*(\sigma^2, \gamma).$$

We start by introducing a bijection from $\Pi_n^{(\sigma^2, \gamma)}(q)$ to $\mathbb{R}^{q+2}$ denoted by $\beta_n(\sigma^2, \gamma)$. The inverse functions are denoted by $\sigma_n^2(\beta)$ and $\gamma_n(\beta)$. Choices of the functional form of β_n will only be specified when necessary and will typically vary across different scenarios. We set $\partial \sigma_n^2 := \partial \sigma_n^2(\beta) / \partial \beta$. Let C_T be the quadratic variation of X scaled by T^{-1} , and let $\widehat{\beta}_n(q), \beta^{(n)}(q) \in \mathbb{R}^{q+2}$ be defined as

$$\widehat{\beta}_n(q) = \beta_n(\widehat{\sigma}_n^2(q), \widehat{\gamma}_n(q)), \quad \beta^{(n)}(q) = \beta_n(\sigma^{(n)}(q)^2, \gamma^{(n)}(q)), \quad \bar{\beta}^{(n)} = \beta_n(C_T, \gamma^{(n)}),$$

Let $\Pi_n^\beta(q) = \{\beta = (\beta_0, \beta_1, \dots, \beta_{q+1})^\top \in \mathbb{R}^{q+2} : \beta = \beta_n(\sigma^2, \gamma) \text{ with } (\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q)\}$. For any $\beta \in \Pi_n^\beta(q)$, and any $S_n \in \{f(\lambda; \cdot, \cdot, \Delta_n), L_n, L_{A,n}, L_{D,n}, \bar{L}_n, \bar{L}_n^*, \Sigma_n, \Omega_n, \Omega_{D,n}, V_n\}$, we let $S_n(\beta) = S_n(\sigma^2, \gamma)$, with (σ^2, γ) satisfying $\beta = \beta_n(\sigma^2, \gamma)$. Furthermore, for $\beta \in \Pi_n^\beta(q)$ and $s \in \{A, D\}$ we define $\Xi_n(\beta), \bar{\Xi}_n(\beta), \Xi_{s,n}(\beta) \in \mathbb{R}^{q+2}$ and $\partial \bar{\Xi}_n^*(\beta) \in \mathcal{M}_{q+2}$ such that

$$(\Xi_n(\beta)_j, \bar{\Xi}_n(\beta)_j, \Xi_{s,n}(\beta)_j) = -\frac{1}{n} \frac{\partial}{\partial \beta_j} (L_n(\beta), \bar{L}_n(\beta), L_{s,n}(\beta)), \quad \partial \bar{\Xi}_n^*(\beta)_{ij} = -\frac{1}{n} \frac{\partial^2}{\partial \beta_i \partial \beta_j} \bar{L}_n^*(\beta), \quad (\text{A.2})$$

and we write $\bar{\eta} := -\partial \bar{\Xi}_n^*(\bar{\beta}^{(n)})^{-1} \partial \sigma_n^2$.

A.2 Proofs of Lemmas

As is typical in the literature, upon using a classical localization procedure (Section 4.4.1 of [Jacod and Protter \(2011\)](#)) we can strengthen the conditions introduced by Assumptions 1, 2, and 3 as follows:

Assumption A1. *There exist a constant $K > 0$ and nonnegative functions Γ and $\tilde{\Gamma}$, such that the processes X , μ , σ , ξ , ξ^{-1} , η , $\tilde{\mu}$, $\tilde{\sigma}$ are bounded by K , and the functions δ and $\tilde{\delta}$ satisfy $|\delta(u)| \leq \Gamma(u) \leq K$ and $\|\tilde{\delta}(u)\| \leq \tilde{\Gamma}(u) \leq K$. The ingredients of ξ and η (not written explicitly) also satisfy the same conditions as above.*

Lemma A1. *For all integers m and h satisfying $0 \leq h \leq m$, it holds that $\mathbb{D}_m^h = O_m \mathbb{F}_m^h O_m$, where $\mathbb{F}_m^h \in \mathcal{M}_m$ given by*

$$(\mathbb{F}_m^h)_{ij} = \mathbb{1}_{\{h=|i-j|\}} - \mathbb{1}_{\{h=i+j\}} - \mathbb{1}_{\{h=2m+2-(i+j)\}}. \quad (\text{A.3})$$

Proof. The lemma can be verified with straightforward algebra. ■

Lemma A2. Suppose $m\Delta_n^{1/2+\alpha} \rightarrow \infty$ for some fixed $\alpha > 0$. Define $\mathbb{F}_m^h$ by (A.3). It holds that for $v \in \{0, 1\}$,

$$V_m^{-1}(\sigma^2, \gamma, \Delta_n) D_m^{-v}(\gamma) = \sum_{h=0}^{m+1} \rho_h(\sigma^2, \gamma, \Delta_n, v) \mathbb{D}_m^h$$

$$\text{and} \quad \Omega_m^{-1}(\sigma^2, \gamma, \Delta_n) O_m D_m^{-v}(\gamma) O_m = \sum_{h=0}^{m+1} \rho_h(\sigma^2, \gamma, \Delta_n, v) \mathbb{F}_m^h.$$

Here $\rho_h(\sigma^2, \gamma, \Delta_n)$ satisfies that, for all sequences of parameters $\{(\sigma_n^2, \gamma_n) \in \Pi_n^{(\sigma^2, \gamma)}(q_n) : n \geq 1\}$ and all $\{q_n\}$, (i) under $\Delta_n^{-1} \chi^2(\sigma_n^2, \gamma_n, \Delta_n) \rightarrow \infty$ and for $v \in \{0, 1\}$,

$$\zeta_n^{2v} \rho_h(\sigma_n^2, \gamma_n, \Delta_n, v) = \frac{1 - z_n^*}{\sigma^2 \Delta_n (1 + z_n^*)} (z_n^*)^{|h|} + O\left((z_n^*)^{|h|} \chi_n^{-2} \log(\Delta_n^{-1/2} \chi_n) + \frac{1}{\chi_n^2} \wedge \frac{1}{h^2 \Delta_n}\right), \quad (\text{A.4})$$

$$\begin{aligned} \zeta_n^{2v} \left(\rho_h(\sigma_n^2, \gamma_n, \Delta_n, v) - \rho_{h+1}(\sigma_n^2, \gamma_n, \Delta_n, v) \right) &= \frac{(1 - z_n^*)^2}{\sigma^2 \Delta_n (1 + z_n^*)} (z_n^*)^h \\ &+ O\left(\Delta_n^{1/2} \chi_n^{-1} \left(\frac{1}{\chi_n^2} \wedge \frac{1}{h^2 \Delta_n} \right) + ((z_n^*)^h \chi_n^{-2} (\Delta_n^{1/2} \chi_n^{-1} \log(\Delta_n^{-1/2} \chi_n) + h^{-1}))\right), \end{aligned} \quad (\text{A.5})$$

$$\begin{aligned} \zeta_n^{2v} \left(2\rho_{h+1}(\sigma_n^2, \gamma_n, \Delta_n, v) - \rho_{h+2}(\sigma_n^2, \gamma_n, \Delta_n, v) - \rho_h(\sigma_n^2, \gamma_n, \Delta_n, v) \right) \\ = -\frac{(1 - z_n^*)^3}{\sigma^2 \Delta_n (1 + z_n^*)} (z_n^*)^h + O\left(\frac{1}{h^2 \chi_n^2} + (z_n^*)^h \Delta_n \chi_n^{-4} \log(\Delta_n^{-1/2} \chi_n)\right), \end{aligned} \quad (\text{A.6})$$

where $(z_n^*, \zeta_n^2, \chi_n^2)$ does not depend on h and is given by

$$z_n^* = 1 - \frac{\sigma_n \Delta_n^{1/2}}{\zeta_n} + o(\Delta_n^{1/2} \chi_n^{-1}), \quad \zeta_n^2 = \sum_{j=-\infty}^{\infty} \gamma_{n, |j|}, \quad \chi_n^2 = \chi^2(\sigma_n^2, \gamma_n, \Delta_n);$$

and (ii) under $\Delta_n^{-1} \chi^2(\sigma_n^2, \gamma_n, \Delta_n) \leq K$,

$$\rho_h(\sigma_n^2, \gamma_n, \Delta_n, 0) = O(h^{-2} \Delta_n^{-1}). \quad (\text{A.7})$$

Proof. Step 1. (Main proof) Given the expression of $V_m^{-1} D_m^{-v}$, that of $\Omega_m^{-1} O_m D_m^{-v} O_m$ directly follows by applying Lemma A1. Hence it suffices to analyze $V_m^{-1} D_m^{-v}$. First, for all $z \in \mathbb{C}$ with $z \neq 0$, we define

$$\mathcal{V}(z; \sigma^2, \gamma, \Delta_n) := \sigma^2 \Delta_n + (2 - z - z^{-1}) f(z; \gamma), \quad \text{with} \quad f(z; \gamma) = \sum_{j=-\infty}^{\infty} \gamma_{|j|} z^j. \quad (\text{A.8})$$

We also define

$$\check{\rho}_h(\sigma^2, \gamma, \Delta_n, v) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{e^{ih\lambda} d\lambda}{\mathcal{V}(e^{i\lambda}; \sigma^2, \gamma, \Delta_n) f^v(z; \gamma)}. \quad (\text{A.9})$$

In the remaining steps, we prove a key property whereby $\check{\rho}_h(\sigma^2, \gamma, \Delta_n)$ satisfies (A.4), (A.5), (A.6), and (A.7) (of course, we will replace ρ_h with $\check{\rho}_h$ in those two equations) for all $\{(\sigma_n^2, \gamma_n) \in \Pi_n^{(\sigma^2, \gamma)}(q_n) : n \geq 1\}$. Now we demonstrate that this property directly leads to what this lemma claims. In view of (A.8) and the definitions of V_m and $\mathbb{D}_m^h$, we have

$$V_m(\sigma^2, \gamma, \Delta_n)_{j,j} D_m^v(\gamma)_{j,j} = \mathcal{V}\left(e^{i\frac{j\pi}{m+1}}; \sigma^2, \gamma, \Delta_n\right) f^v\left(e^{i\frac{j\pi}{m+1}}; \gamma\right), \quad \forall 1 \leq j \leq m.$$

Because $V_m(\sigma^2, \gamma, \Delta_n)$ and $D_m(\gamma)$ are diagonal by construction, we have

$$(V_m^{-1}(\sigma^2, \gamma, \Delta_n) D_m^{-v}(\gamma))_{i,j} = \frac{\delta_{i,j}}{\mathcal{V}\left(e^{i\frac{j\pi}{m+1}}; \sigma^2, \gamma, \Delta_n\right) f^v\left(e^{i\frac{j\pi}{m+1}}; \gamma\right)}.$$

Moreover, since we have $(\mathbb{D}_m^h)_{i,j} = \delta_{i,j}(2 - \delta_{h,0}) \cos \frac{hj\pi}{m+1}$, we only need to show that there exists some $\rho_h(\sigma^2, \gamma, \Delta_n)$ that satisfies

$$\text{that (A.4), (A.5), (A.6), and (A.7) for all } \{(\sigma_n^2, \gamma_n) \in \Pi_n^{(\sigma^2, \gamma)}(q_n) : n \geq 1\} \text{ is true,} \quad (\text{A.10})$$

and that

$$\frac{1}{\mathcal{V}\left(e^{i\frac{j\pi}{m+1}}; \sigma^2, \gamma, \Delta_n\right) f^v\left(e^{i\frac{j\pi}{m+1}}; \gamma\right)} = \sum_{h=0}^{m+1} \rho_h(\sigma^2, \gamma, \Delta_n, v) \cos \frac{hj\pi}{m+1}. \quad (\text{A.11})$$

We now prove this is true. Using the definition of $\check{\rho}_h(\sigma^2, \gamma, \Delta_n, v)$ from (A.9), we can write

$$\begin{aligned} \frac{1}{\mathcal{V}\left(e^{i\frac{j\pi}{m+1}}; \sigma^2, \gamma, \Delta_n\right) f^v\left(e^{i\frac{j\pi}{m+1}}; \gamma\right)} &= \sum_{h=0}^{\infty} \check{\rho}_h(\sigma^2, \gamma, \Delta_n, v) (2 - \delta_{h,0}) \cos \frac{hj\pi}{m+1} \\ &= -\check{\rho}_0(\sigma^2, \gamma, \Delta_n, v) \\ &\quad + 2 \sum_{h=0}^m \sum_{k=0}^{\infty} \check{\rho}_{h+2k(m+1)}(\sigma^2, \gamma, \Delta_n, v) \cos \frac{hj\pi}{m+1} \\ &\quad + 2 \sum_{h=1}^{m+1} \sum_{k=0}^{\infty} \check{\rho}_{m+1-h+(2k+1)(m+1)}(\sigma^2, \gamma, \Delta_n, v) \cos \frac{hj\pi}{m+1}. \end{aligned}$$

Here the last equality comes from basic properties of sine and cosine functions. This indicates that (A.11) indeed holds with $\{\rho_h\}_{h=0}^{m+1}$ given by

$$\rho_0(\sigma^2, \gamma, \Delta_n, v) = \check{\rho}_0 + 2 \sum_{k=1}^{\infty} \check{\rho}_{2k(m+1)}, \quad \rho_{m+1}(\sigma^2, \gamma, \Delta_n, v) = \sum_{k=0}^{\infty} \check{\rho}_{(2k+1)(m+1)}, \quad (\text{A.12})$$

$$\rho_h(\sigma^2, \gamma, \Delta_n, v) = \sum_{k=0}^{\infty} (\check{\rho}_{h+2k(m+1)} + \check{\rho}_{m+1-h+(2k+1)(m+1)}), \quad \forall 1 \leq h \leq m. \quad (\text{A.13})$$

Here we omit the argument $(\sigma^2, \gamma, \Delta_n, v)$ of $\check{\rho}_h$. Suppose $\check{\rho}_h(\sigma^2, \gamma, \Delta_n, v)$ satisfies (A.10). Then, given that $m\Delta_n^{1/2+\alpha} \rightarrow \infty$ for a fixed $\alpha > 0$, we have that $\{\rho_h\}_{h=0}^{m+1}$ defined by (A.12) and (A.13) also satisfies (A.10), which proves the current lemma. Now we move forward to show that $\check{\rho}_h(\sigma^2, \gamma, \Delta_n, v)$ indeed satisfies (A.10).

Step 2. (Characterization of $\check{\rho}$) In this step, we connect the behavior of $\check{\rho}$ with properties of $\mathcal{V}$ using the definition (A.9). We start with a decomposition. We write that for all $p \geq 1$,

$$\mathcal{V}(z; \sigma^2, \gamma, \Delta_n) = \mathcal{V}(z; \sigma^2, \tilde{\gamma}(p, \gamma), \Delta_n) + \mathcal{V}(z; 0, \tilde{\gamma}(-p, \gamma), \Delta_n),$$

where $\tilde{\gamma}(p, \gamma)$ and $\tilde{\gamma}(-p, \gamma)$ are shorthand notation defined by $\tilde{\gamma}(p, \gamma) = (\gamma_0, \gamma_1, \dots, \gamma_p, 0, \dots, 0)^\top$ and $\tilde{\gamma}(-p, \gamma) = \gamma - \tilde{\gamma}(p, \gamma)$. In other words, $\tilde{\gamma}(p, \gamma)$ represents the first $p+1$ components of γ , while $\tilde{\gamma}(-p, \gamma)$ captures the remaining ones. The decomposition directly comes from the fact that $\mathcal{V}$ is linear in γ . In the rest of the proof, for notational simplicity, we write

$$\mathcal{V}(z; \Delta_n) = \mathcal{V}(z; \sigma^2, \gamma, \Delta_n), \quad \mathcal{V}(z; \Delta_n, p) = \mathcal{V}(z; \sigma^2, \tilde{\gamma}(p, \gamma), \Delta_n),$$

$$\mathcal{V}(z; \Delta_n, -p) = \mathcal{V}(z; \sigma^2, \tilde{\gamma}(-p, \gamma), \Delta_n), \quad \text{and} \quad f(z; p) = f(z; \tilde{\gamma}(p, \gamma)).$$

We can now write that for all $|z| = 1$,

$$\mathcal{V}(z; \Delta_n)^{-1} = \mathcal{V}(z; \Delta_n, p_n)^{-1} \left[1 + \sum_{j=1}^{\infty} \left(-\frac{\mathcal{V}(z; \Delta_n, -p_n)}{\mathcal{V}(z; \Delta_n, p_n)} \right)^j \right]. \quad (\text{A.14})$$

For a positive sequence p_n , let $\{\check{\rho}_h(\Delta_n, p_n, v)\}_{h=-\infty}^{\infty}$ and $\{\check{\rho}_h(\Delta_n, -p_n)\}_{h=-\infty}^{\infty}$ be the Fourier coefficients of, respectively, $\frac{1}{\mathcal{V}(e^{i\lambda}; \Delta_n, p_n) f^v(e^{i\lambda}; \gamma)}$ and $\sum_{j=1}^{\infty} \left(-\frac{\mathcal{V}(z; \Delta_n, -p_n)}{\mathcal{V}(z; \Delta_n, p_n)} \right)^j$:

$$\check{\rho}_h(\Delta_n, p_n, v) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{e^{-ih\lambda}}{\mathcal{V}(e^{i\lambda}; \Delta_n, p_n) f^v(e^{i\lambda}; \gamma)} d\lambda, \quad (\text{A.15})$$

$$\check{\rho}_h(\Delta_n, -p_n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-ih\lambda} \sum_{j=1}^{\infty} \left(-\frac{\mathcal{V}(e^{i\lambda}; \Delta_n, -p_n)}{\mathcal{V}(e^{i\lambda}; \Delta_n, p_n)} \right)^j d\lambda. \quad (\text{A.16})$$

In view of (A.9), (A.14), (A.15), and (A.16), we have

$$\check{\rho}_h(\sigma^2, \gamma, \Delta_n, v) = \check{\rho}_h(\Delta_n, p_n, v) + \sum_{j=-\infty}^{\infty} \check{\rho}_j(\Delta_n, p_n, v) \check{\rho}_{h-j}(\Delta_n, -p_n). \quad (\text{A.17})$$

Step 3. (Implication of $(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}$) The definition of $\Pi_n^{(\sigma^2, \gamma)}$ indicates

$$\frac{1}{K} \leq \sigma^2 \leq K, \quad (\text{A.18})$$

where the first inequality is because $\sigma^2 = f(0; \sigma^2, \gamma, \Delta_n)$ and the second is obvious. Given the positivity of σ^2 , the second inequality in (3.11) requires $\frac{\sum_{j=1}^{\infty} j^2 |\gamma_j|}{\inf_{\lambda} |\sigma^2 \Delta_n + f(\lambda; \gamma)|} \leq K$. This further indicates

$$\sup_{\lambda} \left| \frac{d}{d\lambda} \log |\sigma^2 \Delta_n + f(\lambda; \gamma)| \right| \leq K \quad \text{and} \quad \sup_{\lambda} \left| \frac{1}{\sigma^2 \Delta_n + f(\lambda; \gamma)} \frac{d^2}{d\lambda^2} f(\lambda; \gamma) \right| \leq K. \quad (\text{A.19})$$

Because of the periodicity of $\log |\sigma^2 \Delta_n + f(\lambda; \gamma)|$, the first inequality of (A.19) indicates that

$$\sup_{\lambda} \log |\sigma^2 \Delta_n + f(\lambda; \gamma)| - \inf_{\lambda} \log |\sigma^2 \Delta_n + f(\lambda; \gamma)| \leq K. \quad (\text{A.20})$$

Using $\sigma^2 > 0$ and $\sigma^2 \Delta_n + 4f(-\pi; \gamma) > 0$, both of which come from the first inequality in (3.11), we conclude $\sigma^2 \Delta_n + f(-\pi; \gamma) > 0$. This indicates, in view of the fact that $\log x$ diverges as $x \rightarrow 0$ and that $\log |\sigma^2 \Delta_n + f(\lambda; \gamma)|$ has a bounded derivative,

$$\inf_{\lambda} (\sigma^2 \Delta_n + f(\lambda; \gamma)) > 0 \quad \text{and} \quad \inf_{\lambda} (\sigma^2 \Delta_n + f(\lambda; \gamma)) \geq \frac{1}{K} \sup_{\lambda} (\sigma^2 \Delta_n + f(\lambda; \gamma)). \quad (\text{A.21})$$

The two inequalities in (A.21), plus the positivity of σ^2 , indicates that

$$\sup_{\lambda} f(\lambda; \gamma) \leq K \inf_{\lambda} (\sigma^2 \Delta_n + f(\lambda; \gamma)) \leq K \int_{-\pi}^{\pi} \sigma^2 \Delta_n + f(\lambda; \gamma) d\lambda \leq \sigma^2 \Delta_n + \gamma_0 \leq K,$$

where the last inequality comes from the second inequality in (3.11). Furthermore, using (A.21), straightforward algebra shows that uniformly over $-\pi \leq \lambda \leq \pi$,

$$K^{-1} \chi^2 \leq \sigma^2 \Delta_n + f(\lambda; \gamma) \leq K \chi^2 \quad \text{and} \quad \sum_{j=1}^{\infty} j^2 |\gamma_j| \leq K \chi^2, \quad (\text{A.22})$$

where $\chi^2 = \chi^2(\sigma^2, \gamma, \Delta_n)$. We emphasize that all the K s involved in the current step are constants that do not depend on n , and therefore all the bounds here hold uniformly over all $\{(\sigma_n^2, \gamma_n) \in \Pi_n^{(\sigma^2, \gamma)}(q_n) : n \geq 1\}$.

Step 4. (The case $\Delta_n^{-1} \chi_n^2 \rightarrow \infty$: Properties of $\mathcal{V}(z; \Delta_n, p)$, part 1) Throughout the rest of the proof, we suppress the subscript n of (σ_n^2, γ_n) whenever possible. In the case of $\Delta_n^{-1} \chi_n^2 \rightarrow \infty$, we first prove that for each n sufficiently large and p_n satisfying $p_n \Delta_n^{1/2} \chi_n^{-1} \leq K$, there exists a unique complex number z_n^* such that

$$1 - K^{-1} p_n^{-1} \leq |z_n^*| \leq 1 \quad \text{and} \quad \mathcal{V}(z_n^*; \Delta_n, p_n) = 0. \quad (\text{A.23})$$

In other words, asymptotically, z_n^* is the solution of $\mathcal{V}(z; \Delta_n, p_n) = 0$, which is closest to the unit circle in the complex plane. We first show there exists a unique real solution within $[1 - K^{-1}p_n^{-1}, 1]$.

We can calculate

$$\frac{1}{1-z} \frac{d}{dz} \mathcal{V}(z; \Delta_n, p_n) = \frac{1+z}{z^2} f(z; p_n) - \frac{1-z}{z} \frac{d}{dz} f(z; p_n). \quad (\text{A.24})$$

Moreover, we have that for n sufficiently large and uniformly over $z \in (1 - K^{-1}p_n^{-1}, 1)$,

$$f(z; p_n) \geq f(1; p_n) - \sum_{j=1}^{p_n} |\gamma_j| \times |z^j + z^{-j} - 2| \geq K^{-1}\chi_n^2 - Kp_n^{-2} \sum_{j=1}^{p_n} |\gamma_j| j^2 \geq K^{-1}\chi_n^2. \quad (\text{A.25})$$

Here the first inequality comes from the triangular inequality, the second comes from the fact that the highest power terms in $f(z; p_n)$ are z^{p_n} and z^{-p_n} , and the last from the second part of (A.22). Furthermore, for n sufficiently large and uniformly over $z \in (1 - K^{-1}p_n^{-1}, 1)$,

$$\left| \frac{d}{dz} f(z; p_n) \right| \leq \sum_{j=1}^{p_n} j |\gamma_j| |z^{j-1} - z^{-j-1}| \leq Kp_n^{-1} \sum_{j=1}^{p_n} j^2 |\gamma_j| \leq Kp_n^{-1}, \quad (\text{A.26})$$

where once again the first inequality comes from the triangular inequality, the second from the fact that the highest power terms in $f(z; p_n)$ are z^{p_n} and z^{-p_n} , and the last from the second part of (A.22). Plugging (A.25) and (A.26) back into (A.24), we obtain that

$$\inf_{z \in (1 - K^{-1}p_n^{-1}, 1)} \frac{1}{1-z} \frac{d}{dz} \mathcal{V}(z; \Delta_n, p_n) \geq K^{-1}\chi_n^2. \quad (\text{A.27})$$

In view of $\mathcal{V}(1; \Delta_n, p_n) = \sigma^2 \Delta_n$ and $\sigma^2 \in (K^{-1}, K)$, as shown in (A.18), plus applying mean value theorem, (A.27) readily leads to the existence and uniqueness of the real solution within $[1 - K^{-1}p_n^{-1}, 1]$. Now we show that any z_n^* that satisfies (A.23) must be real for large n . Suppose we write $z_n^* = |z_n^*| e^{i\varphi_n^*}$ with $\varphi \in [0, 2\pi)$. We prove $\varphi_n^* = 0$ by contradiction. In view of the fact that for all $|z| = 1$, $\mathcal{V}(z; \Delta_n, p_n) \geq K^{-1}\Delta_n > 0$ by construction, it suffices to show that $\varphi_n^* \neq 0$ indicates that $|z_n^*| = 1$. The imaginary part of $\mathcal{V}(z_n^*; \Delta_n, p_n)$ is

$$\text{Im}(\mathcal{V}(z_n^*; \Delta_n, p_n)) = \mathcal{R}_a(z_n^*) + \mathcal{R}_b(z_n^*),$$

where we use the shorthand notation

$$\mathcal{R}_a(z) = -\sin \varphi \left(|z| - \frac{1}{|z|} \right) \text{Re}(f(z, p_n)) \quad \text{and} \quad \mathcal{R}_b(z) = \cos \varphi \left(2 - |z| - \frac{1}{|z|} \right) \text{Im}(f(z, p_n)),$$

with $\varphi \in [0, 2\pi)$ and $e^{i\varphi} = z/|z|$. We notice that uniformly over $z \in \{z : |z| \in (1 - K^{-1}p_n^{-1}, 1)\}$,

$$\text{Re}(f(z, p_n)) \geq K^{-1} \quad \text{and} \quad |\text{Im}(f(z, p_n))| \leq (1 - |z|) \times |\sin \varphi|, \quad \text{with} \quad e^{i\varphi} = z/|z|,$$

which can be shown by the same argument that justifies (A.25) and (A.26). This result, plus the proximity of $|z_n^*|$ to one by construction, immediately indicates that $\mathcal{R}_a$ dominates $\mathcal{R}_b$ asymptotically:

$$\sup_{z: |z| \in (1-K^{-1}p_n^{-1}, 1)} \left| \frac{\mathcal{R}_b(z)}{\mathcal{R}_a(z)} \right| \rightarrow 0.$$

On the other hand, we obviously have $\text{Im}(\mathcal{V}(z_n^*; \Delta_n, p_n)) = 0$ because $\mathcal{V}(z_n^*; \Delta_n, p_n) = 0$, which further requires $\mathcal{R}_a(z_n^*) = 0$. Therefore $\varphi_n^* \neq 0$ necessarily indicates $|z_n^*| = 1$. Contradiction is established and the fact that z_n^* is real is proved. Finally, we derive the expression of z_n^* . We write $z_n^* = 1 + a\Delta_n^{1/2}\chi_n^{-1} + b\Delta_n\chi_n^{-2} + \dots$ and match the coefficients to let $\mathcal{V}(z_n^*; \Delta_n, p_n) = 0$, which gives the explicit expression of z_n^* , up to $o(\Delta_n^{1/2}\chi_n^{-1})$:

$$z_n^* = 1 - \frac{\sigma_n \Delta_n^{1/2}}{\zeta_n} + o(\Delta_n^{1/2}\chi_n^{-1}). \quad (\text{A.28})$$

Step 5. (The case $\Delta_n^{-1}\chi_n^2 \rightarrow \infty$: Properties of $\mathcal{V}(z; \Delta_n, p)$, part 2) Now we study the properties of $\mathcal{V}(z; \Delta_n, p)$ beyond its closest-to-unit-circle root for the case of $\Delta_n^{-1}\chi_n^2 \rightarrow \infty$. Given (A.28) and noticing that $\mathcal{V}(z; \Delta_n, p_n) = 0$ indicates $\mathcal{V}(z^{-1}; \Delta_n, p_n) = 0$, we can introduce $\tilde{\mathcal{V}}(z; \Delta_n, p_n)$ defined by

$$(z - z_n^*) \left(\frac{1}{z} - z_n^* \right) \tilde{\mathcal{V}}(z; \Delta_n, p_n) = \mathcal{V}(z; \Delta_n, p_n), \quad (\text{A.29})$$

where z can be any nonzero complex number. In other words, $\tilde{\mathcal{V}}(z; \Delta_n, p_n)$ can be understood as capturing the roots of $\mathcal{V}(z; \Delta_n, p_n)$ other than z_n^* and $1/z_n^*$. We now analyze its properties; again, $p_n \Delta_n^{1/2} \chi_n^{-1} \leq K$. We first claim that uniformly over $-\pi \leq \lambda \leq \pi$,

$$K^{-1} \leq \frac{(e^{i\lambda} - z_n^*)(e^{-i\lambda} - z_n^*)}{\Delta_n \chi_n^{-2} + 1 - \cos \lambda} \leq K \quad \text{and} \quad K^{-1} \leq \frac{\mathcal{V}(e^{i\lambda}; \Delta_n, p_n)}{\Delta_n + \chi_n^2(1 - \cos \lambda)} \leq K, \quad (\text{A.30})$$

where the first result is obvious for the expression of z_n^* from (A.28), while the second result can be easily verified using the first part of (A.22). Combined with the construction of $\tilde{\mathcal{V}}(z; \Delta_n, p_n)$ from (A.29), we obtain that uniformly over $-\pi \leq \lambda \leq \pi$,

$$K^{-1}\chi_n^2 \leq \tilde{\mathcal{V}}(e^{i\lambda}; \Delta_n, p_n) \leq K\chi_n^2. \quad (\text{A.31})$$

In other words, $\tilde{\mathcal{V}}(e^{i\lambda})$ is uniformly of the same order of χ_n^2 . Now we bound the derivatives of $\tilde{\mathcal{V}}(e^{i\lambda})$. We can write

$$\tilde{\mathcal{V}}(e^{i\lambda}; \Delta_n, p_n) = h(\lambda; \sigma^2, \gamma, \Delta_n) + (z_n^*)^{-1} f(e^{i\lambda}; p_n),$$

where we introduce the shorthand notation $h(\lambda; \sigma^2, \gamma, \Delta_n) := \frac{\sigma^2 \Delta_n - (1-z^*)^2 (z^*)^{-1} f(e^{i\lambda}; p_n)}{(1-z^*)^2 + (2-2\cos \lambda)z^*}$. Because the second part of (A.22), whereby the first- and second-order derivatives of $f(\lambda; \gamma)$ are bounded by $K\chi_n^2$, plus the fact that $|(z_n^*)^{-1}|$ is bounded because of (A.28), it obviously holds that uniformly over

$$-\pi \leq \lambda \leq \pi,$$

$$\left| \frac{d}{d\lambda} \tilde{\mathcal{V}}(e^{i\lambda}) \right| \leq K\chi_n^2 \quad \text{and} \quad \left| \frac{d^2}{d\lambda^2} \tilde{\mathcal{V}}(e^{i\lambda}) \right| \leq K\chi_n^2, \quad (\text{A.32})$$

as long as we show that uniformly over $-\pi \leq \lambda \leq \pi$,

$$\left| \frac{d}{d\lambda} h(\lambda; \sigma^2, \gamma, \Delta_n) \right| \leq K\chi_n^2 \quad \text{and} \quad \left| \frac{d^2}{d\lambda^2} h(\lambda; \sigma^2, \gamma, \Delta_n) \right| \leq K\chi_n^2. \quad (\text{A.33})$$

We first explicitly calculate these two derivatives of h . Some algebra can show that

$$\frac{d}{d\lambda} h(\lambda; \sigma^2, \gamma, \Delta_n) \quad (\text{A.34})$$

$$= \frac{-(1 - z_n^*)^2 (z_n^*)^{-1}}{(1 - z_n^*)^2 + (2 - 2 \cos \lambda) z_n^*} \frac{d}{d\lambda} f(e^{i\lambda}; p_n) + (\sigma^2 \Delta_n - (1 - z_n^*)^2 (z_n^*)^{-1} f(e^{i\lambda}; p_n)) \frac{d}{d\lambda} \frac{1}{(1 - z_n^*)^2 + (2 - 2 \cos \lambda) z_n^*} \quad (\text{A.35})$$

and

$$\frac{d^2}{d\lambda^2} h(\lambda; \sigma^2, \gamma, \Delta_n) \quad (\text{A.36})$$

$$= \frac{-(1 - z_n^*)^2 (z_n^*)^{-1}}{(1 - z_n^*)^2 + (2 - 2 \cos \lambda) z_n^*} \frac{d^2}{d\lambda^2} f(e^{i\lambda}; p_n) - 2(1 - z_n^*)^2 (z_n^*)^{-1} \left(\frac{d}{d\lambda} \frac{1}{(1 - z_n^*)^2 + (2 - 2 \cos \lambda) z_n^*} \right) \frac{d}{d\lambda} f(e^{i\lambda}; p_n) + (\sigma^2 \Delta_n - (1 - z_n^*)^2 (z_n^*)^{-1} f(e^{i\lambda}; p_n)) \frac{d^2}{d\lambda^2} \frac{1}{(1 - z_n^*)^2 + (2 - 2 \cos \lambda) z_n^*}. \quad (\text{A.37})$$

Using the expression of z^* from (A.28), we have that uniformly over $-\pi \leq \lambda \leq \pi$,

$$\left| \frac{(1 - z_n^*)^2 (z_n^*)^{-1}}{(1 - z_n^*)^2 + (2 - 2 \cos \lambda) z_n^*} \right| \leq K, \quad \left| \frac{d}{d\lambda} \frac{1}{(1 - z_n^*)^2 + (2 - 2 \cos \lambda) z_n^*} \right| \leq K \frac{|\lambda|}{(\Delta_n \chi_n^{-2} + 2 - 2 \cos \lambda)^2},$$

and

$$\left| \frac{d^2}{d\lambda^2} \frac{1}{(1 - z_n^*)^2 + (2 - 2 \cos \lambda) z_n^*} \right| \leq K \frac{1}{(\Delta_n \chi_n^{-2} + 2 - 2 \cos \lambda)^2} + K \frac{\lambda^2}{(\Delta_n \chi_n^{-2} + 2 - 2 \cos \lambda)^3}.$$

Plugging these bounds back into the expressions of the derivatives of $h(\lambda; \sigma^2, \gamma, \Delta_n)$ provided by (A.35) and (A.37), plus the fact that the first- and second-order derivatives of $f(e^{i\lambda}; p_n)$ are bounded by $K\chi_n^2$, as indicated by the second part of (A.22), plus the magnitude of $(1 - z_n^*)$ and $(z_n^*)^{-1}$ indicated by the expression of z_n^* from (A.28), we obtain

$$\left| \frac{d}{d\lambda} h(\lambda; \sigma^2, \gamma, \Delta_n) \right| \leq K\chi_n^2 + K \left| \lambda \frac{\sigma^2 \Delta_n - (1 - z_n^*)^2 (z_n^*)^{-1} f(e^{i\lambda}; p_n)}{(\Delta_n \chi_n^{-2} + 2 - 2 \cos \lambda)^2} \right|$$

and

$$\begin{aligned} \left| \frac{d^2}{d\lambda^2} h(\lambda; \sigma^2, \gamma, \Delta_n) \right| &\leq K\chi_n^2 + K\Delta_n\chi_n^{-2} \frac{|\lambda|}{(\Delta_n\chi_n^{-2} + 2 - 2\cos\lambda)^2} \left| \frac{d}{d\lambda} f(e^{i\lambda}; p_n) \right| \\ &+ K \left| \frac{\sigma^2\Delta_n - (1 - z_n^*)^2(z_n^*)^{-1} f(e^{i\lambda}; p_n)}{(\Delta_n\chi_n^{-2} + 2 - 2\cos\lambda)^2} \right| \\ &+ K \left| \lambda^2 \frac{\sigma^2\Delta_n - (1 - z_n^*)^2(z_n^*)^{-1} f(e^{i\lambda}; p_n)}{(\Delta_n\chi_n^{-2} + 2 - 2\cos\lambda)^3} \right|. \end{aligned}$$

These two results indicate that to prove (A.33), it is sufficient to show that uniformly over $-\pi \leq \lambda \leq \pi$,

$$|\sigma^2\Delta_n - (1 - z_n^*)^2(z_n^*)^{-1} f(e^{i\lambda}; p_n)| \leq K\Delta_n(\lambda^2 + \Delta_n\chi_n^{-2}) \quad \text{and} \quad \left| \frac{d}{d\lambda} f(e^{i\lambda}; p_n) \right| \leq \chi_n^2 |\lambda|. \quad (\text{A.38})$$

The second part of (A.38) comes from the fact that $f(e^{i\lambda}; p_n)$ is a differentiable even function, so $\frac{d}{d\lambda} f(1; p_n) = 0$ and the fact that $\left| \frac{d^2}{d\lambda^2} f(e^{i\lambda}; p_n) \right| \leq K\chi_n^2$ uniformly over $-\pi \leq \lambda \leq \pi$ is indicated by the second part of (A.22). Now we show the first part of (A.38). We recall that the definition of z_n^* requires that

$$\sigma^2\Delta_n - (1 - z_n^*)^2(z_n^*)^{-1} f(z_n^*; p_n) = 0.$$

This indicates that

$$\sigma^2\Delta_n - (1 - z_n^*)^2(z_n^*)^{-1} f(e^{i\lambda}; p_n) = -(1 - z_n^*)^2(z_n^*)^{-1} (f(e^{i\lambda}; p_n) - f(z_n^*; p_n)). \quad (\text{A.39})$$

In view of $(1 - z_n^*)^2 \leq K\Delta_n\chi_n^{-2}$ from the expression of z_n^* given by (A.28), plus the traingular inequality, the first part of (A.38) will then come from (A.39) because uniformly over $-\pi \leq \lambda \leq \pi$,

$$|f(e^{i\lambda}; p_n) - f(1; p_n)| \leq K\lambda^2 \sum_{j=1}^{p_n} j^2 |\gamma_j| \leq K\lambda^2 \chi_n^2, \quad (\text{A.40})$$

$$|f(z_n^*; p_n) - f(1; p_n)| \leq K \sum_{j=1}^{p_n} |\gamma_j| |2 - (z_n^*)^j - (z_n^*)^{-j}| \leq K\Delta_n\chi_n^{-2} \sum_{j=1}^{\infty} j^2 |\gamma_j| \leq K\Delta_n\chi_n^{-2}. \quad (\text{A.41})$$

The first inequality in (A.40) is obvious from the definition of $f(z; p_n)$ given in Step 2, while the second arises from the second part of (A.22). On the other hand, the first inequality in (A.41) is also obvious from the definition of $f(z; p_n)$, the second is from the definition of z_n^* , and the last is once again from the second part of (A.22).

Step 6. (The case of $\Delta_n^{-1}\chi_n^2 \rightarrow \infty$: Properties of $\mathcal{V}(z; \Delta_n, -p)$) Now we study the properties of

$\mathcal{V}(z; \Delta_n, -p)$ for the case of $\Delta_n^{-1} \chi_n^2 \rightarrow \infty$. We notice that

$$\frac{\mathcal{V}(z; \Delta_n, -p)}{\mathcal{V}(z; \Delta_n, p)} = \frac{(2 - z - z^{-1})f(z; -p)}{\sigma^2 \Delta_n + (2 - z - z^{-1})f(z; p)} = \frac{f(z; -p)}{f(z; p)} - \frac{\sigma^2 \Delta_n \frac{f(z; -p)}{f(z; p)}}{\sigma^2 \Delta_n + (2 - z - z^{-1})f(z; p)}.$$

Therefore, we have

$$\sup_{\lambda} \left| \frac{\mathcal{V}(e^{i\lambda}; \Delta_n, -p)}{\mathcal{V}(e^{i\lambda}; \Delta_n, p)} \right| \leq K \chi_n^{-2} \sup_{\lambda} f(e^{i\lambda}; -p) \leq K p^{-2}, \quad (\text{A.42})$$

where the last inequality comes from the second part of (A.22). Further, using the second part of (A.22) and with direct calculations, we obtain that uniformly over $-\pi \leq \lambda \leq \pi$,

$$\left| \frac{d\mathcal{V}(e^{i\lambda}; \Delta_n, p)}{d\lambda} \right| \leq K \chi_n^2 |\lambda|, \quad \left| \frac{d\mathcal{V}(e^{i\lambda}; \Delta_n, -p)}{d\lambda} \right| \leq K \chi_n^2 (\lambda^2 p^{-1} + |\lambda| p^{-2}), \quad (\text{A.43})$$

$$\left| \frac{d^2 \mathcal{V}(e^{i\lambda}; \Delta_n, p)}{d\lambda^2} \right| \leq K \chi_n^2 \quad \text{and} \quad \left| \frac{d^2 \mathcal{V}(e^{i\lambda}; \Delta_n, -p)}{d\lambda^2} \right| \leq K \chi_n^2 (\lambda^2 + |\lambda| p^{-1} + p^{-2}). \quad (\text{A.44})$$

On the other hand, we can calculate

$$\frac{d}{d\lambda} \frac{\mathcal{V}(e^{i\lambda}; \Delta_n, -p)}{\mathcal{V}(e^{i\lambda}; \Delta_n, p)} = -\frac{\mathcal{V}(e^{i\lambda}; \Delta_n, -p)}{\mathcal{V}(e^{i\lambda}; \Delta_n, p)^2} \frac{d}{d\lambda} \mathcal{V}(e^{i\lambda}; \Delta_n, p) + \frac{1}{\mathcal{V}(e^{i\lambda}; \Delta_n, p)} \frac{d}{d\lambda} \mathcal{V}(e^{i\lambda}; \Delta_n, -p) \quad (\text{A.45})$$

and

$$\begin{aligned} \frac{d^2}{d\lambda^2} \frac{\mathcal{V}(e^{i\lambda}; \Delta_n, -p)}{\mathcal{V}(e^{i\lambda}; \Delta_n, p)} &= -\frac{\mathcal{V}(e^{i\lambda}; \Delta_n, -p)}{\mathcal{V}(e^{i\lambda}; \Delta_n, p)^2} \frac{d^2 \mathcal{V}(e^{i\lambda}; \Delta_n, p)}{d\lambda^2} \\ &\quad - \frac{2}{\mathcal{V}(e^{i\lambda}; \Delta_n, p)^2} \frac{d\mathcal{V}(e^{i\lambda}; \Delta_n, p)}{d\lambda} \frac{d\mathcal{V}(e^{i\lambda}; \Delta_n, -p)}{d\lambda} \\ &\quad + \frac{1}{\mathcal{V}(e^{i\lambda}; \Delta_n, p)} \frac{d^2 \mathcal{V}(e^{i\lambda}; \Delta_n, -p)}{d\lambda^2}. \end{aligned} \quad (\text{A.46})$$

Plugging (A.30) (A.42), (A.43), and (A.44) back into (A.45) and (A.46), we obtain

$$\left| \frac{d}{d\lambda} \frac{\mathcal{V}(e^{i\lambda}; \Delta_n, -p)}{\mathcal{V}(e^{i\lambda}; \Delta_n, p)} \right| \leq K p^{-1} \frac{\lambda^2 + \lambda p^{-1}}{\Delta_n \chi_n^{-2} + \lambda^2} \quad (\text{A.47})$$

and

$$\left| \frac{d^2}{d\lambda^2} \frac{\mathcal{V}(e^{i\lambda}; \Delta_n, -p)}{\mathcal{V}(e^{i\lambda}; \Delta_n, p)} \right| \leq K \frac{p^{-2} + \lambda^2}{\Delta_n \chi_n^{-2} + \lambda^2} + K \frac{|\lambda|^3 p^{-1} + \lambda^2 p^{-2}}{(\Delta_n \chi_n^{-2} + \lambda^2)^2}. \quad (\text{A.48})$$

Step 7. (The case of $\Delta_n^{-1} \chi_n^2 \leq K$: Properties of $\mathcal{V}(z; \Delta_n)$) Now we study the properties of $\mathcal{V}(z; \Delta_n)$ for the case of $\Delta_n^{-1} \chi_n^2 \leq K$. We observe that uniformly over $-\pi \leq \lambda \leq \pi$,

$$K^{-1} \Delta_n \leq \mathcal{V}(e^{i\lambda}; \Delta_n) \leq K \Delta_n.$$

Indeed, the first inequality comes from the definition of $\Pi_n^{(\sigma^2, \gamma)}$ specified by (3.11). The second

inequality comes from

$$\mathcal{V}(e^{i\lambda}; \Delta_n) \leq \sigma^2 \Delta_n + K|f(\lambda; \gamma)| \leq K\Delta_n, \quad (\text{A.49})$$

in which the first inequality is obvious, given the definition of $\mathcal{V}(e^{i\lambda}; \Delta_n)$, and the second comes from (A.18), the first part of (A.22), and $\Delta_n^{-1}\chi_n^2 \leq K$. Using the second part of (A.22), we obtain, under the case of $\Delta_n^{-1}\chi_n^2 \leq K$ and uniformly over $-\pi \leq \lambda \leq \pi$,

$$|f(\lambda; \gamma)| \leq K\Delta_n, \quad \left| \frac{df(\lambda; \gamma)}{d\lambda} \right| \leq K\Delta_n, \quad \left| \frac{d^2 f(\lambda; \gamma)}{d^2 \lambda} \right| \leq K\Delta_n.$$

These bounds immediately give that uniformly over $-\pi \leq \lambda \leq \pi$,

$$\left| \frac{d\mathcal{V}(e^{i\lambda}; \Delta_n)}{d\lambda} \right| \leq K|f(\lambda; \gamma)| + K \left| \frac{df(\lambda; \gamma)}{d\lambda} \right| \leq K\Delta_n \quad (\text{A.50})$$

and

$$\left| \frac{d^2 \mathcal{V}(e^{i\lambda}; \Delta_n)}{d\lambda^2} \right| \leq K|f(\lambda; \gamma)| + K \left| \frac{df(\lambda; \gamma)}{d\lambda} \right| + K \left| \frac{d^2 f(\lambda; \gamma)}{d^2 \lambda} \right| \leq K\Delta_n. \quad (\text{A.51})$$

Step 8. (Properties of $\check{\rho}$) In this step we show that $\check{\rho}$ satisfies (A.10). We start with the case of $\Delta_n^{-1}\chi_n^2 \rightarrow \infty$. We first understand that $\check{\rho}_h(\Delta_n, p_n)$ and $\check{\rho}_h(\Delta_n, -p_n)$ are defined by (A.15) and (A.16). We start by introducing $\{\tilde{\rho}_h(\Delta_n, p_n, v)\}_{h=-\infty}^\infty$ as the Fourier coefficients of $\frac{1}{\tilde{\mathcal{V}}(e^{i\lambda}; \Delta_n, p_n) f^v(e^{i\lambda}; \gamma)}$:

$$\tilde{\rho}_h(\Delta_n, p_n, v) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{e^{-ih\lambda}}{\tilde{\mathcal{V}}(e^{i\lambda}; \Delta_n, p_n) f^v(e^{i\lambda}; \gamma)} d\lambda. \quad (\text{A.52})$$

Using the properties of $\tilde{\mathcal{V}}(z; \Delta_n, p_n)$ specified by (A.31) and (A.32), plus the properties of $f(e^{i\lambda}; \gamma)$ provided by (A.22) and (A.18) (notice that $f(e^{i\lambda}; \gamma)$ here is the same as $f(\lambda; \gamma)$ in (A.22) and (A.18), as we slightly abuse the notation), we obtain for $p_n \Delta_n^{1/2} \chi_n^{-1} \leq K$,

$$\sup_{-\pi \leq \lambda \leq \pi} \left| \frac{d^2}{d\lambda^2} \frac{1}{\tilde{\mathcal{V}}(e^{i\lambda}; \Delta_n, p_n) f^v(e^{i\lambda}; \gamma)} \right| \leq K \chi_n^{-2-2v}.$$

Let us emphasize that in view of the previous steps, obviously this result holds uniformly over $\{(\sigma_n^2, \gamma_n) \in \Pi_n^{(\sigma^2, \gamma)}(q_n) : n \geq 1\}$; so do all of the relevant results throughout the proof. We omit mentioning this afterward. According to the well-known results on how the smoothness of a function affects the order of magnitude of its Fourier coefficients (see, e.g., the proof of Theorem II.4.7 on Page 46 of Zygmund (2002)), we have, for $p_n \Delta_n^{1/2} \chi_n^{-1} \leq K$,

$$|\tilde{\rho}_h(\Delta_n, p_n, v)| \leq K \chi_n^{-2-2v} h^{-2}. \quad (\text{A.53})$$

Further, we notice that from (A.29) and (A.52) we have

$$\sum_{j=-\infty}^{\infty} \tilde{\rho}_j(\Delta_n, p_n, v) = \sum_{j=-\infty}^{\infty} \tilde{\rho}_j(\Delta_n, p_n, v) e^{ij0} = \frac{1}{\tilde{\mathcal{V}}(1; \Delta_n, p_n) f^v(1, \gamma)} = \frac{(1 - z_n^*)^2}{\sigma^2 \Delta_n \zeta_n^{2v}}. \quad (\text{A.54})$$

On the other hand, in view of the decomposition of $\mathcal{V}(e^{i\lambda}; \Delta_n, p_n)$ using $\tilde{\mathcal{V}}(z; \Delta_n, p_n)$ and the definition of $\check{\rho}_h(\Delta_n, p_n, v)$ given in (A.15) as the coefficients of the Laurent expansion of $\frac{1}{\mathcal{V}(e^{i\lambda}; \Delta_n, p_n) f^v(e^{i\lambda}; \gamma)}$, we can write

$$\check{\rho}_h(\Delta_n, p_n, v) = \frac{1}{1 - (z_n^*)^2} \sum_{j=-\infty}^{\infty} \tilde{\rho}_j(\Delta_n, p_n, v) (z_n^*)^{|j-h|}. \quad (\text{A.55})$$

Therefore, using (A.53) and (A.54), plus the expression z_n^* from (A.28) and the bound on σ^2 from (A.18), we can write

$$\begin{aligned} \left| \check{\rho}_h(\Delta_n, p_n, v) - \frac{(1 - z_n^*)^2}{\sigma^2 \Delta_n \zeta_n^2} \frac{(z_n^*)^{|h|}}{1 - (z_n^*)^2} \right| &= \frac{1}{1 - (z_n^*)^2} \left| \sum_{j=-\infty}^{\infty} \tilde{\rho}_j(\Delta_n, p_n, v) ((z_n^*)^{|h-j|} - (z_n^*)^{|h|}) \right| \\ &\leq K (z_n^*)^{|h|} \chi_n^{-2-2v} \log(\Delta_n^{-1/2} \chi_n). \end{aligned} \quad (\text{A.56})$$

Now we move to $\check{\rho}_h(\Delta_n, -p_n)$. The properties of $\mathcal{V}(z; \Delta_n, -p)$ provided by Step 6, including (A.42), (A.47), and (A.48), indicate that under $p_n \Delta_n^{-1/2} \chi_n \rightarrow 1$,

$$\int_{-\pi}^{\pi} \left| \frac{d^2}{d\lambda^2} \sum_{j=1}^{\infty} \left(-\frac{\mathcal{V}(e^{i\lambda}; \Delta_n, -p_n)}{\mathcal{V}(e^{i\lambda}; \Delta_n, p_n)} \right)^j \right| d\lambda \leq K.$$

Following the proof of Theorem II.4.7 of Zygmund (2002), plus the definition of $\check{\rho}_h(\Delta_n, -p_n)$ given by (A.16), plus (A.42), this immediately indicates that under $p_n \Delta_n^{-1/2} \chi_n \rightarrow 1$,

$$|\check{\rho}_h(\Delta_n, -p_n)| \leq K((\Delta_n \chi_n^{-2}) \wedge h^{-2}). \quad (\text{A.57})$$

Now we combine (A.56), (A.57), and (A.17) to obtain

$$\check{\rho}_h(\sigma^2, \gamma, \Delta_n, v) = \check{\rho}_h(\Delta_n, p_n, v) + O\left(\frac{1}{\chi_n^{2+2v}} \wedge \frac{1}{h^2 \Delta_n \chi_n^{2v}}\right),$$

which, combined with (A.56) again, is exactly (A.4). Now we prove (A.5). In view of (A.54), we can write

$$\begin{aligned} \check{\rho}_h(\Delta_n, p_n, v) - \check{\rho}_{h+1}(\Delta_n, p_n, v) &= \frac{1}{1 - (z_n^*)^2} \sum_{j=h+1}^{\infty} \tilde{\rho}_j(\Delta_n, p_n, v) (z_n^*)^{j-h-1} (z_n^* - 1) \\ &\quad + \frac{1}{1 - (z_n^*)^2} \sum_{j=-\infty}^h \tilde{\rho}_j(\Delta_n, p_n, v) (z_n^*)^{h-j} (1 - z_n^*). \end{aligned}$$

This indicates, using (A.53), the expression of z_n^* from (A.28), and the bound on σ^2 from (A.18), that

$$\begin{aligned}\zeta_n^{2v} \left(\check{\rho}_h(\Delta_n, p_n, v) - \check{\rho}_{h+1}(\Delta_n, p_n, v) \right) &= \frac{(1 - z_n^*)^2}{\sigma^2 \Delta_n (1 + z_n^*)} (z_n^*)^h + O\left(\Delta_n^{1/2} \chi_n^{-1} \left(\frac{1}{\chi_n^2} \wedge \frac{1}{h^2 \Delta_n} \right) \right) \\ &\quad + O\left((z_n^*)^h \chi_n^{-2} (\Delta_n^{1/2} \chi_n^{-1} \log(\Delta_n^{-1/2} \chi_n) + h^{-1}) \right).\end{aligned}$$

Provided this result, we can immediately verify (A.5) by observing that

$$\begin{aligned}\check{\rho}_h(\sigma^2, \gamma, \Delta_n, v) - \check{\rho}_{h+1}(\sigma^2, \gamma, \Delta_n, v) &= \check{\rho}_h(\Delta_n, p_n, v) - \check{\rho}_{h+1}(\Delta_n, p_n, v) \\ &\quad + \sum_{j=-\infty}^{\infty} \left(\check{\rho}_j(\Delta_n, p_n, v) - \check{\rho}_{j+1}(\Delta_n, p_n, v) \right) \check{\rho}_{h-j}(\Delta_n, -p_n)\end{aligned}$$

and using the bound on $\check{\rho}_h(\Delta_n, -p_n)$ given by (A.57). We can prove (A.6) in the same way. Indeed, (A.54) and (A.55) indicate that

$$\begin{aligned}2\check{\rho}_{h+1}(\Delta_n, p_n, v) - \check{\rho}_{h+2}(\Delta_n, p_n, v) - \check{\rho}_h(\Delta_n, p_n, v) \\ = -\frac{(1 - z_n^*)^3}{\sigma^2 \Delta_n (1 + z_n^*)} (z_n^*)^h - \frac{1 - z_n^*}{1 + z_n^*} (z_n^*)^h \sum_{j=h+2}^{\infty} \tilde{\rho}_j(\Delta_n, p_n, v) \left((z_n^*)^{j-2h-1} - 1 \right) \\ - \frac{1 - z_n^*}{1 + z_n^*} (z_n^*)^h \tilde{\rho}_{h+1}(\Delta_n, p_n, v) \left((z_n^*)^{1-h} \frac{2}{z_n^* - 1} - 1 \right) \\ - \frac{1 - z_n^*}{1 + z_n^*} (z_n^*)^h \sum_{j=-\infty}^h \tilde{\rho}_j(\Delta_n, p_n, v) \left((z_n^*)^{1-j} - 1 \right).\end{aligned}$$

Using (A.53), the expression of z_n^* from (A.28), and the bound on σ^2 from (A.18), we obtain

$$\begin{aligned}\zeta_n^{2v} \left(2\check{\rho}_{h+1}(\Delta_n, p_n, v) - \check{\rho}_{h+2}(\Delta_n, p_n, v) - \check{\rho}_h(\Delta_n, p_n, v) \right) \\ = -\frac{(1 - z_n^*)^3}{\sigma^2 \Delta_n (1 + z_n^*)} (z_n^*)^h + O\left((z_n^*)^h \Delta_n \chi_n^{-4} \log(\Delta_n^{-1/2} \chi_n) + \chi_n^{-2} h^{-2} \right).\end{aligned}$$

Combining this result with the equality

$$\begin{aligned}2\check{\rho}_{h+1}(\sigma^2, \gamma, \Delta_n, v) - \check{\rho}_h(\sigma^2, \gamma, \Delta_n, v) - \check{\rho}_{h+1}(\sigma^2, \gamma, \Delta_n, v) \\ = 2\check{\rho}_{h+1}(\Delta_n, p_n, v) - \check{\rho}_{h+2}(\Delta_n, p_n, v) - \check{\rho}_h(\Delta_n, p_n, v) \\ + \sum_{j=-\infty}^{\infty} \left(2\check{\rho}_{h+1}(\Delta_n, p_n, v) - \check{\rho}_{h+2}(\Delta_n, p_n, v) - \check{\rho}_h(\Delta_n, p_n, v) \right) \check{\rho}_{h-j}(\Delta_n, -p_n),\end{aligned}$$

which comes immediately from (A.17), and the bound on $\check{\rho}_h(\Delta_n, -p_n)$ given by (A.57), we readily obtain (A.6). We move forward to the case of $\Delta_n^{-1} \chi_n^2 \leq K$ and prove (A.7). This is relatively

straightforward. Given (A.49), (A.50), and (A.51), it follows that

$$\sup_{\lambda} \left| \frac{d^2}{d\lambda^2} \frac{1}{\mathcal{V}(e^{i\lambda}; \Delta_n)} \right| \leq K \Delta_n^{-1}.$$

Following the proof of Theorem II.4.7 of Zygmund (2002), we immediately obtain

$$\check{\rho}_h(\sigma^2, \gamma, \Delta_n, 0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{e^{-ih\lambda}}{\mathcal{V}(e^{i\lambda}; \Delta_n)} d\lambda = O\left(\frac{1}{h^2 \Delta_n}\right). \quad (\text{A.58})$$

■

Lemma A3. Suppose $m\Delta_n^{1/2+\alpha} \rightarrow \infty$ for some fixed $\alpha > 0$. Also suppose $(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q)$ with q fixed and $\frac{1}{K} \leq \inf_{\lambda} f(\lambda; \gamma) \leq \sup_{\lambda} f(\lambda; \gamma) \leq K$. Let $\zeta^2 = f(0; \gamma)$. As $n \rightarrow \infty$, it holds that

$$V_n(\sigma^2, \gamma, \Delta_n) = \frac{1}{\tilde{\zeta}^2} V_n(\tilde{\sigma}^2, \tilde{\zeta}^2, \Delta_n) O_n D_n(\tilde{\gamma}) O_n, \text{ with } |\sigma^2 - \tilde{\sigma}^2| + \|\tilde{\gamma} - \gamma\| + |\tilde{\zeta}^2 - \zeta^2| \lesssim n^{-1/2},$$

and

$$\Omega_n^{-1}(\sigma^2, \zeta^2, 0)_{ij} = b_n(z_n^{|i-j|} - z_n^{i+j} - z_n^{2n+2-i-j}) + O(n^{-\infty}),$$

where b_n and z_n do not depend on i or j and satisfy

$$b_n = \frac{1}{2\sigma\zeta\Delta_n^{1/2}} + O(1), \quad z_n = 1 - \frac{\sqrt{\sigma^2\Delta_n}}{\zeta} + O(n^{-1}).$$

Proof. The desired results follow from a simplified version of the proof of Lemma A2. Concretely, the fact that q is finite allows us to solve explicitly the $q+1$ zeros of $\mathcal{V}(z; \sigma^2, \gamma, \Delta_n)$, up to $O(n^{-1})$. Hence, we directly obtain the first result using steps 1, 3, and 4 of the proof of A2. The second result follows by also applying Lemma A1. ■

Lemma A4. Suppose $m\Delta_n^{1/2+\alpha} \rightarrow \infty$ for some fixed $\alpha > 0$. Let $\zeta^2 = f(0; \gamma)$. Omitting the argument $(\sigma^2, \gamma, \Delta_n)$ of V_m and D_m , it holds that, for all $(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q_n)$ satisfying $\sup_{\lambda} f(\lambda; \gamma) \sim \inf_{\lambda} f(\lambda; \gamma)$ and $\Delta_n^{-1}\gamma_0 \rightarrow \infty$,

(i) For $j \in \{1, 2, 3, 4\}$,

$$\text{tr}(V_m^{-j}) = \frac{\lambda_j m}{\zeta(\sigma^2 \Delta_n)^{j-1/2}} + o(m\Delta_n^{-j+1/2}), \quad \text{with } \lambda_1 = \frac{1}{2}, \quad \lambda_2 = \frac{1}{4}, \quad \lambda_3 = \frac{3}{16}, \quad \lambda_4 = \frac{5}{32}.$$

(ii) For $0 \leq i, j \leq q_n$, and $v \in \{0, 1\}$,

$$\frac{1}{m} \text{tr} \left(V_m^{-1} \frac{\partial V_m}{\partial \gamma_i} \left(V_m^{-1} \frac{\partial V_m}{\partial \gamma_j} \right)^v \right) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\partial \log f(\lambda; \gamma)}{\partial \gamma_i} \left(\frac{\partial \log f(\lambda; \gamma)}{\partial \gamma_j} \right)^v d\lambda + o(1).$$

Proof. The current lemma follows from straightforward algebra using Lemma A2. ■

Lemma A5. Suppose Assumptions 1 - 4 hold. For all sequences $\{q_n\}$ and under $\iota^{(n)} \Delta_n^{-1/2} \rightarrow \infty$, it holds that for all $0 \leq j \leq q_n$,

$$|\sigma^{(n)}(q_n)^2 - C_T| \lesssim K \|\gamma^{(n)}\|_{1,(q_n)} / (\iota^{(n)})^2, \quad |\gamma^{(n)}(q_n)_j - \gamma_j^{(n)}| \lesssim \left(\frac{\Delta_n^{3/4} q_n}{(\iota^{(n)})^{3/2}} + 1 \right) \|\gamma^{(n)}\|_{(q_n)}.$$

Proof. Step 1. (Characterization of $\sigma^{(n)}(q_n)^2 - C_T$) Throughout the proof, we omit writing the subscript n of q_n . The inequality obviously holds, by the definition of $\Pi_n^{(\sigma^2, \gamma)}(q)$, if $\|\gamma^{(n)}\|_{(q_n)} / (\iota^{(n)})^2 \geq \frac{1}{K}$. The subsequence argument indicates that we only need to consider the case in which $\|\gamma^{(n)}\|_{1,(q_n)} / (\iota^{(n)})^2 = o(1)$. The definitions of $\Pi_n^{(\sigma^2, \gamma)}(q)$ and of $(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))$ and Assumption 4 indicate that $\frac{\partial \bar{L}_n^*}{\partial \gamma_j} = 0$, $\forall 0 \leq j \leq q$ and $\frac{\partial \bar{L}_n^*}{\partial \sigma^2} = 0$. Now we solve these first-order conditions explicitly. From the definition of $\bar{L}_n^*(\sigma^2, \gamma)$, we have that for $0 \leq j \leq q$,

$$-\frac{2}{n_T} \frac{\partial \bar{L}_n^*(\sigma^2, \gamma)}{\partial \sigma^2} = \frac{\Delta_n}{2\pi} \int_{-\pi}^{\pi} \frac{f(\lambda; C_T, \gamma^{(n)}, \Delta_n) - f(\lambda; \sigma^2, \gamma, \Delta_n)}{f^2(\lambda; \sigma^2, \gamma, \Delta_n)} d\lambda, \quad (\text{A.59})$$

$$-\frac{2}{n_T} \frac{1}{2 - \delta_{0,j}} \frac{\partial \bar{L}_n^*(\sigma^2, \gamma)}{\partial \gamma_j} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{f(\lambda; C_T, \gamma^{(n)}, \Delta_n) - f(\lambda; \sigma^2, \gamma, \Delta_n)}{f^2(\lambda; \sigma^2, \gamma, \Delta_n)} (2 - 2 \cos \lambda) \cos j \lambda d\lambda. \quad (\text{A.60})$$

Substituting the definition of f back into (A.59) and (A.60), we can write that for $0 \leq j \leq q$,

$$-\frac{2}{n_T} \frac{\partial \bar{L}_n^*(\sigma^2, \gamma)}{\partial \sigma^2} = c(\sigma^2, \gamma)_{1,1} (C_T - \sigma^{(n)}(q)^2) + \sum_{k=0}^{\infty} c(\sigma^2, \gamma)_{1,2+k} (\gamma_k^{(n)} - \gamma_k), \quad (\text{A.61})$$

$$-\frac{2}{n_T} \frac{\partial \bar{L}_n^*(\sigma^2, \gamma)}{\partial \gamma_j} = c(\sigma^2, \gamma)_{j+2,1} (C_T - \sigma^{(n)}(q)^2) + \sum_{k=0}^{\infty} c(\sigma^2, \gamma)_{j+2,2+k} (\gamma_k^{(n)} - \gamma_k). \quad (\text{A.62})$$

Here we use the shorthand notation

$$\begin{aligned} c(\sigma^2, \gamma)_{1,1} &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\Delta_n^2}{f^2(\lambda; \sigma^2, \gamma, \Delta_n)} d\lambda, \\ c(\sigma^2, \gamma)_{j+2,1} = c(\sigma^2, \gamma)_{1,2+j} &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\Delta_n (2 - 2 \cos \lambda) (2 - \delta_{j,0}) \cos j \lambda}{f^2(\lambda; \sigma^2, \gamma, \Delta_n)} d\lambda, \\ c(\sigma^2, \gamma)_{j+2,k+2} &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{(2 - 2 \cos \lambda)^2 (2 - \delta_{j,0}) \cos j \lambda (2 - \delta_{k,0}) \cos k \lambda}{f^2(\lambda; \sigma^2, \gamma, \Delta_n)} d\lambda. \end{aligned}$$

Recall that equations (A.61) and (A.62), evaluated at $(\sigma^2, \gamma) = (\sigma^{(n)}(q)^2, \gamma^{(n)}(q))$ are the first-order conditions we are solving. In view of the form of (A.61) and (A.62), plus the fact that $\gamma^{(n)}(q)_j = 0$ for $j \geq q+1$, we introduce a $(q+2)$ -dimensional vector A defined as the first $q+2$ components of $(C_T - \sigma^{(n)}(q)^2, \gamma^{(n)} - \gamma^{(n)}(q))$. We can write the first-order conditions in a compact form:

$$\sum_{k=1}^{q+2} c_{j,k} A_k + \sum_{k=q+3}^{\infty} c_{j,k} \gamma_{k-2}^{(n)} = 0$$

for all $1 \leq j \leq q+2$. Here we omit the argument $(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))$ of c for cleaner exposition. This system of $q+2$ equations can be regarded as a matrix equation satisfied by vector A . Indeed, we can write $CA + B = 0$, where C is a $(q+2) \times (q+2)$ matrix and B is a $(q+2)$ -dimensional vector and their entries are given by $C_{j,k} = c(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))_{j,k}$, and $B_j = \sum_{k=q+3}^{\infty} c(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))_{j,k} \gamma_{k-2}^{(n)}$. We can invert the equation $CA + B = 0$ and obtain an explicit characterization of $(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))$:

$$\sigma^{(n)}(q)^2 - C_T = \sum_{k=1}^{q+2} (C^{-1})_{1,k} B_k, \quad \gamma^{(n)}(q)_j - \gamma_j^{(n)} = \sum_{k=1}^{q+2} (C^{-1})_{j+2,k} B_k. \quad (\text{A.63})$$

We note that here we have only solved $(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))$ partially, because vector B and matrix C depend on $(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))$. But the goal here is to show that $\sigma^{(n)}(q)^2 - C_T$ is small, and such partial solution turns out to be sufficient.

Step 2. (Behavior of χ^2) It is clear from (A.63) that we understand the properties of c and C^{-1} to provide a bound on $\sigma^{(n)}(q_n)^2 - C_T$. Before that, we first understand the behavior of

$$\chi^{(n)}(q)^2 := \chi^2(\sigma^{(n)}(q)^2, \gamma^{(n)}(q), \Delta_n),$$

which is technically necessary. We introduce shorthand $\gamma_{(q)}^{(n)}$ to be the $(q+1)$ -dimensional vector consisting of the first $q+1$ components of $\gamma^{(n)}$. Because of the properties of $\theta^{(n)}$ specified by Assumption 4 and the definitions of $\gamma^{(n)}$ and $\Pi_n^{(\sigma^2, \gamma)}(q)$, we have $(C_T, \gamma_{(q)}^{(n)}) \in \Pi_n^{(\sigma^2, \gamma)}(q)$. Therefore the first part of (A.22) indicates that uniformly over $-\pi \leq \lambda \leq \pi$,

$$C_T \Delta_n + f(\lambda; \gamma_{(q)}^{(n)}) \sim \chi^2(C_T, \gamma_{(q)}^{(n)}, \Delta_n). \quad (\text{A.64})$$

In view of the definition of $\chi^2(\sigma^2, \gamma, \Delta_n)$, we conclude that under $\Delta_n^{-1/2} \iota^{(n)} \rightarrow \infty$,

$$\chi^2(C_T, \gamma_{(q)}^{(n)}, \Delta_n) \sim (\iota^{(n)})^2. \quad (\text{A.65})$$

Because we are considering $\|\gamma^{(n)}\|_{1,(q)} (\iota^{(n)})^{-2} = o(1)$, as we discussed in Step 1, and the fact that (A.64) and (A.65) jointly indicate $f(\lambda; \gamma^{(n)}) \geq K^{-1} (\iota^{(n)})^2$ under $\Delta_n^{-1/2} \iota^{(n)} \rightarrow \infty$ and uniformly over $-\pi \leq \lambda \leq \pi$, $f(\lambda; \gamma_{(q)}^{(n)}) = f(\lambda; \gamma^{(n)}) - \sum_{j=q+1}^{\infty} 2\gamma_j^{(n)} \cos \lambda \geq K^{-1} (\iota^{(n)})^2$, which further gives

$$\frac{f(\lambda; C_T, \gamma_{(q)}^{(n)}, \Delta_n)}{f(\lambda; C_T, \gamma_{(q)}^{(n)}, \Delta_n)} = 1 + \frac{(2 - 2 \cos \lambda) \sum_{j=q+1}^{\infty} 2\gamma_j^{(n)} \cos \lambda}{\sigma^2 \Delta_n + (2 - 2 \cos \lambda) f(\lambda; \gamma_{(q)}^{(n)})} = 1 + o(1).$$

Given this result, and in view of the definition of $\bar{L}_n^*(\sigma^2, \gamma)$, we can write $2n_T^{-1} \bar{L}_n^*(C_T, \gamma_{(q)}^{(n)}) = -\log \chi^2(C_T, \gamma_{(q)}^{(n)}, \Delta_n) - 1 + o(1)$. Since $(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))$ is constructed as the maximizer of $\bar{L}_n^*$ over

$\Pi_n^{(\sigma^2, \gamma)}(q)$, we have

$$2n_T^{-1} \bar{L}_n^*(\sigma^{(n)}(q)^2, \gamma^{(n)}(q)) \geq -\log \chi^2(C_T, \gamma_{(q)}^{(n)}, \Delta_n) - 1 + o(1). \quad (\text{A.66})$$

On the other hand, starting from the definition of $\bar{L}_n^*$, we can write

$$2n_T^{-1} \bar{L}_n^*(\sigma^{(n)}(q)^2, \gamma^{(n)}(q)) \leq -\log \chi^2(\sigma^{(n)}(q)^2, \gamma^{(n)}(q), \Delta_n) - K^{-1} \frac{\chi^2(C_T, \gamma_{(q)}^{(n)}, \Delta_n)}{\chi^2(\sigma^{(n)}(q)^2, \gamma^{(n)}(q), \Delta_n)}, \quad (\text{A.67})$$

where the last step comes from the first inequality in (A.64) and the fact that it holds uniformly over $-\pi \leq \lambda \leq \pi$ that $0 < f(\lambda; \sigma^{(n)}(q)^2, \gamma^{(n)}(q), \Delta_n) \leq K \chi^2(\sigma^{(n)}(q)^2, \gamma^{(n)}(q), \Delta_n)$, which is indicated by the first part of (A.22), since we obviously have $(\sigma^{(n)}(q)^2, \gamma^{(n)}(q)) \in \Pi_n^{(\sigma^2, \gamma)}(q)$. Combining (A.66) and (A.67) indicates that $\log \frac{\chi^2(\sigma^{(n)}(q)^2, \gamma^{(n)}(q), \Delta_n)}{\chi^2(C_T, \gamma_{(q)}^{(n)}, \Delta_n)} + K^{-1} \frac{\chi^2(C_T, \gamma_{(q)}^{(n)}, \Delta_n)}{\chi^2(\sigma^{(n)}(q)^2, \gamma^{(n)}(q), \Delta_n)} \leq 1 + o(1)$. Combined with (A.66) and (A.67), this indicates that we must have

$$\chi^2(\sigma^{(n)}(q)^2, \gamma^{(n)}(q), \Delta_n) \sim \chi^2(C_T, \gamma_{(q)}^{(n)}, \Delta_n). \quad (\text{A.68})$$

Step 3. (Properties of c) In this step we analyze the properties of c . We first express c in terms of Fourier coefficients of various functions. Fourier analysis states that $\frac{1}{f(\lambda; \sigma^2, \gamma, \Delta_n)} = \sum_{j=-\infty}^{\infty} \rho_j e^{ij\lambda}$, with $\rho_j := \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{e^{-ij\lambda} d\lambda}{f(\lambda; \sigma^2, \gamma, \Delta_n)}$. This allows us to write $c(\sigma^2, \gamma)_{1,1} = \Delta_n^2 \sum_{j=-\infty}^{\infty} \rho_j^2$. Now we move to $c(\sigma^2, \gamma)_{1,2+j}$ and $c(\sigma^2, \gamma)_{j+2,k+2}$. We first show some auxiliary results. We introduce for $p \in \{1, 2\}$, $\rho_j^{(\gamma, p)} := \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{e^{-ij\lambda} d\lambda}{f^p(\lambda; \gamma)}$. We can calculate with some algebra that for $p \in \{1, 2\}$,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{e^{ii\lambda}}{f^p(\lambda; \gamma)} d\lambda = \rho_i^{(\gamma, p)}, \quad (\text{A.69})$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1}{f(\lambda; \sigma^2, \gamma, \Delta_n)} \frac{e^{ii\lambda}}{f^p(\lambda; \gamma)} d\lambda = \sum_{j=-\infty}^{\infty} \rho_j^{(\gamma, p)} \rho_{i+j}, \quad (\text{A.70})$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1}{f^2(\lambda; \sigma^2, \gamma, \Delta_n)} \frac{e^{ii\lambda}}{f^p(\lambda; \gamma)} d\lambda = \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \rho_j^{(\gamma, p)} \rho_k \rho_{i+j+k}. \quad (\text{A.71})$$

Next, from the definition of $f(\lambda; \sigma^2, \gamma, \Delta_n)$ we notice that

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{(2 - 2 \cos \lambda) \cos i\lambda}{f^2(\lambda; \sigma^2, \gamma, \Delta_n)} d\lambda = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{f(\lambda; \sigma^2, \gamma, \Delta_n) - \sigma^2 \Delta_n}{f^2(\lambda; \sigma^2, \gamma, \Delta_n)} \frac{e^{ii\lambda}}{f(\lambda; \gamma)} d\lambda,$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{(2 - 2 \cos \lambda)^2 \cos j\lambda \cos k\lambda}{f^2(\lambda; \sigma^2, \gamma, \Delta_n)} d\lambda = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{(f(\lambda; \sigma^2, \gamma, \Delta_n) - \sigma^2 \Delta_n)^2}{f^2(\lambda; \sigma^2, \gamma, \Delta_n)} \frac{e^{i(i+j)\lambda} + e^{i(i-j)\lambda}}{2f^2(\lambda; \gamma)} d\lambda.$$

With these two equalities, plus (A.69) - (A.71), we are able to write

$$\begin{aligned}
c(\sigma^2, \gamma)_{1,2+j} &= \Delta_n(2 - \delta_{j,0}) \left(\sum_{k=-\infty}^{\infty} \rho_j^{(\gamma,1)} \rho_{j+k} - \sigma^2 \Delta_n \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} \rho_k^{(\gamma,1)} \rho_l \rho_{j+k+l} \right), \\
c(\sigma^2, \gamma)_{i+2,j+2} &= \frac{1}{2}(2 - \delta_{i,0})(2 - \delta_{j,0}) \left((\rho_{i+j}^{(\gamma,2)} + \rho_{i-j}^{(\gamma,2)}) - 2\sigma^2 \Delta_n \sum_{l=-\infty}^{\infty} \rho_l^{(\gamma,2)} (\rho_{i+j+l} + \rho_{i-j+l}) \right. \\
&\quad \left. + (\sigma^2 \Delta_n)^2 \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} \rho_k^{(\gamma,2)} \rho_l (\rho_{i+j+k+l} + \rho_{i-j+k+l}) \right).
\end{aligned}$$

At this stage, it is quite clear that in order to bound c , we only need to control the behavior of ρ and $\rho^{(\gamma,p)}$. To emphasize the dependence, we write $\rho_h(\sigma^2, \gamma, \Delta_n)$ and $\rho_h^{(\gamma,p)}(\gamma)$. Scrutiny of the proof of Lemma A2 reveals that $\rho_h(\sigma^2, \gamma, \Delta_n)$ is exactly $\check{\rho}_h(\sigma^2, \gamma, \Delta_n)$ defined in (A.9). Therefore, we have under $\{(\sigma_n^2, \gamma_n) \in \Pi_n^{(\sigma^2, \gamma)}(q) : n \geq 1\}$ and $\Delta_n^{-1} \chi^2(\sigma_n^2, \gamma_n, \Delta_n) \rightarrow \infty$, $\rho_h(\sigma_n^2, \gamma_n, \Delta_n)$ satisfies (A.4). It is worth pointing out that ρ_h here and ρ_h in Lemma A2 are not exactly the same, but are very close and both satisfy (A.4). See the ending part of step 1 of the proof of Lemma A2 for details. On the other hand, from the construction of $\Pi_n^{(\sigma^2, \gamma)}(q)$, we know under $\{(\sigma_n^2, \gamma_n) \in \Pi_n^{(\sigma^2, \gamma)}(q) : n \geq 1\}$ and $\Delta_n^{-1} \chi^2(\sigma_n^2, \gamma_n, \Delta_n) \rightarrow \infty$ that uniformly over $-\pi \leq \lambda \leq \pi$,

$$K^{-1} \chi_n^2 \leq f(\lambda; \gamma_n) \leq K \chi_n^2 \quad \text{and} \quad \sum_{h=0}^{\infty} h^2 |(\gamma_n)_h| \leq K \chi_n^2, \quad \text{with} \quad \chi_n^2 := \chi^2(\sigma_n^2, \gamma_n, \Delta_n),$$

where both results come from (A.18) and (A.22). An immediate result is that under the same condition and for $p \in \{1, 2\}$, we can control the order of magnitude of the Fourier coefficients $f^{-p}(\lambda; \gamma_n)$ (see, e.g., the proof of Theorem II.4.7 of Zygmund (2002)), i.e., $\rho_h^{(\gamma,p)}(\gamma_n)$ as

$$\sum_{h=0}^{\infty} (h+1)^2 \chi_n^{2p} |\rho_h^{(\gamma,p)}(\gamma_n)| \leq K. \quad (\text{A.72})$$

Given the bounds on $\rho_h(\sigma_n^2, \gamma_n, \Delta_n)$ and $\rho_h^{(\gamma,p)}(\gamma_n)$, plus $\frac{1}{K} \leq \sigma_n^2 \leq K$ as indicated by (A.18), some algebra leads to the following technical results, again under the condition $\{(\sigma_n^2, \gamma_n) \in \Pi_n^{(\sigma^2, \gamma)}(q) : n \geq 1\}$ and $\Delta_n^{-1} \chi^2(\sigma_n^2, \gamma_n, \Delta_n) \rightarrow \infty$ and for $p \in \{1, 2\}$,

$$\begin{aligned}
\sum_{l=-\infty}^{\infty} |\rho_l \rho_{i+l}| &\lesssim \frac{e^{-|i| \Delta_n^{1/2} \chi_n^{-1}}}{\Delta_n^{3/2} \chi_n} + \frac{1}{\Delta_n} \left(\frac{1}{\chi_n^2} \wedge \frac{1}{i^2 \Delta_n} \right), \\
\chi_n^{2p} \sum_{j=-\infty}^{\infty} |\rho_j^{(\gamma,p)} \rho_{i+j}| &\lesssim \frac{1}{\Delta_n^{1/2} \chi_n} \left(e^{-|i| \Delta_n^{1/2} \chi_n^{-1}} + \frac{\Delta_n^{-1/2} \chi_n}{i^2} \wedge 1 \right), \\
\chi_n^{2p} \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} |\rho_k^{(\gamma,p)} \rho_l \rho_{i+k+l}| &\lesssim \frac{1}{\Delta_n^{3/2} \chi_n} \left(e^{-|i| \Delta_n^{1/2} \chi_n^{-1}} + \frac{\Delta_n^{-1/2} \chi_n}{i^2} \wedge 1 \right),
\end{aligned}$$

where $\chi_n^2 := \chi^2(\sigma_n^2, \gamma_n, \Delta_n)$. Therefore, using the previous expressions of c in terms of ρ and $\rho^{(\gamma, p)}$, plus that $\Delta_n^{-1} \chi^2(\sigma^{(n)}(q)^2, \gamma^{(n)}(q), \Delta_n) \rightarrow \infty$ as required by (A.68) and $\Delta_n^{-1/2} \iota^{(n)} \rightarrow \infty$, we are able to write

$$\begin{aligned} |c(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))_{1,1}| &\lesssim \Delta_n^{1/2} (\iota^{(n)})^{-1}, \\ (\iota^{(n)})^2 |c(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))_{1,2+j}| &\lesssim \frac{\Delta_n^{1/2}}{\iota^{(n)}} e^{-|j|\Delta_n^{1/2} (\iota^{(n)})^{-1}} + \frac{1}{j^2} \wedge \frac{\Delta_n^{1/2}}{\iota^{(n)}}, \end{aligned} \quad (\text{A.73})$$

$$\begin{aligned} (\iota^{(n)})^4 |c(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))_{i+2,j+2}| &\lesssim |\rho_{i+j}^{(\gamma,2)} + \rho_{i-j}^{(\gamma,2)}| \\ &\quad + \frac{\Delta_n^{1/2}}{\iota^{(n)}} \left(e^{-|i+j|\Delta_n^{1/2} (\iota^{(n)})^{-1}} + e^{-|i-j|\Delta_n^{1/2} (\iota^{(n)})^{-1}} \right). \end{aligned} \quad (\text{A.74})$$

Step 4. (Properties of C^{-1} : Special case) Now we invert matrix C . We define a $(q+2) \times (q+2)$ matrix $C(\sigma^2, \gamma)$ whose entries are

$$C(\sigma^2, \gamma)_{i,j} = c(\sigma^2, \gamma)_{i,j}, \quad 1 \leq i, j \leq q+2. \quad (\text{A.75})$$

Obviously, the matrix C appearing in (A.63) is just $C(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))$. We note that by definition $(\sigma^{(n)}(q)^2, \gamma^{(n)}(q)) \in \Pi_n^{(\sigma^2, \gamma)}(q)$. Also, we have $\Delta_n^{-1/2} \chi(\sigma^{(n)}(q)^2, \gamma^{(n)}(q), \Delta_n) \rightarrow \infty$, as indicated by (A.68) and $\Delta_n^{-1/2} \iota^{(n)} \rightarrow \infty$. It is hence more than enough to calculate $C^{-1}(\sigma_n^2, \gamma_n)$ for all $\{(\sigma_n^2, \gamma_n) \in \Pi_n^{(\sigma^2, \gamma)}(q) : n \geq 1\}$ satisfying $\Delta_n^{-1/2} \chi_n \rightarrow \infty$, where we recall that $\chi_n^2 := \chi^2(\sigma_n^2, \gamma_n, \Delta_n)$. The current step considers this problem under the following restriction:

$$(\gamma_n)_j = 0 \text{ for all } n \geq 1 \text{ and } j \geq \lceil \Delta_n^{-1/2} \chi_n \rceil + 1. \quad (\text{A.76})$$

The next step will show that restriction (A.76) is innocuous. To be able to invert C , an ∞ -dimensional matrix, we introduce a reparameterization scheme. Concretely, with a scalar z and a $(q+1)$ -dimensional vector ϕ , we define $f(\lambda; \phi) = \sum_{j=0}^q (2 - \delta_{j,0}) \phi_j \cos j\lambda$ and $f(\lambda; z, \phi) = |1 - ze^{i\lambda}|^2 f(\lambda; \phi)$. We connect the parameterization (z, ϕ) to (σ_n^2, γ_n) , which belongs to $\Pi_n^{(\sigma^2, \gamma)}(q)$, by requiring

$$z = z(\sigma_n^2, \gamma_n) := z_n^* \quad \text{and} \quad \phi_j = \phi_j(\sigma_n^2, \gamma_n) := \tilde{\rho}_h(\Delta_n, \lceil \Delta_n^{-1/2} \chi_n \rceil), \quad (\text{A.77})$$

where z_n^* is defined in (A.23), with the argument p_n in $\mathcal{V}(z_n^*; \Delta_n, p_n)$ set as $\lceil \Delta_n^{-1/2} \chi_n \rceil$, and $\tilde{\rho}_h$ is defined by (A.52), both appearing in the proof of Lemma A2. Because p_n satisfies $p_n \leq K \Delta_n^{-1/2} \chi_n$, plus $(\sigma_n^2, \gamma_n) \in \Pi_n^{(\sigma^2, \gamma)}(q)$ and $\Delta_n^{-1/2} \chi_n \rightarrow \infty$, the results obtained in step 4 and step 5 of the proof of Lemma A2 hold. Of these, we use the expression z_n^* provided by (A.28) and the properties of $\tilde{\mathcal{V}}$ given by (A.31) and (A.32). Because $(\gamma_n)_j = 0$ for all $j \geq p_n$ as required by (A.76), we have, as made obvious by step 2 of the proof of Lemma A2 and the definition of (z, ϕ) given in (A.77), that

$$f(\lambda; z, \phi) = f(\lambda; \sigma_n^2, \gamma_n, \Delta_n). \quad (\text{A.78})$$

We define $C(z, \phi)$ as a $(q+2) \times (q+2)$ matrix given by

$$C(z, \phi) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{\partial \log f(\lambda; z, \phi)}{\partial(z, \phi)} \right)^{\top} \frac{\partial \log f(\lambda; z, \phi)}{\partial(z, \phi)} d\lambda. \quad (\text{A.79})$$

We further partition matrix $C(z, \phi)$ into four submatrices:

$$C(z, \phi) = \begin{pmatrix} C_{zz} & C_{z\phi} \\ C_{\phi z} & C_{\phi\phi} \end{pmatrix}. \quad (\text{A.80})$$

Here we require C_{zz} to be a scalar. Such partition leads to uniquely defined submatrices. We can write, for all $0 \leq i, j \leq q$,

$$\begin{aligned} c(z, \phi)_{1,1} &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{\partial \log |1 - ze^{i\lambda}|^2}{\partial z} \right)^2 d\lambda = \frac{2}{1 - z^2}, \\ c(z, \phi)_{1,j+2} &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\partial \log |1 - ze^{i\lambda}|^2}{\partial z} \frac{\partial \log f(\lambda; \phi)}{\partial \phi_j} d\lambda, \\ c(z, \phi)_{i+2,j+2} &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\partial \log f(\lambda; \phi)}{\partial \phi_i} \frac{\partial \log f(\lambda; \phi)}{\partial \phi_j} d\lambda = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{(2 - \delta_{i,0}) \cos i\lambda}{f(\lambda; \phi)} \frac{(2 - \delta_{j,0}) \cos j\lambda}{f(\lambda; \phi)} d\lambda. \end{aligned}$$

Because of $f(\lambda; \phi) = \tilde{\mathcal{V}}(e^{i\lambda}; \Delta_n, q)$ as indicated by (A.77) and the properties of $\tilde{\mathcal{V}}$ given by (A.31) and (A.32), we have

$$K^{-1} \chi_n^2 \leq f(\lambda; \phi) \leq K \chi_n^2 \quad \text{and} \quad \left| \frac{d^2 f(\lambda; \phi)}{d\lambda^2} \right| \leq K \chi_n^2. \quad (\text{A.81})$$

Therefore, in view of the expression of z_n^* provided by (A.28), all the entries of C can be rewritten, for all $0 \leq i, j \leq q$, as

$$C_{zz} = \frac{2}{1 - z^2}, \quad (C_{z\phi})_{j+1} = -\frac{2z^j}{f(0; \phi)} + o(\chi_n^{-2}), \quad (\text{A.82})$$

$$(C_{\phi\phi})_{i+1,j+1} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{(2 - \delta_{i,0}) \cos i\lambda}{f(\lambda; \phi)} \frac{(2 - \delta_{j,0}) \cos j\lambda}{f(\lambda; \phi)} d\lambda. \quad (\text{A.83})$$

We introduce an auxiliary $(q+1) \times (q+1)$ matrix $\tilde{C}_{\phi\phi}$:

$$(\tilde{C}_{\phi\phi})_{i+1,j+1} := \frac{1}{2\pi} \int_{-\pi}^{\pi} f^2(\lambda; \phi) \cos i\lambda \cos j\lambda d\lambda. \quad (\text{A.84})$$

In view of the expressions of $C_{\phi\phi}$ provided by (A.83), we have

$$(C_{\phi\phi} \tilde{C}_{\phi\phi})_{i+1,j+1} = \frac{(2 - \delta_{i,0})}{(2\pi)^2} \sum_{k=-q}^q \int_{-\pi}^{\pi} f^2(\lambda; \phi) \cos i\lambda e^{ik\lambda} d\lambda \int_{-\pi}^{\pi} \frac{\cos j\lambda'}{f^2(\lambda'; \phi)} e^{-ik\lambda'} d\lambda'.$$

It is straightforward to calculate, using the orthogonality of complex exponentials, that

$$\frac{(2 - \delta_{i,0})}{(2\pi)^2} \sum_{k=-\infty}^{\infty} \int_{-\pi}^{\pi} f^2(\lambda; \phi) \cos i\lambda e^{ik\lambda} d\lambda \int_{-\pi}^{\pi} \frac{\cos j\lambda'}{f^2(\lambda'; \phi)} e^{-ik\lambda'} d\lambda' = \delta_{i,j}.$$

On the other hand, according to (A.81) and the proof of Theorem II.4.7 of Zygmund (2002), we have for (z, ϕ) satisfying (A.77), under $\{(\sigma_n^2, \gamma_n) \in \Pi_n^{(\sigma^2, \gamma)}(q) : n \geq 1\}$ and $\Delta_n^{-1/2} \chi_n \rightarrow \infty$, and for $k \geq q+1$ and $0 \leq i \leq q$,

$$\left| \chi_n^{-4} \int_{-\pi}^{\pi} f^2(\lambda; \phi) \cos i\lambda e^{ik\lambda} d\lambda \right| + \left| \chi_n^4 \int_{-\pi}^{\pi} f^{-2}(\lambda; \phi) \cos i\lambda e^{ik\lambda} d\lambda \right| \leq \frac{K}{(k-i)^2}.$$

We hence are able to write $C_{\phi\phi} \tilde{C}_{\phi\phi} = \mathbb{I}_{q+1} + A$, with $|A_{i,j}| \leq K \frac{q+1-i \vee j}{(q+1-i)^2 (q+1-j)^2}$, which immediately gives

$$C_{\phi\phi}^{-1} = \tilde{C}_{\phi\phi} - \tilde{C}_{\phi\phi} (\mathbb{I}_{q+1} + A)^{-1} A. \quad (\text{A.85})$$

Because of (A.81) and Proposition 4.5.3 in Brockwell and Davis (1991), we have $\chi_n^4 C_{\phi\phi} \sim \mathbb{I}_{q+1}$ and $\chi_n^{-4} \tilde{C}_{\phi\phi} \sim \mathbb{I}_{q+1}$, where $\sim$ is defined based on Loewner partial order. We hence have $(\mathbb{I}_{q+1} + A)(\mathbb{I}_{q+1} + A)^{\top} \sim \mathbb{I}_{q+1}$ and therefore $((\mathbb{I}_{q+1} + A)(\mathbb{I}_{q+1} + A)^{\top})^{-1} \sim \mathbb{I}_{q+1}$. This allows us to conclude that

$$\|\chi_n^{-4} \tilde{C}_{\phi\phi} (\mathbb{I}_{q+1} + A)^{-1}\| \leq K,$$

where $\|\cdot\|$ stands for the matrix operator norm. We hence have, by Cauchy-Schwarz inequality,

$$\sum_{i=0}^q |\chi_n^{-4} (\tilde{C}_{\phi\phi} (\mathbb{I}_{q+1} + A)^{-1} A)_{i+1, j+1}| \leq K \left(q \sum_{i=0}^q A_{i+1, j+1}^2 \right)^{1/2} \leq \frac{K\sqrt{q}}{(q+1-j)^2}. \quad (\text{A.86})$$

Because of (A.80), block matrix inversion states that for $0 \leq i, j \leq q$,

$$\begin{aligned} C^{-1}(z, \phi)_{1,1} &= (C_{zz} - C_{z\phi} C_{\phi\phi}^{-1} C_{\phi z})^{-1}, \\ C^{-1}(z, \phi)_{1, j+2} &= -(C_{zz} - C_{z\phi} C_{\phi\phi}^{-1} C_{\phi z})^{-1} (C_{z\phi} C_{\phi\phi}^{-1})_{j+1}, \\ C^{-1}(z, \phi)_{i+2, j+2} &= (C_{\phi\phi}^{-1})_{i+1, j+1} + (C_{z\phi} C_{\phi\phi}^{-1})_{i+1} (C_{zz} - C_{z\phi} C_{\phi\phi}^{-1} C_{\phi z})^{-1} (C_{z\phi} C_{\phi\phi}^{-1})_{j+1}. \end{aligned}$$

We therefore have obtained all the elements needed for calculation of $C^{-1}(z, \phi)$. Indeed, we have C_{zz} and $C_{z\phi}$, provided by (A.82), and $C_{\phi\phi}^{-1}$, provided by (A.85) and (A.86). It is straightforward to calculate that for (z, ϕ) satisfying (A.77) and under $\{(\sigma_n^2, \gamma_n) \in \Pi_n^{(\sigma^2, \gamma)}(q) : n \geq 1\}$ and $\Delta_n^{-1/2} \chi_n \rightarrow \infty$,

$$\begin{aligned} C^{-1}(z, \phi)_{1,1} &= (C_{zz} - C_{z\phi} C_{\phi\phi}^{-1} C_{\phi z})^{-1} = \frac{1 - z^2}{2z^{2q}} + o(\Delta_n^{1/2} \chi_n^{-1}), \\ C^{-1}(z, \phi)_{1, j+2} &= \frac{1 - z^2}{2z^{2q}} f(0; \phi) z^j + O\left(\frac{\Delta_n^{1/2} \chi_n^{-1} \sqrt{q}}{(q+1-j)^2}\right), \end{aligned}$$

$$C^{-1}(z, \phi)_{i+2, j+2} = \tilde{C}_{\phi\phi} - \tilde{C}_{\phi\phi}(\mathbb{I}_{q+1} + A)^{-1}A + \frac{1-z^2}{2z^{2q}}f^2(0; \phi)z^{i+j} + O\left(\frac{\Delta_n^{1/2}\chi_n^{-1}\sqrt{q}}{(q+1-i \vee j)^2}\right).$$

Now we move on to calculate $C^{-1}(\sigma^2, \gamma)$. In view of the definition of c specified in step 1, we realize that

$$C(\sigma^2, \gamma) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{\partial \log f(\lambda; \sigma^2, \gamma, \Delta_n)}{\partial(\sigma^2, \gamma)} \right)^\top \frac{\partial \log f(\lambda; \sigma^2, \gamma, \Delta_n)}{\partial(\sigma^2, \gamma)} d\lambda. \quad (\text{A.87})$$

According to the inverse function theorem, plus the equality $f(\lambda; z, \phi) = f(\lambda; \sigma^2, \gamma, \Delta_n)$ from (A.78) and the definition of $C(z, \phi)$ specified in (A.79), we have that under the restriction (A.76),

$$C^{-1}(\sigma_n^2, \gamma_n) = \frac{\partial(\sigma_n^2, \gamma_n)}{\partial(z, \phi)} C^{-1}(z, \phi) \left(\frac{\partial(\sigma_n^2, \gamma_n)}{\partial(z, \phi)} \right)^\top.$$

The mapping between (σ_n^2, γ_n) and (z, ϕ) specified in (A.77) indicates that

$$\frac{\partial \sigma_n^2}{\partial z} = -2(1-z)f(0; \phi), \quad \frac{\partial \sigma_n^2}{\partial \phi_k} = (2 - \delta_{k,0})(1-z)^2,$$

$$\frac{\partial(\gamma_n)_j}{\partial \phi_k} = z\delta_{k,j} - (1-z)^2(k-j)_+, \quad \frac{\partial(\gamma_n)_j}{\partial z} = \phi_j + (1-z) \sum_{k=0}^q \phi_k(k-j)_+.$$

Here and below, we use $x_+ = \max\{x, 0\}$. Some algebra yields that under $\{(\sigma_n^2, \gamma_n) \in \Pi_n^{(\sigma^2, \gamma)}(q) : n \geq 1\}$, $\Delta_n^{-1/2}\chi_n \rightarrow \infty$, and the restriction (A.76), and for all $0 \leq j \leq q$,

$$C^{-1}(\sigma_n^2, \gamma_n)_{1,1} = 4\Delta_n^{-2}(1-z)^3f^2(0; \phi) + 2\Delta_n^{-2}(1-z)^4qf^2(0; \phi) + o\left(\Delta_n^{-1/2}\chi_n + q\right), \quad (\text{A.88})$$

$$\begin{aligned} C^{-1}(\sigma_n^2, \gamma_n)_{1,j+2} &= -\Delta_n^{-1}(1-z)^2f^2(0; \phi) \left(1 + 2(1-z)(q-j) + \frac{1}{2}(1-z)^2(q-j)^2 \right) \\ &\quad + O\left(\chi_n^2(j+1)^{-2}\right) + o\left(\chi_n^2 + \Delta_n(q-j)^2\right), \end{aligned} \quad (\text{A.89})$$

$$\begin{aligned} C^{-1}(\sigma_n^2, \gamma_n)_{i+2, j+2} &= (1+o(1)) \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} f^2(\lambda; \phi) \cos i\lambda \cos j\lambda d\lambda \right. \\ &\quad + f^2(0; \phi) \left(\frac{1}{2}(1-z)^4 \sum_{k=0}^q (k-i)_+(k-j)_+ - \frac{1}{2}z(1-z)^2|i-j| \right. \\ &\quad \left. \left. + (1-z^2)(z+(q-j+1)(1-z))(z+(q-i+1)(1-z)) \right) \right]. \end{aligned} \quad (\text{A.90})$$

Step 5. (Properties of C^{-1} : General case) This step shows that the restriction (A.76) is not needed to obtain (A.88), (A.89), and (A.90), given $\{(\sigma_n^2, \gamma_n) \in \Pi_n^{(\sigma^2, \gamma)}(q) : n \geq 1\}$ and $\Delta_n^{-1/2}\chi_n \rightarrow \infty$. Clearly, we only need to consider the case in which $q > \lceil \Delta_n^{-1/2}\chi_n \rceil$, otherwise the restriction (A.76) is automatically satisfied from the definition of $\Pi_n^{(\sigma^2, \gamma)}(q)$. Using the fact that $\frac{\partial f(\lambda; \sigma^2, \gamma, \Delta_n)}{\partial(\sigma^2, \gamma)}$ does not depend on (σ^2, γ) , we can write

$$C(\sigma^2, \gamma) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{\partial \log f(\lambda; \sigma^2, \tilde{\gamma}(p, \gamma), \Delta_n)}{\partial(\sigma^2, \gamma)} \right)^\top \frac{\partial \log f(\lambda; \sigma^2, \tilde{\gamma}(p, \gamma), \Delta_n)}{\partial(\sigma^2, \gamma)} d\lambda$$

$$\times \left[1 + \sum_{j=1}^{\infty} \left(- \frac{\mathcal{V}(e^{i\lambda}; \Delta_n, -p)}{\mathcal{V}(e^{i\lambda}; \Delta_n, p)} \right)^j \right]^2 d\lambda. \quad (\text{A.91})$$

Here we recall that $\tilde{\gamma}(p, \gamma) = (\gamma_0, \gamma_1, \dots, \gamma_p, 0, \dots, 0)^T$, $\mathcal{V}(z; \Delta_n, p)$, and $\mathcal{V}(z; \Delta_n, -p)$ are all introduced in step 2 of the proof of Lemma A2 and we use (A.14). We observe the apparent fact that without the multiplicative term $\left[1 + \sum_{j=1}^{\infty} \left(- \frac{\mathcal{V}(e^{i\lambda}; \Delta_n, -p)}{\mathcal{V}(e^{i\lambda}; \Delta_n, p)} \right)^j \right]^2$, the right-hand side of (A.91) would just be $C(\sigma^2, \tilde{\gamma}(p, \gamma))$. We can then conclude, following the proof of Proposition 4.5.3 in Brockwell and Davis (1991) and using the bound on $\left| \frac{\mathcal{V}(e^{i\lambda}; \Delta_n, -p)}{\mathcal{V}(e^{i\lambda}; \Delta_n, p)} \right|$ provided by (A.42), that

$$(1 - Kp^{-2})C(\sigma^2, \tilde{\gamma}(p, \gamma)) \leq C(\sigma^2, \gamma) \leq (1 + Kp^{-2})C(\sigma^2, \tilde{\gamma}(p, \gamma)), \quad (\text{A.92})$$

where $\leq$ is the Loewner partial order. We let $p_n = \lceil \Delta_n^{-1/2} \chi_n \rceil$. Then $\tilde{\gamma}(p_n, \gamma_n)$ satisfies (A.76). Since $\Delta_n^{-1/2} \chi_n \rightarrow \infty$, we have $p_n \rightarrow \infty$ and hence $\{(\sigma_n^2, \tilde{\gamma}(p_n, \gamma_n)) \in \Pi_n^{(\sigma^2, \gamma)}(q) : n \geq 1\}$ and $\Delta_n^{-1} \chi^2(\sigma_n^2, \tilde{\gamma}(p_n, \gamma_n), \Delta_n) \rightarrow \infty$. Therefore, the expressions (A.88), (A.89), and (A.90) in step 4 apply to $C^{-1}(\sigma_n^2, \tilde{\gamma}(p_n, \gamma_n))$. If we further apply (A.92), plus that $f(\lambda; \tilde{\gamma}(p_n, \gamma_n))/f(\lambda; \gamma_n) \rightarrow 1$ uniformly over λ , we have $C(\sigma^2, \gamma)^{-1} = C(\sigma^2, \tilde{\gamma}(p, \gamma))^{-1} \sum_{j=0}^{\infty} ([C(\sigma^2, \gamma) - C(\sigma^2, \tilde{\gamma}(p, \gamma))] C(\sigma^2, \gamma)^{-1})^j$ and direct calculation leads to the fact that (A.88), (A.89), and (A.90) indeed hold, for all parameter sequences $\{(\sigma_n^2, \gamma_n) \in \Pi_n^{(\sigma^2, \gamma)}(q) : n \geq 1\}$ satisfying $\Delta_n^{-1/2} \chi_n \rightarrow \infty$.

Step 6. (Bound on $\sigma^{(n)}(q_n)^2 - C_T$) Given the relation (A.68) and $\Delta_n^{-1/2} \iota^{(n)} \rightarrow \infty$, we immediately obtain $\Delta_n^{-1} \chi^2(\sigma^{(n)}(q)^2, \gamma^{(n)}(q), \Delta_n) \rightarrow \infty$. Also, we have $(\sigma^{(n)}(q)^2, \gamma^{(n)}(q)) \in \Pi_n^{(\sigma^2, \gamma)}(q)$ by definition. Thus, in view of step 5, the relations (A.88), (A.89), and (A.90) apply to $C^{-1}(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))$ (of course, now with $z = z(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))$ and $\phi_j = \phi_j(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))$). Using the expression of z_n^* given by (A.23) (see explanation after (A.77)), the bound on $f(\lambda; \phi)$ provided by (A.81), and the relation (A.68), we can write for $0 \leq j \leq q$,

$$C^{-1}(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))_{1,1} = 4\Delta_n^{-1/2} C_T^{3/2} f^{1/2}(0; \phi) + 2C_T^2 q + o\left(\Delta_n^{-1/2} \iota^{(n)} + q\right), \quad (\text{A.93})$$

$$\begin{aligned} C^{-1}(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))_{1,j+2} &= -C_T f(0; \phi) \left(1 + 2(1-z)(q-j) + \frac{1}{2}(1-z)^2(q-j)^2 \right) \\ &\quad + O\left((\iota^{(n)})^2(j+1)^{-2}\right) + o\left((\iota^{(n)})^2 + \Delta_n(q-j)^2\right), \end{aligned} \quad (\text{A.94})$$

$$\begin{aligned} C^{-1}(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))_{i+2,j+2} &= (1 + o(1)) \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} f^2(\lambda; \phi) \cos i\lambda \cos j\lambda d\lambda \right. \\ &\quad + f^2(0; \phi) \left(\frac{1}{2}(1-z)^4 \sum_{k=0}^q (k-i)_+(k-j)_+ - \frac{1}{2}z(1-z)^2|i-j| \right. \\ &\quad \left. \left. + (1-z^2)(z + (q-j+1)(1-z))(z + (q-i+1)(1-z)) \right) \right]. \end{aligned} \quad (\text{A.95})$$

In view of the bounds on c provided by (A.73), (A.74), and (A.72) and the relation (A.68), we have

for all $2 \leq i \leq q+2$ and $k \geq q+3$,

$$|C_{1,1}^{-1}c_{1,k}| + \sum_{j=2}^{q+2} |C_{1,j}^{-1}c_{j,k}| \lesssim (\iota^{(n)})^{-2}, \quad |C_{i,1}^{-1}| \left(\sum_{l=q+3}^{\infty} |c_{1,l}|^2 \right)^{1/2} + \sum_{j=2}^{q+2} |C_{i,j}^{-1}| \left(\sum_{l=q+3}^{\infty} |c_{j,l}|^2 \right)^{1/2} \lesssim \frac{\Delta_n^{3/4} q}{(\iota^{(n)})^{3/2}} + 1,$$

where C^{-1} and c are evaluated at $(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))$. Substituting this result back into (A.63) immediately proves the lemma. ■

Lemma A6. Suppose Assumptions 1 - 4 hold. For all sequences $\{q_n\}$ and under $\Delta_n^{-1/2} \iota^{(n)} \leq K$, it holds that for all $0 \leq j \leq q_n$,

$$|\sigma^{(n)}(q_n)^2 - C_T| \leq K \Delta_n^{-1} \|\gamma^{(n)}\|_{1,(q_n)}, \quad |\gamma^{(n)}(q_n)_j - \gamma_j^{(n)}| \lesssim (q_n + 1) \|\gamma^{(n)}\|_{(q_n)}.$$

Proof. Step 1. (Characterization of $\sigma^{(n)}(q_n)^2 - C_T$) Throughout the proof, we omit writing the subscript n of q_n . We set the bijection β_n as

$$\beta_n(\sigma^2, \gamma)_j = \frac{\Delta_n^{-1}}{2\pi} \int_{-\pi}^{\pi} f(\lambda; \sigma^2, \gamma, \Delta_n) e^{ij\lambda} d\lambda, \quad 0 \leq j \leq q+1. \quad (\text{A.96})$$

In view of (A.96), we have $\sigma^{(n)}(q)^2 = \sum_{j=-q-1}^{q+1} \beta^{(n)}(q)_{|j|}$ and $\gamma^{(n)}(q)_j = -\Delta_n \sum_{i=j+1}^{q+1} (i-j) \beta^{(n)}(q)_i$. The current lemma, therefore, would follow from

$$\|\beta^{(n)}(q) - \bar{\beta}^{(n)}\|_1 \leq K \Delta_n^{-1} \|\gamma^{(n)}\|_{(q)}. \quad (\text{A.97})$$

Trivially, (A.97) holds under $\Delta_n^{-1} \|\gamma^{(n)}\|_{1,(q_n)} \geq \frac{1}{K}$, given $\Delta_n^{-1/2} \iota^{(n)} \leq K$ and Assumption 4. The subsequence argument indicates that we only need to consider the case in which $\|\gamma^{(n)}\|_{1,(q)} = o(\Delta_n)$. In view of the definitions of $\bar{L}_n^*(\beta)$, $\beta^{(n)}(q)$, and $\Pi_n^\beta(q)$, and Assumption 4, we have for all $0 \leq j \leq q+1$,

$$\frac{\partial}{\partial \beta_j} \bar{L}_n^*(\beta^{(n)}(q)) = 0. \quad (\text{A.98})$$

On the other hand, using (A.96), we obtain that for all $0 \leq j \leq q+1$,

$$-\frac{2}{n_T} \frac{1}{2 - \delta_{0,j}} \frac{\partial \bar{L}_n^*(\beta)}{\partial \beta_j} = A(\beta)_j + B(\beta)_j, \quad (\text{A.99})$$

where $A(\beta)_j = \sum_{k=0}^{q+1} (2 - \delta_{k,0}) c(\beta)_{j,k} (\bar{\beta}_k^{(n)} - \beta_k)$ and $B(\beta)_j = \sum_{k=q+2}^{\infty} 2c(\beta)_{j,k} \bar{\beta}_k^{(n)}$. Here we use the shorthand notation $c(\beta)_{j,k} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\cos j\lambda \cos k\lambda}{f^2(\lambda; \beta)} d\lambda$, where $f(\lambda; \beta) = \sum_{j=-\infty}^{\infty} \beta_{|j|} e^{ij\lambda}$. In view of (A.99), if we let $C(\beta)$ be the $(q+2) \times (q+2)$ matrix whose entries are $C(\beta)_{i,j} = c(\beta)_{i,j}$ for all

$0 \leq i, j \leq q+1$, the first-order condition (A.98) can be rewritten as

$$\bar{\beta}_j^{(n)} - \beta^{(n)}(q)_j = \frac{2}{2 - \delta_{j,0}} \sum_{i=0}^{q+1} C^{-1}(\beta^{(n)}(q))_{i,j} \sum_{k=q+2}^{\infty} c(\beta^{(n)}(q))_{i,k} \beta_k^{(n)}. \quad (\text{A.100})$$

Step 2. (Properties of c and C^{-1}) In this step we provide bounds on c and C^{-1} , from which we would immediately prove the current lemma. We first note the connection between $\beta^{(n)}(q)$ and $(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))$ and that $(\sigma^{(n)}(q)^2, \gamma^{(n)}(q)) \in \Pi_n^{(\sigma^2, \gamma)}(q)$. Then according to (A.18) and (A.22), Assumption 4 and the construction of $\Pi_n^{(\sigma^2, \gamma)}(q)$, which we recall comes from (3.11), indicate that under $\Delta_n^{-1/2} \iota^{(n)} \leq K$ and $\|\gamma^{(n)}\|_{1,(q)} = o(\Delta_n)$, for n large enough and uniformly over λ ,

$$f(\lambda; \beta^{(n)}(q)) \sim 1 \quad \text{and} \quad \frac{d^2 f(\lambda; \beta^{(n)}(q))}{d\lambda^2} \leq K. \quad (\text{A.101})$$

An immediate result of (A.101) is that, following the proof of Theorem II.4.7 of Zygmund (2002), the definition of $c(\beta)_{j,k}$ as Fourier coefficients of $f^{-2}(\lambda; \beta)$ indicates that

$$|c(\beta^{(n)}(q))_{j,k}| \leq K(|j - k| + 1)^{-2}. \quad (\text{A.102})$$

Now we analyze C^{-1} . We introduce an auxiliary $(q+2) \times (q+2)$ matrix $\tilde{C}(\beta)$:

$$\tilde{C}(\beta)_{i,j} := \frac{1}{2\pi} \int_{-\pi}^{\pi} f^2(\lambda; \beta) (2 - \delta_{i,0})(2 - \delta_{j,0}) \cos i\lambda \cos j\lambda d\lambda, \quad \text{with } 0 \leq i, j \leq q+1.$$

If we compare (A.101) with (A.81), we can repeat the derivation of (A.85) and (A.86) and conclude that $\sum_{j=0}^{q+1} |\tilde{C}(\beta^{(n)}(q))_{i,j}| \leq K$ for all $0 \leq i \leq q+1$ and that

$$C^{-1}(\beta^{(n)}(q)) = \tilde{C}(\beta^{(n)}(q)) - \tilde{C}(\beta^{(n)}(q))(\mathbb{I}_{q+2} + A)^{-1}A, \quad (\text{A.103})$$

where the $(q+2) \times (q+2)$ matrix A satisfies $\sum_{i=0}^{q+1} |(\tilde{C}(\beta^{(n)}(q))(\mathbb{I}_{q+1} + A)^{-1}A)_{i,j}| \leq \frac{K\sqrt{q}}{(q+2-j)^2}$. We then immediately have for all $0 \leq i \leq q+1$,

$$\sum_{j=0}^{q+1} |C^{-1}(\beta^{(n)}(q))_{i,j}| \leq K. \quad (\text{A.104})$$

Applying Hölder's inequality to the expression of $\bar{\beta}_j^{(n)} - \beta^{(n)}(q)_j$ provided in (A.100) leads to

$$\sum_{j=0}^{q+1} |\bar{\beta}_j^{(n)} - \beta^{(n)}(q)_j| \leq 2\|\bar{\beta}^{(n)}\|_{(q+1)} \sum_{j=0}^{q+1} \sum_{i=0}^{q+1} |C^{-1}(\beta^{(n)}(q))_{i,j}| \left(\sum_{k=q+2}^{\infty} |c(\beta^{(n)}(q))_{i,k}|^2 \right)^{1/2}.$$

Therefore, in view of (A.102) and (A.104), we have already proved (A.97), and thereby the current lemma, by observing that $\|\bar{\beta}^{(n)}\|_{1,(q+1)} = O(\Delta_n^{-1} \|\gamma^{(n)}\|_{1,(q)})$. ■

Here we use $\log \det A = \text{tr} \log A$. The challenge we face is that we do not have an analytical expression for Σ_n^{-1} . However, we observe that

$$\Sigma_n^{-1} = \Omega_n^{-1} - \Omega_n^{-1} R_n \Omega_n^{-1} + \Omega_n^{-1} R_n \Sigma_n^{-1} R_n \Omega_n^{-1}, \quad \text{with} \quad R_n(\beta) := \Sigma_n(\beta) - \Omega_n(\beta). \quad (\text{A.112})$$

Although in the last term on the right-hand side, Σ_n^{-1} still appears; later we show that we can replace it with Ω_n^{-1} for the purpose of bounding $\mathcal{R}_a$ and $\mathcal{R}_b$. Now we apply (A.112) to (A.111). Introduce simplifying notation

$$\begin{aligned} \mathcal{R}_{a1}(\beta) &:= \text{tr}(\Omega_n^{-1} R_n \Omega_n^{-1}), \quad \mathcal{R}_{a2}(\beta) := \text{tr}(\Omega_n^{-1} R_n \Sigma_n^{-1} R_n \Omega_n^{-1}), \quad \mathcal{R}_{b1}(\beta) := Y_n^\top \Omega_n^{-1} R_n \Omega_n^{-2} Y_n, \\ \mathcal{R}_{b2}(\beta) &:= Y_n^\top \Omega_n^{-1} R_n \Sigma_n^{-1} R_n \Omega_n^{-2} Y_n, \quad \text{and} \quad \mathcal{R}_{b3}(\beta) := Y_n^\top \Omega_n^{-1} R_n \Sigma_n^{-2} R_n \Omega_n^{-1} Y_n. \end{aligned} \quad (\text{A.113})$$

Here we drop the argument β of Ω_n and Σ_n . Then we can rewrite (A.111) as

$$\Delta_n^{-1} \mathcal{R}_a = \mathbb{1}_{\Omega'_n} (-\mathcal{R}_{a1}(\beta^{(n)}) + \mathcal{R}_{a2}(\beta^{(n)})), \quad \Delta_n^{-1} \mathcal{R}_b = \mathbb{1}_{\Omega'_n} (2\mathcal{R}_{b1}(\beta^{(n)}) - 2\mathcal{R}_{b2}(\beta^{(n)}) - \mathcal{R}_{b3}(\beta^{(n)})).$$

In view of the triangle inequality, the desired result (A.110) follows from the fact that for all $A \in \{\mathcal{R}_{a1}, \mathcal{R}_{a2}, \mathcal{R}_{b1}, \mathcal{R}_{b2}, \mathcal{R}_{b3}\}$, $\mathbb{E}|A(\beta^{(n)})| = o(n^{5/4})$.

Step 2. (Bounds of Σ) In this step we prove

$$\Sigma_n((\sigma^{(n)})^2, \gamma_0^{(n)}) \leq K \Sigma_n((\sigma^{(n)})^2, \gamma^{(n)}). \quad (\text{A.114})$$

Namely, we bound $\Sigma_n((\sigma^{(n)})^2, \gamma^{(n)})$ from below by $\Sigma_n((\sigma^{(n)})^2, \gamma_0^{(n)})$. For all $x = (x_1, x_2, \dots, x_{n_T})^\top \in \mathbb{R}^{n_T}$, define $\tilde{x} = (\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_{n_T+1})^\top \in \mathbb{R}^{n_T+1}$ by $\tilde{x}_j = x_{j-1} - x_j$ with $x_0 = x_{n_T+1} = 0$. We deduce (A.114) from

$$x^\top \Sigma_n((\sigma^{(n)})^2, \gamma^{(n)}) x = (\sigma^{(n)})^2 \Delta_n \|x\|^2 + \tilde{x}^\top \Gamma(\gamma^{(n)})_{n+1} \tilde{x} \quad \text{and} \quad \Gamma(\gamma^{(n)})_{n+1} \sim \gamma_0^{(n)} \mathbb{I}_{n+1}. \quad (\text{A.115})$$

Here we define $\Gamma(\gamma^{(n)})_{n+1} \in \mathcal{M}_{n_T+1}$ by $(\Gamma(\gamma^{(n)})_{n+1})_{ij} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\lambda; \gamma^{(n)}) e^{i(i-j)\lambda} d\lambda$. $\mathbb{I}_{n+1}$ is the $(n_T+1) \times (n_T+1)$ identity matrix. The first claim in (A.115) holds by definition of Σ_n . The second claim in (A.115) comes from Proposition 4.5.3 in Brockwell and Davis (1991) by applying (A.18) and (A.22).

Step 3. (Useful estimates) We provide some estimates in this step. We start by introducing some notation:

$$A(1)_j := (\Omega_n^{-1} Y_n)_j, \quad A(2)_{ij} := (\Omega_n^{-1})_{i,j} - (\Omega_n^{-1})_{i,j+1}, \quad \text{and} \quad A(3)_{ij} := A(2)_{ij} - A(2)_{i+1,j}.$$

We have the following four estimates:

$$\begin{cases} \mathbb{E}(\mathbb{1}_{\Omega'_n} A(1)_{h_n}^2) \leq K h_n, & \mathbb{E}(\mathbb{1}_{\Omega'_n} (A(1)_{h_n} - A(1)_{h_n+1})^2) \leq K, \\ |\mathbb{1}_{\Omega'_n} (\Omega_n^{-1})_{1h_n}| \leq K, & |\mathbb{1}_{\Omega'_n} A(3)_{h_n, l_n}| \leq K n^{-1/2} + K \mathbb{1}_{\{|h_n - l_n| \leq 2\}}. \end{cases}$$

All of the estimates are direct results of Lemma A2. Note $A(3)_{i,j}$ is a linear combination of four entries of Ω_n^{-1} . Due to such a combination, for $|h_n - l_n| \geq 3$, the magnitude of $A(3)_{h_n, l_n}$ is reduced by a factor of n compared with $(\Omega_n^{-1})_{i,j} \sim \sqrt{n}$.

Step 4. (Bound on $\mathcal{R}_{b2}(\beta^{(n)})$) In this step we show that $\mathbb{E}|\mathcal{R}_{b2}(\beta^{(n)})| = o(n^{5/4})$. We obtain results in other cases following the same reasoning. Note that Σ_n is positive definite, as it is a covariance matrix. Then the definition of $\mathcal{R}_{b2}(\beta)$ given by (A.113) and the Cauchy-Schwarz inequality indicate that $\mathbb{E}|\mathcal{R}_{b2}(\beta^{(n)})| = o(n^{5/4})$ follows from

$$\mathbb{E}|\mathbb{1}_{\Omega'_n} Y_n^\top \Omega_n^{-1} R_n \Sigma_n^{-1} R_n \Omega_n^{-1} Y_n| = o(n^{1/4}) \quad \text{and} \quad \mathbb{E}|\mathbb{1}_{\Omega'_n} Y_n^\top \Omega_n^{-2} R_n \Sigma_n^{-1} R_n \Omega_n^{-2} Y_n| = o(n^{9/4}). \quad (\text{A.116})$$

Here Ω_n , R_n , and Σ_n are evaluated at $\beta^{(n)}$. Now we show the first claim in (A.116). The second comes from the same reasoning. In Step 1 we state that we can replace Σ_n^{-1} with Ω_n^{-1} . Obviously, we can do so if $\Sigma_n^{-1} \leq K \Omega_n^{-1}$. Here $\leq$ stands for Loewner partial order. Indeed, Corollary 7.7.4 in Horn and Johnson (2013) states that for any two Hermitian matrices A and B , if $A < B$, then $A^{-1} > B^{-1}$. Hence, we only need $\Omega_n \leq K \Sigma_n$. On the one hand, we have (A.114). On the other hand, in view of the definitions of Ω_n and V_n , we conclude that

$$\Omega_n((\sigma^{(n)})^2, \gamma^{(n)}) \lesssim O_n V_n((\sigma^{(n)})^2, \gamma_0^{(n)}) O_n = \Omega_n((\sigma^{(n)})^2, \gamma_0^{(n)}) = \Sigma_n((\sigma^{(n)})^2, \gamma_0^{(n)}). \quad (\text{A.117})$$

Here the first equality can be verified using Lemma A1. Given (A.114) and (A.117), we have $\Omega_n((\sigma^{(n)})^2, \gamma^{(n)}) \leq K \Sigma_n((\sigma^{(n)})^2, \gamma^{(n)})$. Therefore, the first claim in (A.116) follows from

$$\mathbb{E}|\mathbb{1}_{\Omega'_n} Y_n^\top \Omega_n^{-1} R_n \Omega_n^{-1} R_n \Omega_n^{-1} Y_n| = o(n^{1/4}), \quad (\text{A.118})$$

with Ω_n , R_n and Σ_n evaluated at $\beta^{(n)}$.

Step 5. (Proof of (A.118)) First, we derive the expression of R_n . Using Lemma A1 and the definition of $\{\mathbb{F}_n^h : 0 \leq h \leq n\}$ given by (A.3) therein, we write

$$\Omega_n(\sigma^2, \gamma) = \sigma^2 \Delta_n \mathbb{I}_n + \sum_{h=0}^{q_n} \gamma_h (2\mathbb{F}_n^h - \mathbb{F}_n^{h+1} - \mathbb{F}_n^{h-1}). \quad (\text{A.119})$$

Here $\mathbb{F}_n^h = 0_{n \times n}$ for $h = -1$ by convention. On the other hand, we rewrite Σ_n defined by (3.7) as

$$\Sigma_n(\sigma^2, \gamma) = \sigma^2 \Delta_n \mathbb{I}_n + \sum_{h=0}^{q_n} \gamma_h (2\mathbb{G}_n^h - \mathbb{G}_n^{h+1} - \mathbb{G}_n^{h-1}). \quad (\text{A.120})$$

To write R_n in a more compact form, define $\mathbb{K}_n^h, \tilde{\mathbb{K}}_n^h \in \mathcal{M}_{n_T}$ by $(\mathbb{K}_n^h)_{ij} = \mathbb{1}_{\{h=i+j\}} - \mathbb{1}_{\{h+1=i+j\}}$ and $(\tilde{\mathbb{K}}_n^h)_{ij} = (\mathbb{K}_n^h)_{n+1-i, n+1-j}$. Obviously, $\mathbb{K}_n^h + \tilde{\mathbb{K}}_n^h = \mathbb{G}_n^h - \mathbb{G}_n^{h+1} - \mathbb{F}_n^h + \mathbb{F}_n^{h+1}$; hence (A.119) and (A.120) lead to

$$R_n(\beta^{(n)}) = \Sigma_n(\beta^{(n)}) - \Omega_n(\beta^{(n)}) = \sum_{h=0}^{q_n-1} (\gamma_h^{(n)} - \gamma_{h+1}^{(n)}) (\mathbb{K}_n^h + \tilde{\mathbb{K}}_n^h). \quad (\text{A.121})$$

Here $\gamma_{q_n+1}^{(n)} = 0$ by convention. Next, we apply (A.121) to (A.118). In view of the symmetry between $\mathbb{K}_n^h$ and $\tilde{\mathbb{K}}_n^h$, we can replace $R_n(\beta^{(n)})$ in (A.118) with $\tilde{R}_n(\beta^{(n)}) := \sum_{h=0}^{q_n-1} (\gamma_h^{(n)} - \gamma_{h+1}^{(n)}) \mathbb{K}_n^h$. Then (A.118) becomes

$$\mathbb{E} \left| \mathbb{1}_{\Omega'_n} \sum_{h=0}^{q_n-1} \sum_{l=0}^{q_n-1} (\gamma_h^{(n)} - \gamma_{h+1}^{(n)}) (\gamma_l^{(n)} - \gamma_{l+1}^{(n)}) Y_n^\top \Omega_n^{-1} \mathbb{K}_n^h \Omega_n^{-1} \mathbb{K}_n^l \Omega_n^{-1} Y_n \right| = o(n^{1/4}). \quad (\text{A.122})$$

From the definition of $\gamma^{(n)}$ and applying Hölder's inequality, we only need prove for all $n^{1/3}h_n \leq K$ and $n^{1/3}l_n \leq K$ that

$$\mathbb{E} \left| \mathbb{1}_{\Omega'_n} Y_n^\top \Omega_n^{-1} \mathbb{K}_n^{h_n} \Omega_n^{-1} \mathbb{K}_n^{l_n} \Omega_n^{-1} Y_n \right| \leq K n^{1/4} h_n l_n. \quad (\text{A.123})$$

In view of the notation introduced in Step 3, plus the definition of $\mathbb{K}_n^h$, we have that

$$\begin{aligned} Y_n^\top \Omega_n^{-1} \mathbb{K}_n^h \Omega_n^{-1} \mathbb{K}_n^l \Omega_n^{-1} Y_n &= \sum_{i=1}^{h-1} \sum_{j=1}^{l-1} A(1)_i A(1)_j A(3)_{h-i, h-j} - A(1)_h \sum_{j=1}^{l-1} A(1)_j A(2)_{1, l-j} \\ &\quad - A(1)_l \sum_{i=1}^{h-1} A(1)_i A(2)_{1, h-j} + A(1)_h A(1)_l (\Omega_n^{-1})_{11}. \end{aligned} \quad (\text{A.124})$$

We hence deduce (A.123) by applying the Cauchy-Schwarz inequality and the four estimates provided in Step 3 to (A.124). ■

Lemma A8. Suppose Assumptions 1 - 4 hold and $q_n \leq K n^{1/3}$. It holds that under either $n^{1/2} \iota^{(n)} \rightarrow \infty$ or $n^{1/2} \iota^{(n)} \leq K$,

$$\bar{\eta}^\top \Xi_{A,n}(\beta^{(n)}(q_n)) - \bar{\eta}^\top \Xi_{D,n}(\beta^{(n)}(q_n)) = o_P \left(n^{-1/2} \sqrt{q_n + 1} + n^{-1/4} \sqrt{\iota^{(n)}} \right). \quad (\text{A.125})$$

Proof. All of the contents of the proof of Lemma A7 until (A.111) remain valid if we replace Σ_n with Ω_n , Ω_n with $\Omega_{D,n}$, and (A.105) with (A.125). Unlike the situation there, we do know the analytical expressions of both Ω_n^{-1} and $\Omega_{D,n}^{-1}$, as given by Lemma A2. Note that $\Omega_{D,n}$ is a block-diagonal matrix and we apply Lemma A2 to each block. Instead of (A.112), we use

$$\Omega_n^{-1} = \Omega_{D,n}^{-1} - \Omega_n^{-1} R_n \Omega_{D,n}^{-1} \quad \text{and} \quad R_n := \Omega_n - \Omega_{D,n}.$$

The justification for doing so is the same as the one stated at the end of the proof of Lemma A7. Indeed, by definition, R_n here has only nonzero entries near the top-left or right-bottom corners of

the blocks $\Omega_{D,n}$ consists of. According to Lemma A2, $(\Omega_{D,n}^{-1})_{i,j}$ shrinks when either i or j approaches the borders of those blocks. Moreover, locally R_n also maintains a structure similar to (A.121); See the comment at the end of the proof of Lemma A7. Hence, we obtain (A.125) following the same reasoning as in the proof of Lemma A7. Note that we will skip Steps 2 and 4 there, since we know both $\Omega_{D,n}^{-1}$ and Ω_n^{-1} . ■

Lemma A9. *Suppose Assumptions 1 - 4 hold. It holds that for all sequences $q_n \leq Kn^{1/3}$ and under either $n^{1/2}\iota^{(n)} \rightarrow \infty$ or $n^{1/2}\iota^{(n)} \leq K$,*

$$\sup_{(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q_n)} \left| \frac{L_n(\sigma^2, \gamma) - \bar{L}_n^*(\sigma^2, \gamma)}{\bar{L}_n^*(C_T, \gamma^{(n)}) - \bar{L}_n^*(\sigma^2, \gamma) + n} \right| = o_P(1). \quad (\text{A.126})$$

Proof. Step 1. (Technical results) We start by defining a family of subsets of $\Pi_n^{(\sigma^2, \gamma)}(q)$ indexed by α_1 and α_2 :

$$\Pi_n^{(\sigma^2, \gamma)}(q, \alpha_1, \alpha_2) = \{(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q) : \alpha_1 \leq \chi^2(\sigma^2, \gamma, \Delta_n) \leq \alpha_2\}.$$

Obviously, $\Pi_n^{(\sigma^2, \gamma)}(q) = \Pi_n^{(\sigma^2, \gamma)}(q, 0, \alpha) \cup \Pi_n^{(\sigma^2, \gamma)}(q, \alpha, \infty)$ for all α . In this step we aim to prove that for all $\alpha_n \rightarrow \infty$ and all K fixed,

$$\sup_{(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q_n, \alpha_n \Delta_n, \infty)} \left| \frac{L_{A,n}(\sigma^2, \gamma) - \bar{L}_n^*(\sigma^2, \gamma)}{\bar{L}_n^*(C_T, \gamma^{(n)}) - \bar{L}_n^*(\sigma^2, \gamma) + n} \right| = o_P(1), \quad (\text{A.127})$$

$$\text{and} \quad \sup_{(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q_n, 0, K \Delta_n)} \left| \frac{L_{A,n}(\sigma^2, \gamma) - \bar{L}_n^*(\sigma^2, \gamma)}{\bar{L}_n^*(C_T, \gamma^{(n)}) - \bar{L}_n^*(\sigma^2, \gamma) + n} \right| = o_P(1). \quad (\text{A.128})$$

We consider the case in which $n^{1/2}\iota^{(n)} \leq K$. The case in which $n^{1/2}\iota^{(n)} \rightarrow \infty$ follows from the same reasoning. Straightforwardly, it holds that in restriction to Ω'_n ,

$$\sup_{(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q_n, \alpha_n \Delta_n, \infty)} \left| \bar{L}_n^*(C_T, \gamma^{(n)}) - \bar{L}_n^*(\sigma^2, \gamma) - \frac{n_T}{2} \log \frac{\chi^2(\sigma^2, \gamma, \Delta_n)}{\chi^2(C_T, \gamma^{(n)}, \Delta_n)} \right| \leq Kn.$$

Because $\chi^2(\sigma^2, \gamma, \Delta_n) \geq \alpha_n \Delta_n$ and $\chi^2(C_T, \gamma^{(n)}, \Delta_n) \in (K^{-1} \Delta_n, K \Delta_n)$, it holds that in restriction to Ω'_n ,

$$\inf_{(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q_n, \alpha_n \Delta_n, \infty)} \frac{1}{n_T} |\bar{L}_n^*(C_T, \gamma^{(n)}) - \bar{L}_n^*(\sigma^2, \gamma)| \rightarrow \infty.$$

Hence, in view of the triangle inequality and the definitions of $L_{A,n}$ and $\bar{L}_n^*$, plus using (A.107), to obtain (A.127) it is enough to show that

$$\sup_{(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q_n, \alpha_n \Delta_n, \infty)} Y_n^\top \Omega_n(\sigma^2, \gamma)^{-1} Y_n = O_P(n) \quad (\text{A.129})$$

$$\text{and} \quad \sup_{(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q_n, \alpha_n \Delta_n, \infty)} \frac{\mathbb{1}_{\Omega'_n}}{2\pi} \int_{-\pi}^{\pi} \frac{f(\lambda; C_T, \gamma^{(n)}, \Delta_n)}{f(\lambda; \sigma^2, \gamma, \Delta_n)} d\lambda \leq K.$$

The second bound is obviously true and we now show the first bound (A.129). Lemma A2 states that we can write

$$Y_n^\top \Omega_n(\sigma^2, \gamma)^{-1} Y_n = Y_n^\top \sum_{h=0}^{\infty} \rho_h(\sigma^2, \gamma, \Delta_n) \mathbb{F}_n^h Y_n, \quad (\text{A.130})$$

where $\rho_h(\sigma^2, \gamma, \Delta_n)$ satisfies (A.4) uniformly over $(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q_n, \alpha_n \Delta_n, \infty)$. On the other hand, with $n^{1/2} \iota^{(n)} \leq K$ and Assumption 4, we conclude that

$$\mathbb{E}|Y_n^\top \mathbb{F}_n^h Y_n| \leq K h^{-2} + K n^{-1/2}. \quad (\text{A.131})$$

The combination of (A.130), (A.4), (A.131), and (A.107), plus Hölder's inequality, readily yields (A.129) and hence (A.127) is proved. We now prove (A.128). Since $\bar{L}_n^*(C_T, \gamma^{(n)}) - \bar{L}_n^*(\sigma^2, \gamma)$ is always nonnegative, in view of the definitions of $L_{A,n}$ and $\bar{L}_n^*$, (A.128) directly comes from

$$\sup_{(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q_n, 0, K \Delta_n)} \left| Y_n^\top \Omega_n(\sigma^2, \gamma)^{-1} Y_n - \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{f(\lambda; C_T, \gamma^{(n)}, \Delta_n)}{f(\lambda; \sigma^2, \gamma, \Delta_n)} d\lambda \right| = o_P(n). \quad (\text{A.132})$$

The uniform convergence (A.132) comes from establishing pointwise convergence and stochastic equicontinuity, following the same reasoning as for Theorem 2.1 and Corollary 2.2 in Newey (1991). Applying steps 1 and 7 of the proof of Lemma A2, we have, for all deterministic $\{(\sigma_n^2, \gamma_n) \in \Pi_n^{(\sigma^2, \gamma)}(q_n, 0, K \Delta_n) : n \geq 1\}$ and under $n^{1/2} \iota^{(n)} \leq K$,

$$Y_n^\top \Omega_n(\sigma_n^2, \gamma_n)^{-1} Y_n - \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{f(\lambda; C_T, \gamma^{(n)}, \Delta_n)}{f(\lambda; \sigma_n^2, \gamma_n, \Delta_n)} d\lambda = o_P(n).$$

On the other hand, using (A.131) and (A.7), plus Assumption 2, we can repeat the reasoning for deriving (A.129) to conclude that for all $0 \leq j \leq q_n$ and with Ω'_n introduced above (A.107),

$$\mathbb{E} \left(\mathbb{1}_{\Omega'_n} \sup_{(\sigma^2, \gamma)} \left| \frac{\partial}{\partial \sigma^2} Y_n^\top \Omega_n(\sigma^2, \gamma)^{-1} Y_n \right| \right) \leq K n, \quad \mathbb{E} \left(\mathbb{1}_{\Omega'_n} \sup_{(\sigma^2, \gamma)} \left| \Delta_n \frac{\partial}{\partial \gamma_j} Y_n^\top \Omega_n(\sigma^2, \gamma)^{-1} Y_n \right| \right) \leq K n,$$

$$\sup_{(\sigma^2, \gamma)} \left| \mathbb{1}_{\Omega'_n} \frac{\partial}{\partial \sigma^2} \int_{-\pi}^{\pi} \frac{f(\lambda; C_T, \gamma^{(n)}, \Delta_n)}{f(\lambda; \sigma_n^2, \gamma_n, \Delta_n)} d\lambda \right| \leq K, \quad \sup_{(\sigma^2, \gamma)} \left| \mathbb{1}_{\Omega'_n} \Delta_n \frac{\partial}{\partial \gamma_j} \int_{-\pi}^{\pi} \frac{f(\lambda; C_T, \gamma^{(n)}, \Delta_n)}{f(\lambda; \sigma_n^2, \gamma_n, \Delta_n)} d\lambda \right| \leq K,$$

from which the stochastic equicontinuity follows. Here the range over which the supremums are taken is $(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q_n, 0, K \Delta_n)$ and the additional factor Δ_n compared with Assumption 3A in Newey (1991) arises because of Assumption 4 and the definition of $\Pi_n^{(\sigma^2, \gamma)}(q_n, 0, K \Delta_n)$.

Step 2. (Conclusion) In view of (A.127) and (A.128), plus using the subsequence argument, we

obtain

$$\sup_{(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q_n)} \left| \frac{L_{A,n}(\sigma^2, \gamma) - \bar{L}_n^*(\sigma^2, \gamma)}{\bar{L}_n^*(C_T, \gamma^{(n)}) - \bar{L}_n^*(\sigma^2, \gamma) + n} \right| = o_P(n). \quad (\text{A.133})$$

Further, we have, following the reasoning in the proof of Lemma A7 and using Lemma A2, that

$$\sup_{(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q_n)} \left| \frac{L_n(\sigma^2, \gamma) - L_{A,n}(\sigma^2, \gamma)}{\bar{L}_n^*(C_T, \gamma^{(n)}) - \bar{L}_n^*(\sigma^2, \gamma) + n} \right| = o_P(1). \quad (\text{A.134})$$

Note that the bound we require here is less sharp and that only Σ_n^{-1} and Ω_n^{-1} themselves are involved, as we do not take derivatives here. The lemma is a direct result of (A.133) and (A.134). ■

Lemma A10. *Suppose Assumptions 1 - 4 hold. For all sequences $\{q_n\}$ and $\{q'_n\}$, it holds that with probability approaching one,*

$$\frac{1}{n_T} \bar{L}_n^*(\sigma^{(n)}(q_n)^2, \gamma^{(n)}(q_n)) - \frac{1}{n_T} \bar{L}_n^*(\sigma^{(n)}(q'_n)^2, \gamma^{(n)}(q'_n)) \sim \psi_n^4 (\|\tilde{\kappa}^{(n)}\|_{(q'_n)}^2 - \|\tilde{\kappa}^{(n)}\|_{(q_n)}^2).$$

Proof. We define $C_{i,j} = \pi^{-1} \int_{-\pi}^{\pi} |1 - \psi_n e^{i\lambda}|^{-4} \cos i\lambda \cos j\lambda d\lambda$. It holds that

$$(1 - \psi_n^2)^2 C_{i,j} = \psi_n^{|i-j|} |i-j| + \psi_n^{|i+j|} |i+j| + \frac{1 + \psi_n^2}{1 - \psi_n^2} (\psi_n^{|i-j|} + \psi_n^{|i+j|}).$$

We introduce m -dimensional matrices C_m , $\mathbb{J}_m^h$, and $\mathbb{K}_m^h$ with entries given by $C_{i,j}$, $(\mathbb{J}_m^h)_{i,j} = \mathbb{1}_{\{|i-j|=h\}}$, and $(\mathbb{K}_m^h)_{i,j} = \mathbb{1}_{\{i+j=h\}} + \mathbb{1}_{\{2m+2-i-j=h\}}$. We further define $\bar{C}_m = (1 - \psi_n^2)^2 C_m$, $\tilde{C}_m = \sum_{h=0}^{\infty} \mathbb{J}_m^h \psi_n^h (h + (1 + \psi_n^2)(1 - \psi_n^2)^{-1})$, and $\check{C}_m = \sum_{h=0}^{\infty} \mathbb{K}_m^h \psi_n^h (h + (1 + \psi_n^2)(1 - \psi_n^2)^{-1})$. From Lemma A1, it follows that

$$(O_m(\tilde{C}_m - \check{C}_m)O)_{i,j} = \delta_{i,j} \sum_{h=-\infty}^{\infty} \psi_n^{|h|} (|h| + (1 + \psi_n^2)(1 - \psi_n^2)^{-1}) \cos \frac{hj\pi}{m+1}.$$

By direct calculations, we obtain $\tilde{C}_m \geq \check{C}_m$ for m sufficiently large. On the other hand, it holds that for all m -dimensional vector β ,

$$\beta^\top \tilde{C}_m \beta = (1 - \psi_n^2)^2 \sum_{j=-\infty}^{\infty} \left| \sum_{k=0}^{\infty} (k+1) \psi_n^k \beta_{j+k} \right|^2, \quad (\text{A.135})$$

$$\beta^\top \check{C}_m \beta = (1 - \psi_n^2)^2 \sum_{j=-\infty}^{\infty} \left| \sum_{k=0}^{\infty} (k+1) \psi_n^k \beta_{|j+k|} (1 + \delta_{j+k,0}) \right|^2. \quad (\text{A.136})$$

Here we take $\beta_j = 0$ for all $j < 0$. Because (A.135) and (A.136) hold for arbitrary m , and comparing $\tilde{C}_m - \check{C}_m$ with $\bar{C}_m$, we prove the lemma by using the fact that $f(\lambda; \sigma^{(n)}(q_n)^2, \gamma^{(n)}(q_n), \Delta_n) \sim ((\iota^{(n)})^2 + n^{-1}) |1 - \psi_n e^{i\lambda}|^2$ with probability approaching one. ■

Appendix B Proof of Main Results

B.1 Proof of Theorem 1

Proof. The assumptions of Theorem 1 lead to the fact that $\bar{L}_n^*(C_T, \gamma^{(n)}) - \bar{L}_n^*(\sigma^{(n)}(q)^2, \gamma^{(n)}(q)) = 0$ for all $q \geq q^*$. Then, in view of the proof of Lemma B4 in the online appendix of Da and Xiu (2021), we have

$$\bar{L}_n^*(C_T, \gamma^{(n)}) - \bar{L}_n^*(\sigma^{(n)}(\hat{q}_n)^2, \gamma^{(n)}(\hat{q}_n)) \leq \frac{\log n}{2}(q^* - \hat{q}_n) + O(\hat{q}_n + \log n). \quad (\text{B.1})$$

As an immediate result, we obtain that for n sufficiently large, $\hat{q}_n \leq q^*$. On the other hand, we have $(\log n)^{-1}(\bar{L}_n^*(C_T, \gamma^{(n)}) - \bar{L}_n^*(\sigma^{(n)}(q)^2, \gamma^{(n)}(q))) \rightarrow \infty$ for all $q \leq q^* - 1$, according to the assumption that $\sqrt{n}(\log n)^{-1}|\theta_{q^*}^{(n)}| \rightarrow \infty$. Then (B.1) indicates that for n sufficiently large, $\hat{q}_n \geq q^*$ and we conclude the proof. ■

B.2 Proofs of Theorem 2, Corollary 1, and Proposition 1

Lemma B1. *Suppose the same assumptions as those in Theorem 2 hold. Then it holds that*

$$\hat{\mathcal{R}}_n(q^*) = o_P(1) \quad \text{and} \quad \mathcal{R}^{(n)}(q^*) = o_P(1).$$

Proof. We start by proving the convergence of $\hat{\mathcal{R}}_n(q_n^*)$. From Lemma A9 it directly follows that

$$\bar{L}_n^*(C_T, \gamma^{(n)}) - \bar{L}_n^*(\hat{\sigma}_n^2(\hat{q}_n), \hat{\gamma}_n(\hat{q}_n)) = o_P(n). \quad (\text{B.2})$$

Since both (σ_n^2, γ_n) and $(C_T, \gamma^{(n)})$ belong to $\Pi_n^{(\sigma^2, \gamma)}$, according to Theorem 4.1.1, Proposition 4.5.3, Proposition 3.2.1, and Theorem 3.1.2 in Brockwell and Davis (1991), there exist unique (χ_n^2, ϕ_n) and $((\chi^{(n)})^2, \phi^{(n)})$ such that for all $-\pi \leq \lambda \leq \pi$,

$$f(\lambda; \hat{\sigma}_n^2(\hat{q}_n), \hat{\gamma}_n(\hat{q}_n), \Delta_n) = \chi_n^2 g(\lambda; \phi_n) \quad \text{and} \quad f(\lambda; C_T, \gamma^{(n)}, \Delta_n) = (\chi^{(n)})^2 g(\lambda; \phi^{(n)}), \quad (\text{B.3})$$

$$1 + \inf_{z \in \mathbb{C}, |z| \leq 1} \sum_{j=1}^{\infty} \phi_{n,j} z^j > 0 \quad \text{and} \quad 1 + \inf_{z \in \mathbb{C}, |z| \leq 1} \sum_{j=1}^{\infty} \phi_j^{(n)} z^j > 0. \quad (\text{B.4})$$

In view of (B.3) and the definition of $\bar{L}_n^*$, the bound (B.2) can be rewritten in terms of (χ_n^2, ϕ_n) and $((\chi^{(n)})^2, \phi^{(n)})$, which leads to

$$\log \frac{\chi_n^2}{(\chi^{(n)})^2} = o_P(1) \quad \text{and} \quad \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{f(\lambda; C_T, \gamma^{(n)}, \Delta_n)}{f(\lambda; \hat{\sigma}_n^2, \hat{\gamma}_n, \Delta_n)} d\lambda - 1 = o_P(1). \quad (\text{B.5})$$

Here we use (A.107) and the fact that $(2\pi)^{-1} \int_{-\pi}^{\pi} g(\lambda; \phi^{(n)})/g(\lambda; \phi_n) d\lambda \geq 1$, indicated by (B.4). With $\chi^{(n)}$ calculated using Assumption 4, the first part of (B.5) indicates that $\log \chi_n^2 = \log(\iota^{(n)})^2 + o_P(1)$. Substituting the estimate of χ_n^2 back into (A.22), plus using the second part of (B.5), plus (A.107), immediately allows us to prove the convergence of $\hat{\mathcal{R}}_n(q_n^*)$. On the other hand, the convergence of

$\mathcal{R}^{(n)}(q^*) = o_P(1)$ follows directly from Lemma A5. We conclude the proof. ■

Proofs of Theorem 2 and Corollary 1. Step 1. (Preparation) In this proof the dependence of $\beta^{(n)}$ on q is suppressed and here we take $q = q^*$. We set $\beta_n(\sigma^2, \gamma) = (\sigma^2, \gamma)$. We start by introducing $(q^* + 2) \times (q^* + 2)$ matrices $\partial \Xi_n(\beta_n, \beta'_n, k)$ with $k \in \{1, 2\}$ and $\beta_n, \beta'_n \in \Pi_n^\beta(q^*)$, defined, for $0 \leq i, j \leq q^* + 1$, by

$$\partial \Xi_n(\beta_n, \beta'_n; 1)_{i,j} = \frac{1}{2n} \text{tr} \left(\frac{\partial \log \Omega_n(\beta_n)}{\partial \beta_i} \frac{\partial \log \Omega_n(\beta'_n)}{\partial \beta_j} \right), \quad (\text{B.6})$$

$$\partial \Xi_n(\beta_n, \beta'_n; 2)_{i,j} = \frac{1}{4n} \text{tr} \left(\frac{\partial \log \Omega_n(\beta_n)}{\partial \beta_i} \frac{\partial \log \Omega_n(\beta'_n)}{\partial \beta_j} (\Omega_n(\beta_n)^{-1} + \Omega_n(\beta'_n)^{-1}) Y_n Y_n^\top \right). \quad (\text{B.7})$$

We further denote $\partial \Xi_n(\beta_n; j) := \partial \Xi_n(\beta_n, \beta_n; j)$. In addition, we use $\partial \Xi_n(\bar{\beta}^{(n)}, q^*; j)$ and $\partial \bar{\Xi}_n^*(\bar{\beta}^{(n)}, q^*)$, respectively, to denote the $(q^* + 2) \times (q^* + 2)$ matrices with entries defined by (B.6) and (B.7) and with entries defined by (A.2). On the other hand, we let $\{\check{\beta}_n \in \Pi_n^\beta(q^*) : n \geq 1\}$ be a sequence of $(q^* + 2)$ -dimensional random vectors that satisfies the equation $\Xi_n(\check{\beta}_n) = 0_{q^*+2}$, and the condition whereby $\sup_\lambda |f(\lambda; \check{\beta}_n, \Delta_n) f(\lambda; \bar{\beta}^{(n)}, \Delta_n)^{-1} - 1| = o_P(1)$ holds. In view of the definition of $\partial \Xi_n(\beta_n, \beta'_n; j)$ introduced in (B.6) and (B.7), plus applying the rules of matrix differentiation, in particular that $\Omega_n(\beta)$ and $\Omega_n(\beta')$ commute for all (β, β') , we observe that

$$\check{\beta}_n - \beta^{(n)} = (2\partial \Xi_n(\check{\beta}_n, \beta^{(n)}; 2) - \partial \Xi_n(\check{\beta}_n, \beta^{(n)}; 1))^{-1} (\Xi_{A,n}(\check{\beta}_n) - \Xi_{A,n}(\beta^{(n)})). \quad (\text{B.8})$$

On the other hand, using $\mathbb{D}_m^j = O_m \mathbb{F}_m^j O_m$ and the connection between matrix V_m and spectral density $f(\lambda; \beta, \Delta_n)$ and the positivity of both following the reasoning of step 1 of the proof of Lemma A2, plus the fact that $\partial f(\lambda; \beta, \Delta_n) / \partial \beta$ does not depend on β , we have, for all $\alpha_n \rightarrow 0$ and $j \in \{1, 2\}$, and under that $\sup_\lambda |f(\lambda; b_n, \Delta_n) f(\lambda; \bar{\beta}^{(n)}, \Delta_n)^{-1} - 1| \rightarrow 0$ for $b_n \in \{\check{\beta}_n, \beta^{(n)}\}$,

$$\begin{cases} (1 - \alpha_n) \partial \Xi_n(\bar{\beta}^{(n)}, q^*; j) \leq \partial \Xi_n(\check{\beta}_n, \beta^{(n)}; j) \leq (1 + \alpha_n) \partial \Xi_n(\bar{\beta}^{(n)}, q^*; j) \\ (1 - \alpha_n) \partial \bar{\Xi}_n^*(\bar{\beta}^{(n)}, q^*) \leq \partial \Xi_n(\bar{\beta}^{(n)}, q^*; 1) \leq (1 + \alpha_n) \partial \bar{\Xi}_n^*(\bar{\beta}^{(n)}, q^*) \end{cases} \quad (\text{B.9})$$

Furthermore, using Lemma A2, we can derive $\mathbb{E} |\mathbb{1}_{\Omega'_n} \left(\text{tr}(\Omega_n(\bar{\beta}^{(n)})^{-1} Y_n Y_n^\top - \mathbb{I}_n) \right)^2| \leq Kn$, which, combined with (A.22) and (A.107), leads to the fact that for some $\alpha_n \rightarrow 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}((1 - \alpha_n) \partial \Xi_n(\bar{\beta}^{(n)}, q^*; 1) \leq \partial \Xi_n(\bar{\beta}^{(n)}, q^*; 2) \leq (1 + \alpha_n) \partial \Xi_n(\bar{\beta}^{(n)}, q^*; 1)) = 1. \quad (\text{B.10})$$

Step 2. (Main Proof) We define, for $1 \leq j \leq J_d + 2$ and $1 \leq i \leq q^* + 1$,

$$\begin{aligned} \mathcal{V}_n(j)_i &= \frac{1}{2n} \frac{\partial}{\partial \beta_i} \text{tr}(\Omega_{n_d}(\tilde{\beta}^{(n)})^{-1} Y_n^C(j) Y_n^C(j)^\top), \\ \mathcal{V}_n(j)_i &= \frac{1}{2n} \frac{\partial}{\partial \beta_i} \text{tr}(\Omega_{n_d}(\tilde{\beta}^{(n)})^{-1} \Omega_n^{U,C}(j)), \end{aligned}$$

where $\tilde{\beta}^{(n)} := (1, (\iota^{(n)})^2 \kappa^{(n)})$, $Y_n^C(j) := (Y_{n,1}^C(j), \dots, Y_{n,n_d}^C(j))^\top$ with $Y_{n,k}^C(j) = \Delta_k^n U^C(j)$. Under Assumption A1, for any two $(\mathcal{F}_t)$ -stopping times $S \leq S'$, any $p \geq 0$, and any process A among $\mu, \sigma, \xi, \xi^{-1}, \eta$ (Section 2.1.5 of Jacod and Protter (2011)):

$$\begin{cases} |\mathbb{E}(A_{S'} - A_S | \mathcal{F}_S)| \leq K \mathbb{E}(S' - S | \mathcal{F}_S), \\ \mathbb{E}(\sup_{S \leq s \leq S'} \|A_s - A_S\|^p | \mathcal{F}_S) \leq K \mathbb{E}((S' - S)^{1 \wedge (p/2)} | \mathcal{F}_S). \end{cases} \quad (\text{B.11})$$

By Lemmas A3 and A4 and (B.11), we obtain $\Xi_{D,n}(\beta^{(n)})_0 = o_P(n^{-1/2})$ and, for all $1 \leq i \leq q^* + 1$,

$$\Xi_{D,n}(\beta^{(n)})_i - \bar{\Xi}_n(\beta^{(n)})_i = \sum_{j=1}^{J_d+2} \bar{\eta}^{-4} (\mathcal{V}_n(j)_i - \bar{\mathcal{V}}_n(j)_i) + o_P(n^{-1/2}), \quad (\text{B.12})$$

where we write $\bar{\eta}^2 := T^{-1} \bar{\xi}_T \int_0^T \eta_s^2 \xi_s^{-1} ds$ with $\bar{\xi}_T^{-1} := \frac{1}{T} \int_0^T \xi_s^{-1} ds$, and $\bar{\Xi}_n(\beta)$ and $\bar{\Xi}_n(\beta^{(n)})$ are defined in (A.2). We intentionally let the range of summation goes to $J_d + 2$, the reason of which will become more clear when we later on establish the stable CLT for triangular arrays.

Now we analyze $\bar{\Xi}_n(\beta^{(n)})$. Using the short-hand notation $n(j) = n_d + \delta_{j,J_d+1}(n'_d - n_d)$, we have

$$\bar{\Xi}_n(\beta^{(n)})_i = \frac{1}{2n} \sum_{j=1}^{J_d+1} \text{tr} \left(\frac{\partial \Omega_{n(j)}^{-1}(\beta^{(n)})}{\partial \beta_i} (\Omega_{n(j)}(\beta^{(n)}) - \Omega_{n(j)}^Y(j)) \right). \quad (\text{B.13})$$

Since $\Omega_{n(j)}^Y(j)$ varies over j , we define, for $j = 1, \dots, J_d + 1$,

$$\eta_{(j)}^2 := \frac{n_T}{n(j)T\bar{\xi}_T^{-1}} \int_{t(j)_0}^{t(j)_{n(j)}} \eta_s^2 \xi_s^{-1} ds.$$

By construction, we have

$$\sum_{j=1}^{J_d+1} n(j) (\eta_{(j)}^2 - \bar{\eta}^2) = 0. \quad (\text{B.14})$$

We further set

$$\gamma_{(j),h} := \eta_{(j)}^2 (\iota^{(n)})^2 \kappa_h^{(n)}, \quad V_{n(j),(j)} := V_{n(j)}(0, \gamma_{(j)}), \quad \Omega_{n(j),(j)} := O_{n(j)} V_{n(j),(j)} O_{n(j)}.$$

Recall $V_{n(j)}(\beta^{(n)}) = V_{n(j)}(\beta_0^{(n)}, \gamma_h^{(n)})$ with $\gamma_h^{(n)} = \bar{\eta}^2 (\iota^{(n)})^2 \kappa_h^{(n)}$. Therefore, we have

$$V_{n(j),(j)} - V_{n(j)}(\beta^{(n)}) = V_{n(j)}(-\beta_0^{(n)}, (\eta_{(j)}^2 - \bar{\eta}^2) (\iota^{(n)})^2 \kappa^{(n)}). \quad (\text{B.15})$$

Combining (B.14) and (B.15), and applying Lemma A4, we obtain

$$\frac{1}{2n} \sum_{j=1}^{J_d+1} \text{tr}((\Omega_{n(j)}(\beta^{(n)}) - \Omega_{n(j),(j)})) = o_P(n^{-1/2}). \quad (\text{B.16})$$

Inserting the identity $\Omega_{n(j)} - \Omega_{n(j)}^Y(j) = (\Omega_{n(j)} - \Omega_{(j),n(j)}) + (\Omega_{(j),n(j)} - \Omega_{n(j)}^Y(j))$ into (B.13), and applying (B.16), we obtain, for all $1 \leq i \leq q^* + 1$,

$$\bar{\Xi}_n(\beta^{(n)})_i = \frac{1}{2n} \sum_{j=1}^{J_d+1} \text{tr} \left(\frac{\partial \Omega_{n(j)}^{-1}(\beta^{(n)})}{\partial \beta_i} (\Omega_{(j),n(j)} - \Omega_{n(j)}^Y(j)) \right) + o_P(n^{-1/2}), \quad (\text{B.17})$$

Next, we further analyze $\bar{\Xi}_n(\beta^{(n)})$ and introduce

$$\tilde{\eta}_{(j)}^2 := \frac{1}{n(j)} \sum_{k=1}^{n(j)} \eta_{t(j)_{k-1}}^2, \quad \tilde{\gamma}_{(j),h} := \tilde{\eta}_{(j)}^2 (\iota^{(n)})^2 \kappa_h^{(n)}, \quad \tilde{\Omega}_{(j),n(j)} := O_{n(j)} V_{n(j)}(0, \tilde{\gamma}_{(j)}) O_{n(j)}.$$

By Lemma A1, and Assumption A1, we have, for all $1 \leq i \leq q^* + 1$,

$$\sum_{j=1}^{J_d+1} \text{tr} \left(\frac{\partial \Omega_{n(j)}^{-1}(\beta^{(n)})}{\partial \beta_i} (\tilde{\Omega}_{(j),n(j)} - \Omega_{n(j)}^Y(j)) \right) = o_P(n^{1/2}). \quad (\text{B.18})$$

On the other hand, it holds by definition that

$$V_{n(j),n(j)} - V_{n(j)}(0, \tilde{\gamma}_{(j)}) = V_{n(j)}(0, (\eta_{(j)}^2 - \tilde{\eta}_{(j)}^2)(\iota^{(n)})^2 \kappa^{(n)}). \quad (\text{B.19})$$

Inserting the identity $\Omega_{(j),n(j)} - \Omega_{n(j)}^Y(j) = (\tilde{\Omega}_{(j),n(j)} - \Omega_{n(j)}^Y(j)) + (\Omega_{(j),n(j)} - \tilde{\Omega}_{(j),n(j)})$ into (B.17), and applying (B.19) and Lemma A4, we obtain, for all $1 \leq i \leq q^* + 1$,

$$\bar{\Xi}_n(\beta^{(n)})_i = \frac{w_i}{n} \sum_{j=1}^{J_d+1} n(j)(\eta_{(j)}^2 - \tilde{\eta}_{(j)}^2) + o_P(n^{-1/2}), \quad (\text{B.20})$$

where

$$w_i := -\frac{1}{4\pi\bar{\eta}^2} \int_{-\pi}^{\pi} \frac{\partial \log f(\lambda; \gamma^*)}{\partial \gamma_{i-1}} d\lambda.$$

Now we investigate $\sum_{j=1}^{J_d+1} n(j)(\eta_{(j)}^2 - \tilde{\eta}_{(j)}^2)$ in (B.20). By Assumption A1 and (A.107), we have

$$\frac{n_T}{n\bar{\xi}_T^{-1}} = 1 - \frac{1}{n\bar{\xi}_T^{-1}} \sum_{i=1}^{n_T} (\chi_i - 1) + o_P(n^{-1/2}). \quad (\text{B.21})$$

We can write

$$\begin{aligned} & \frac{1}{n} \sum_{j=1}^{J_d+1} n(j)(\eta_{(j)}^2 - \tilde{\eta}_{(j)}^2) = \frac{n_T}{nT\bar{\xi}_T^{-1}} \int_0^T \eta_s^2 \xi_s^{-1} ds - \frac{1}{n} \sum_{j=1}^{J_d+1} \sum_{k=1}^{n(j)} \eta_{t(j)_{k-1}}^2 \\ &= \left(\frac{n_T}{n\bar{\xi}_T^{-1}} - 1 \right) \frac{1}{T} \int_0^T \eta_s^2 \xi_s^{-1} ds + \frac{1}{T} \sum_{k=1}^{n_T} \int_{t_{k-1}}^{t_k} (\eta_s^2 \xi_s^{-1} - \eta_{t_{k-1}}^2 \xi_{t_{k-1}}^{-1}) ds + \frac{1}{n} \sum_{k=1}^{n_T} \eta_{t_{k-1}}^2 (\chi_k - 1) \end{aligned}$$

$$= -\frac{\bar{\eta}^2}{n} \sum_{i=1}^{n_T} (\chi_i - 1) + \frac{1}{n} \sum_{j=1}^{J_d+1} \sum_{k=1}^{n(j)} \eta_C^2(j) (\chi(j)_k - 1) + o_P(n^{-1/2}) \quad (\text{B.22})$$

The third equality follows from (B.11) and (B.21). Combining (B.12), (B.20), and (B.22), we obtain for all $1 \leq i \leq q^* + 1$, (again that the summation range goes to $J_d + 2$ is intentional and adding $B_n(J_d + 2)$ and $\tilde{B}_n(J_d + 2)$ is asymptotically negligible)

$$\Xi_n(\beta^{(n)})_i - \sum_{j=1}^{J_d+2} (\bar{\eta}^{-4} (\mathcal{V}_n(j)_i - \bar{\mathcal{V}}_n(j)_i) + w_i B_n(j) - \bar{\eta}^2 w_i \tilde{B}_n(j)) = o_P(n^{-1/2}). \quad (\text{B.23})$$

where we use short-hand notation

$$B_n(j) := \frac{n_d}{n} \eta_C^2(j) (\bar{\chi}(j) - 1), \quad \tilde{B}_n(j) = \frac{n_d}{n} (\bar{\chi}(j) - 1), \quad \text{with} \quad \bar{\chi}(j) := \frac{1}{n_d} \sum_{k=1}^{n_d} \chi(j)_k.$$

Now we establish the stable CLT of $\sum_{j=1}^{J_d+2} d_n(j)$, where $d_n(j) := ((\mathcal{V}_n(j) - \bar{\mathcal{V}}_n(j))^\top, B_n(j), \tilde{B}_n(j))^\top$. We set $\mathcal{F}^\varepsilon(j) = \sigma(\varepsilon^C(k)_i : i \leq n_d, k \leq j-1)$, $\mathcal{F}^\chi(j) = \sigma(\chi_i : i \leq (j-1)n_d)$, and

$$\mathcal{F}(j) = \mathcal{F}_{t(j)_0} \otimes \mathcal{F}^\chi(j) \otimes \mathcal{F}^\varepsilon(j).$$

By definition $d_n(j)$ is $\mathcal{F}(j+1)$ -measurable for each j . Block j 's noise innovations $\varepsilon^C(j)$ are $\mathcal{F}^\varepsilon(j+1)$ -measurable and independent of $\mathcal{F}(j)$; block j 's sampling variables $(\chi(j)_1, \dots, \chi(j)_{n_d})$ are $\mathcal{F}^\chi(j+1)$ -measurable and independent of $\mathcal{F}(j)$. Furthermore, we define $n_t = \max\{i : t_i \leq t\}$ and $J(t) = \lfloor n_t/n_d \rfloor + 1$. By definition we have $J(T) = \lfloor n_T/n_d \rfloor + 1 = J_d + 2$ and that $J(t)$ is a $(\mathcal{F}(j))$ -stopping time. Therefore, we can establish the stable CLT of $\sum_{j=1}^{J_d+2} d_n(j)$ by verifying the conditional mean condition, conditional covariance condition, higher-order moment condition and orthogonality condition as required by Theorem 2.2.15 of [Jacod and Protter \(2011\)](#).

The conditional mean condition $\left\| \sum_{1 \leq j \leq J(t)} \mathbb{E}(d_n(j) \mid \mathcal{F}(j)) \right\| = o_P(n^{-1/2})$ for all $t > 0$ follows from that for all $1 \leq i \leq q^* + 1$ and all $1 \leq j \leq J(t)$,

$$\mathbb{E}(B_n(j) \mid \mathcal{F}(j)) = \mathbb{E}(\tilde{B}_n(j) \mid \mathcal{F}(j)) = 0, \quad \sup_{1 \leq j' \leq J(t)} |\mathbb{E}(\mathcal{V}_n(j')_i - \bar{\mathcal{V}}_n(j')_i \mid \mathcal{F}(j'))| \leq K n^{-3/4}.$$

The first part follows from $\mathbb{E}(\chi(j)_i \mid \mathcal{F}(j)) = 1$ for all $0 \leq i \leq n_d$, the second part comes from Lemma A3 and that $\varepsilon^C(j)$ is independent of $\mathcal{F}(j)$. On the other hand, the higher-order moment condition $\sum_{1 \leq j \leq J_d+2} \mathbb{E}(\|d_n(j)\|^4 \mid \mathcal{F}(j)) = o_P(n^{-2})$ follows from Assumption A1, Lemma A3, and Lemma A4. Next, we study the conditional covariances of $d_n(j)$. We observe that $\bar{\mathcal{V}}_n(j)$ is $\mathcal{F}(j)$ -measurable, that, conditionally on $\mathcal{F}(j)$, $\mathcal{V}_n(j)$ is independent of both $B_n(j)$ and $\tilde{B}_n(j)$, and that, conditionally on $\mathcal{F}(j)$, $B_n(j)$ and $\tilde{B}_n(j)$ are perfectly correlated. Hence, we only need to calculate the conditional

variances of $\mathcal{V}_n(j)$ and $\tilde{B}_n(j)$. Using Lemma A3, we can write, for all $1 \leq i, i' \leq q^*$ and all $t > 0$,

$$\sup_{1 \leq j \leq J(t)} \|\text{Var}(\mathcal{V}_n(j) \mid \mathcal{F}(j))_{i,i'} - \tilde{\mathcal{V}}_n(j)_{i,i'}\| = o_P(n^{-3/2}), \quad (\text{B.24})$$

where we use the short-hand notation

$$\tilde{\mathcal{V}}_n(j)_{i,i'} := \frac{\eta_C^4(j)}{2n^2} \text{tr} \left(\Omega_{n_d}^{-2} \frac{\partial \Omega_{n_d}}{\partial \beta_i} \frac{\partial \Omega_{n_d}}{\partial \beta_{i'}} \right) + \frac{\eta_C^4(j) \text{cum}_4(\varepsilon)}{4n^2} \sum_{k=1}^{n_d} \left(\Omega_{n_d}^{-1} \frac{\partial \Omega_{n_d}}{\partial \beta_i} \right)_{kk} \left(\Omega_{n_d}^{-1} \frac{\partial \Omega_{n_d}}{\partial \beta_{i'}} \right)_{kk},$$

with all the Ω_{n_d} matrices evaluated at $\tilde{\beta}^{(n)}$. By Lemma A4, we can calculate, with Ω_{n_d} evaluated at $\tilde{\beta}^{(n)}$ and for all $1 \leq i, i' \leq q^*$,

$$\begin{aligned} \frac{1}{n_d} \text{tr} \left(\Omega_{n_d}^{-2} \frac{\partial \Omega_{n_d}}{\partial \beta_i} \frac{\partial \Omega_{n_d}}{\partial \beta_{i'}} \right) &= \frac{(\iota^{(n)})^{-4}}{2\pi} \int_{-\pi}^{\pi} \frac{\partial \log f(\lambda; \kappa^{(n)})}{\partial \gamma_{i-1}} \frac{\partial \log f(\lambda; \kappa^{(n)})}{\partial \gamma_{i'-1}} d\lambda + o(1) \\ &= (\iota^{(n)})^{-4} W(\kappa^{(n)})_{i,i'} + o(1), \\ \frac{1}{n_d} \sum_{k=1}^{n_d} \left(\Omega_{n_d}^{-1} \frac{\partial \Omega_{n_d}}{\partial \beta_i} \right)_{kk} \left(\Omega_{n_d}^{-1} \frac{\partial \Omega_{n_d}}{\partial \beta_{i'}} \right)_{kk} &= \frac{(\iota^{(n)})^{-4}}{(2\pi)^2} \int_{-\pi}^{\pi} \frac{\partial \log f(\lambda; \kappa^{(n)})}{\partial \gamma_{i-1}} d\lambda \\ &\quad \times \int_{-\pi}^{\pi} \frac{\partial \log f(\lambda'; \kappa^{(n)})}{\partial \gamma_{i'-1}} d\lambda' + o(1) \\ &= (\iota^{(n)})^{-4} \sum_{k,k'} W(\kappa^{(n)})_{i,k} \kappa_{k-1}^{(n)} W(\kappa^{(n)})_{i',k'} \kappa_{k'-1}^{(n)} + o(1). \end{aligned}$$

Note that the subscripts of $W(\kappa^{(n)})$ starts from 1 whereas the subscripts of κ starts from zero. Substituting these two results into (B.24), together with the Riemann sum argument and Assumption A1, we obtain that, for all $t > 0$,

$$\begin{aligned} n \sum_{j=1}^{J(t)} \text{Var}(\mathcal{V}_n(j) \mid \mathcal{F}(j)) &= \frac{n_d \sum_{j=1}^{J(t)} \eta_C^4(j)}{2n(\iota^{(n)})^4} \left(W(\kappa^{(n)}) + \frac{\text{cum}_4(\varepsilon)}{2} W(\kappa^{(n)}) \kappa^{(n)} \kappa^{(n)\top} W(\kappa^{(n)}) \right) + o_P(1) \\ &= \frac{\bar{\eta}^4 \int_0^t \eta_s^4 \xi_s^{-1} ds}{2T} \left(W(\gamma^*) + \frac{\text{cum}_4(\varepsilon)}{2} W(\gamma^*) \gamma^* \gamma^{*\top} W(\gamma^*) \right) + o_P(1). \end{aligned}$$

Next, we analyze $\tilde{B}_n(j)$, based on which we can fully characterize $n \sum_{j=1}^{J(t)} \text{Var}(d_n(j) \mid \mathcal{F}(j))$. Since $\chi(j)_k$ are i.i.d. with $\text{Var}(\chi) = m_2 - 1$, independent of $\mathcal{F}(j)$, the per-block conditional variance is

$$\text{Var}(\tilde{B}_n(j) \mid \mathcal{F}(j)) = \frac{n_d(m_2 - 1)}{n^2}. \quad (\text{B.25})$$

Therefore, together with Assumption A1, we can write

$$n \sum_{j=1}^{J(t)} \text{Var}(\tilde{B}_n(j) \mid \mathcal{F}(j)) = \frac{m_2 - 1}{T} \int_0^t \xi_s^{-1} ds + o_P(1),$$

$$\begin{aligned}
n \sum_{j=1}^{J(t)} \text{Var}(B_n(j) \mid \mathcal{F}(j)) &= \frac{m_2 - 1}{T} \int_0^t \eta_s^4 \xi_s^{-1} ds + o_P(1), \\
n \sum_{j=1}^{J(t)} \text{Cov}(\tilde{B}_n(j), B_n(j) \mid \mathcal{F}(j)) &= \frac{m_2 - 1}{T} \int_0^t \eta_s^2 \xi_s^{-1} ds + o_P(1).
\end{aligned}$$

Now we verify the orthogonality condition. For any bounded $\mathcal{F}_t$ -martingale M and for all j , we have

$$\mathbb{E}(d_n(j)(M_{t(j+1)_0} - M_{t(j)_0}) \mid \mathcal{F}(j)) = 0.$$

This comes from that all the entries of $d_n(j)$ are constructed from $\Delta_k^n U^C(j)$, $\bar{\chi}(j)$ and $\eta_C(j)$, each of which, conditionally on $\mathcal{F}(j)$, is independent of the $(\mathcal{F}_t)$ -martingale increment $M_{t(j+1)_0} - M_{t(j)_0}$, and that $M_{t(j+1)_0} - M_{t(j)_0}$ conditionally on $\mathcal{F}(j)$ is mean zero. Hence, we obtain that, for all $t > 0$,

$$n^{1/2} \sum_{j=1}^{J(t)} d_n(j) \xrightarrow{\mathcal{L}_{s-\mathcal{F}_\infty}} \tilde{d}(t), \quad (\text{B.26})$$

where $\tilde{d}(t) := (\tilde{d}(t)_1, \dots, \tilde{d}(t)_{q^*+3})$ is a continuous process defined on an extended probability space $(\bar{\Omega}, \bar{\mathcal{F}}, (\bar{\mathcal{F}}_t)_{t \geq 0}, \bar{\mathbb{P}})$ and which, conditionally on $\mathcal{F}_\infty$, is a centered Gaussian process with independent increments whose components satisfy

$$\begin{aligned}
\mathbb{E}(\tilde{d}(t)_i \tilde{d}(t)_{i'} \mid \mathcal{F}_\infty) &= \mathbb{1}_{\{i, i' \leq q^*+1\}} \frac{\bar{\eta}^4 \int_0^t \eta_s^4 \xi_s^{-1} ds}{2T} \left(W(\gamma^*) + \frac{\text{cum}_4(\varepsilon)}{2} W(\gamma^*) \gamma^* \gamma^{*\top} W(\gamma^*) \right)_{i, i'} \\
&+ \mathbb{1}_{\{i=i'=q^*+2\}} \frac{\int_0^t \eta_s^4 \xi_s^{-1} ds}{T} + \mathbb{1}_{\{i=i'=q^*+3\}} \frac{1}{T} \int_0^t \xi_s^{-1} ds \\
&+ (\mathbb{1}_{\{i=q^*+2, i'=i+1\}} + \mathbb{1}_{\{i'=q^*+2, i=i+1\}}) \frac{1}{T} \int_0^t \eta_s^2 \xi_s^{-1} ds.
\end{aligned}$$

Combining (B.23) and (B.26), together with that $\bar{\eta}$ and w_i are $\mathcal{F}_\infty$ -measurable, we obtain

$$n^{1/2} (\Xi_n(\beta^{(n)})_1, \dots, \Xi_n(\beta^{(n)})_{q^*+1}) \xrightarrow{\mathcal{L}_{s-\mathcal{F}_\infty}} \tilde{\Xi}, \quad (\text{B.27})$$

where $\tilde{\Xi} := (\tilde{\Xi}_1, \dots, \tilde{\Xi}_{q^*+1})$ is defined on the extended probability space $(\bar{\Omega}, \bar{\mathcal{F}}, (\bar{\mathcal{F}}_t)_{t \geq 0}, \bar{\mathbb{P}})$ and which, conditionally on $\mathcal{F}_\infty$, is a centered Gaussian variable whose components satisfy

$$\begin{aligned}
\mathbb{E}(\tilde{\Xi}^\top \tilde{\Xi} \mid \mathcal{F}_\infty) &= (2W(\gamma^*) + \text{cum}_4(\varepsilon) W(\gamma^*) \gamma^* \gamma^{*\top} W(\gamma^*)) \frac{\int_0^T \eta_s^4 \xi_s^{-1} ds}{4T \bar{\eta}^4} \\
&+ W(\gamma^*) \gamma^* \gamma^{*\top} W(\gamma^*) (m_2 - 1) \left(\frac{1}{4T \bar{\eta}^4} \int_0^T \eta_s^4 \xi_s^{-1} ds - \frac{1}{4\xi_T} \right).
\end{aligned}$$

On the other hand, straightforward algebra leads to

$$\begin{aligned}
n^{-1}n_T\partial\Xi_n(\bar{\beta}^{(n)}, q^*)_{1,1}^{-1} &= 2(\bar{\beta}_0^{(n)})^{3/2}\Delta_n^{-1/2}\sum_{r=0}^{q^*}(2-\delta_{r,0})\gamma^* + O(1), \\
n^{-1}n_T\partial\Xi_n(\bar{\beta}^{(n)}, q^*)_{1,k+1}^{-1} &= -\frac{2\bar{\beta}_0^{(n)}}{\sum_{r=0}^{q^*}(2-\delta_{r,0})\gamma^*}\sum_{r=1}^{q^*+1}(2-\delta_{r,1})W(\gamma^*)_{r,k}^{-1} + O(n^{-1/2}), \\
n^{-1}n_T\partial\Xi_n(\bar{\beta}^{(n)}, q^*)_{j+1,k+1}^{-1} &= 2W(\gamma^*)_{j,k}^{-1} + O(n^{-1/2}).
\end{aligned}$$

Hence, (B.9) and (B.10) jointly indicate that for all $2 \leq j \leq q^* + 2$,

$$\check{\beta}_{n,j} - \beta_j^{(n)} = -(0_{j-1}, 1, 0_{q^*+2-j})^\top \partial\Xi_n(\bar{\beta}^{(n)}, q^*)^{-1} \Xi_{D,n}(\beta^{(n)}) + o_P(n^{-1/2}). \quad (\text{B.28})$$

Here we also use Lemmas A7 and A8, (A.109), and the relation (B.8). At this stage, in view of the fact that by definition $(\hat{\sigma}_n^2(q_n), \hat{\gamma}_n(q_n))$ maximizes $L_n(\sigma^2, \gamma)$ over $\Pi_n^{(\sigma, \gamma^2)}(q_n)$ and the definition of $\check{\beta}_n$, plus Lemma B1, the combination of (B.27) and (B.28) proves the theorem. Applying continuous mapping theorem, we obtain the corollary. ■

Proof of Proposition 1. Step 1. (Efficient return removal in $\hat{B}'_n$ and $\hat{B}''_n$) Since $Y_{n,i} = \Delta_i^n X + \Delta_i^n U$, expanding $Y_{n,i}^2 Y_{n,i+j}^2$ and subtracting the noise-only leading term $(\Delta_i^n U)^2 (\Delta_{i+j}^n U)^2$ produces cross terms grouped by the total power of $\Delta^n X$. These are controlled by the bounds

$$\sum_{i=1}^{n_T} (\Delta_i^n X)^4 = O_P(1), \quad (\text{B.29})$$

$$\sum_{i=1}^{n_T} (\Delta_i^n U)^{2p} = O_P(n_T) \quad \text{for any finite } p. \quad (\text{B.30})$$

Bound (B.29) is Theorem 3.3.1 of Jacod and Protter (2011) under Assumption A1. Bound (B.30) follows from Assumptions 2 and 4 combined with the boundedness of η_t and ξ_t under Assumption A1. Applying Cauchy–Schwarz to each group of cross terms (with the total power of $\Delta^n X$ determining the rate), all remainders are $o_P(1)$, giving

$$\hat{B}''_n = \frac{1}{4n_T k_n} \sum_{i=1+k_n}^{n_T-k_n} (\Delta_i^n U)^2 \sum_{j=-k_n}^{k_n} (\Delta_{i+j}^n U)^2 + o_P(1), \quad (\text{B.31})$$

$$\hat{B}'_n = \frac{1}{4n_T k_n} \sum_{i=1}^{n_T-2k_n} (\Delta_i^n U)^2 \sum_{j=k_n+1}^{2k_n} (\Delta_{i+j}^n U)^2 + o_P(1). \quad (\text{B.32})$$

Step 2. (Noise-only Riemann sums) With the signal removed, we expand the noise differences in

terms of the noise levels. Since $\Delta_i^n U = U_i - U_{i-1}$,

$$(\Delta_i^n U)^2 = U_i^2 - 2U_i U_{i-1} + U_{i-1}^2.$$

The product $(\Delta_i^n U)^2 (\Delta_{i+j}^n U)^2$ therefore expands as a finite linear combination (with fixed integer coefficients) of fourth-order products $U_{i+\alpha_1} U_{i+\alpha_2} U_{i+j+\beta_1} U_{i+j+\beta_2}$ with $\alpha_k, \beta_k \in \{0, -1\}$. Each noise level satisfies $U_i = \iota \eta_{t_i} \theta(B) \varepsilon_i$ (Assumption 3), where $\theta(B) \varepsilon_i = \varepsilon_i + \theta_1 \varepsilon_{i-1} + \dots + \theta_{q^*} \varepsilon_{i-q^*}$ is the stationary noise component. Since η_{t_i} and $\eta_{t_{i-1}}$ are not equal in general, the noise difference $\Delta_i^n U$ does not factor as a single η times a stationary increment; this is why we expand into the noise levels themselves.

For each fixed $j \in \{-2k_n, \dots, 2k_n\}$ and each fixed $\alpha_1, \alpha_2, \beta_1, \beta_2 \in \{0, -1\}$, the sum $\sum_{i=1}^{n_T} U_{i+\alpha_1} \times U_{i+\alpha_2} U_{i+j+\beta_1} U_{i+j+\beta_2}$ is a noise-level statistic of the type handled by Theorems 3.1 and 3.3 of [Jacod et al. \(2017\)](#). Since all indices lie within an $O(k_n)$ -window of index i and $k_n/n_T \rightarrow 0$, the slowly varying diurnal pattern η is approximately constant within each local block. The technique of [Jacod et al. \(2017\)](#) then gives, with $\mathbb{E}\kappa[\cdot]$ denoting expectation over the stationary component $\theta(B)\varepsilon$:

$$\begin{aligned} & \frac{1}{n_T} \sum_{i=1}^{n_T} U_{i+\alpha_1} U_{i+\alpha_2} U_{i+j+\beta_1} U_{i+j+\beta_2} \\ &= \iota^4 \frac{\int_0^T \eta_s^4 \xi_s^{-1} ds}{\int_0^T \xi_s^{-1} ds} \cdot \mathbb{E}\kappa[(\theta(B)\varepsilon)_{\alpha_1} (\theta(B)\varepsilon)_{\alpha_2} (\theta(B)\varepsilon)_{j+\beta_1} (\theta(B)\varepsilon)_{j+\beta_2}] + o_P(1), \end{aligned}$$

under Assumption A1 and (B.11). For $\widehat{B}'_n$, all lags satisfy $j > k_n \gg q^* + 1$, so the stationary noise components near index i are independent of those near index $i+j$ (since $\theta(B)\varepsilon$ has $\text{MA}(q^*)$ support), and the fourth moment factors. The expansion coefficients give

$$\kappa_0 - 2\kappa_{-1} + \kappa_0 = 2(\kappa_0 - \kappa_1),$$

where $\kappa_h = \frac{1}{2\pi} \int g(\lambda; \theta) e^{ih\lambda} d\lambda$ is the autocovariance of $\theta(B)\varepsilon$ at lag h , since $\mathbb{E}[(\theta(B)\varepsilon)_a (\theta(B)\varepsilon)_b] = \kappa_{|a-b|}$. Since $\gamma_j^* = \iota^2 \frac{\int \eta^2 \xi^{-1} ds}{\int \xi^{-1} ds} \cdot \kappa_j$, the factored product is $\iota^4 [2(\kappa_0 - \kappa_1)]^2 = 4(\gamma_0^* - \gamma_1^*)^2 \left(\frac{\int \xi^{-1} ds}{\int \eta^2 \xi^{-1} ds} \right)^2$, and the 1/4 prefactor in $\widehat{B}'_n$ cancels the leading 4. For $\widehat{B}''_n$, the lags $|j| \leq q^* + 1$ produce cross terms. Applying the moment-cumulant relation to expand $\mathbb{E}\kappa[(\theta(B)\varepsilon)_{\alpha_1} (\theta(B)\varepsilon)_{\alpha_2} (\theta(B)\varepsilon)_{j+\beta_1} (\theta(B)\varepsilon)_{j+\beta_2}]$ in the display above, and summing over $\alpha_1, \alpha_2, \beta_1, \beta_2 \in \{0, -1\}$ with the expansion coefficients from $(\Delta_i^n U)^2 (\Delta_{i+j}^n U)^2$, the first pairing reproduces $[2(\kappa_0 - \kappa_1)]^2$ (as above), while the second and third pairings each contribute

$$(2\kappa_j - \kappa_{j-1} - \kappa_{j+1})^2,$$

which vanishes for $|j| > q^* + 1$. Summing over j and applying Parseval gives $2 \sum_j (2\kappa_j - \kappa_{j-1} - \kappa_{j+1})^2 = \frac{4}{\pi} \int_{-\pi}^{\pi} (1 - \cos \lambda)^2 g(\lambda; \theta)^2 d\lambda$. For the cumulant term, using $\text{cum}_4((\theta(B)\varepsilon)_a, (\theta(B)\varepsilon)_b, (\theta(B)\varepsilon)_c, (\theta(B)\varepsilon)_d) = \text{cum}_4(\varepsilon) \times \sum_m \theta_{a-m} \theta_{b-m} \theta_{c-m} \theta_{d-m}$ and summing over both the expansion coefficients

and j yields $\text{cum}_4(\varepsilon) \cdot [2(\kappa_0 - \kappa_1)]^2$.

Carrying out this calculation:

$$\begin{aligned} \frac{1}{4n_T} \sum_{i=1+k_n}^{n_T-k_n} (\Delta_i^n U)^2 \sum_{j=-k_n}^{k_n} (\Delta_{i+j}^n U)^2 &= \frac{(\int_0^T \eta_s^4 \xi_s^{-1} ds)(\int_0^T \xi_s^{-1} ds)}{(\int_0^T \eta_s^2 \xi_s^{-1} ds)^2} \left((2k_n + \text{cum}_4(\varepsilon))(\gamma_0^* - \gamma_1^*)^2 \right. \\ &\quad \left. + \frac{1}{\pi} \int_{-\pi}^{\pi} f(\lambda; \gamma^*)^2 (1 - \cos \lambda)^2 d\lambda \right) + o_P(1), \end{aligned} \quad (\text{B.33})$$

$$\frac{1}{4n_T} \sum_{i=1}^{n_T-2k_n} (\Delta_i^n U)^2 \sum_{j=k_n+1}^{2k_n} (\Delta_{i+j}^n U)^2 = \frac{(\int_0^T \eta_s^4 \xi_s^{-1} ds)(\int_0^T \xi_s^{-1} ds)}{(\int_0^T \eta_s^2 \xi_s^{-1} ds)^2} k_n (\gamma_0^* - \gamma_1^*)^2 + o_P(1). \quad (\text{B.34})$$

Finally, $\hat{m}$ targets $m_2 - 1$; it is constructed from local time-gap ratios in the same way as the estimator $U_t^{1,n}$ of [Jacod et al. \(2017\)](#) (eq. 3.15, Theorem 3.3 eq. 3.17), so their result gives $\hat{m} \xrightarrow{P} m_2 - 1$ directly.

Step 3. (Conclusion) Using (B.31), (B.32), (B.33), (B.34), and $\hat{m} \xrightarrow{P} m_2 - 1$, together with $\|\hat{\gamma}_n - \gamma^*\| = o_P(1)$ (Theorem 2), the continuous mapping theorem gives $\widehat{\text{AVAR}}_1 \xrightarrow{P} n_T n^{-1} \text{AVAR}_1$. The scaling factor $n_T n^{-1}$ arises as $n_T/n \xrightarrow{P} T/\int_0^T \xi_s^{-1} ds$ due to Assumption 2, which completes the proof. ■

B.3 Proof of Proposition 2

Proof. Step 1. (Limit of $G_n(x)$) We construct a family of DGP sequences indexed by $b \in \mathbb{R}$. Fix a probability measure $\mathbb{P}$ satisfying Assumptions 1–4 with equispaced observation times $t_i - t_{i-1} = T/n$ (i.e., $\xi_t = 1$ and $\chi_i = 1$), constant spot volatility $\sigma_t = \sigma_0 > 0$, constant $\eta_t = 1$, and Gaussian innovations $\varepsilon \sim \mathcal{N}(0, 1)$. Set the noise magnitude $\iota^{(n)} = \iota_0$ for a fixed constant $\iota_0 \geq K$, and define the MA(1) parameter

$$\theta^{(n)} = \frac{\sqrt{\log n} + b}{\sqrt{n}}, \quad b \in \mathbb{R}. \quad (\text{B.35})$$

This DGP sequence satisfies Assumptions 1–4 for each $b \in \mathbb{R}$, with $\iota^{(n)} = \iota_0 \geq K$. We denote this DGP sequence by $\{\mathbb{P}_b^{(n)}\}_{n \geq 1}$.

Under $\mathbb{P}_b^{(n)}$, the noise autocovariances are

$$\gamma_0^{(n)} = \iota_0^2 (1 + (\theta^{(n)})^2), \quad \gamma_1^{(n)} = \iota_0^2 \theta^{(n)} = \frac{\iota_0^2 (\sqrt{\log n} + b)}{\sqrt{n}}. \quad (\text{B.36})$$

Since $\theta^{(n)} \rightarrow 0$, the parameter sequence $\gamma^{(n)} = (\gamma_0^{(n)}, \gamma_1^{(n)})$ converges to the MA(0) boundary $\gamma^* = (\iota_0^2, 0)$.

We now establish the joint asymptotic behavior of the QMLE under both $q = 0$ and $q = 1$. Following the same reasoning as in the proof of Theorem 2, the standard asymptotic analysis yields

the joint convergence

$$n^{1/2} \begin{pmatrix} \hat{\gamma}_{n,0}(0) - \gamma_0^{(n)}(0) \\ \hat{\gamma}_{n,0}(1) - \gamma_0^{(n)}(1) \\ \hat{\gamma}_{n,1}(1) - \gamma_1^{(n)}(1) \end{pmatrix} \xrightarrow{\mathcal{L}} \mathcal{N} \left(0, \iota_0^4 \begin{pmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right). \quad (\text{B.37})$$

The covariance matrix has the following structure. The diagonal entries come from specializing the AVAR_1 formula in Theorem 2 to the simplified DGP ($\text{cum}_4(\varepsilon) = 0$, $m_2 = 1$, $\eta_t = \xi_t = 1$): the AVAR reduces to $\text{AVAR}_1 = 2W(\gamma^*)^{-1}$, and since $W(\gamma^*) = \iota_0^{-4} \text{diag}(1, 2)$ at the MA(0) boundary, we get $\text{AVAR}_1 = \text{diag}(2\iota_0^4, \iota_0^4)$. The off-diagonal zero blocks arise from $W_{01}(\gamma^*) = \frac{1}{2\pi} \int_{-\pi}^{\pi} 2 \cos \lambda d\lambda = 0$, which implies that the Fisher information is block-diagonal between γ_0 and γ_1 . Since MA(1) is the correctly specified model, $\gamma^{(n)}(1) = \gamma^{(n)}$, and the corresponding approximation errors are exactly zero. For the MA(0) model, the approximation error satisfies $\|\gamma_0^{(n)}(0) - \gamma_0^{(n)}\| = o(n^{-1/2})$.

On the other hand, in view of the definition of $\hat{q}_n$ by BIC and the fact that $\theta^{(n)} \rightarrow 0$, the Hessian of the log-likelihood satisfies

$$-\frac{2}{n} \frac{\partial^2 L_n(\hat{\gamma}_n(1))}{\partial \gamma_i \partial \gamma_j} = W_{ij}(\gamma^*) + o_P(1) = \frac{1}{\iota_0^4} \text{diag}(1, 2)_{ij} + o_P(1). \quad (\text{B.38})$$

We use the mean value theorem to expand the BIC criterion:

$$\begin{aligned} \text{BIC}_n(1) - \text{BIC}_n(0) &= \log n + \sum_{i,j} \frac{\partial^2 L_n(\hat{\gamma}_n(1))}{\partial \gamma_i \partial \gamma_j} (\hat{\gamma}_n(0)_i - \hat{\gamma}_n(1)_i) (\hat{\gamma}_n(0)_j - \hat{\gamma}_n(1)_j) \\ &\quad + o_P(n \|\hat{\gamma}_n(0) - \hat{\gamma}_n(1)\|^2), \end{aligned} \quad (\text{B.39})$$

where $\hat{\gamma}_n(0) - \hat{\gamma}_n(1)$ is understood as the $(q+2)$ -dimensional vector $(\hat{\gamma}_{n,0}(0) - \hat{\gamma}_{n,0}(1), 0 - \hat{\gamma}_{n,1}(1))$. From (B.37), the first component satisfies $n^{1/2}(\hat{\gamma}_{n,0}(0) - \hat{\gamma}_{n,0}(1)) = o_P(1)$, while the second component satisfies $n^{1/2}\hat{\gamma}_{n,1}(1) = n^{1/2}\gamma_1^{(n)} + \iota_0^2 Z + o_P(1)$ where $Z \sim \mathcal{N}(0, 1)$. Combining (B.38) with (B.39) and using $W_{01} = 0$, the leading term of the BIC difference is determined by the γ_1 component alone:

$$\text{BIC}_n(1) - \text{BIC}_n(0) = \log n - \frac{n}{\iota_0^4} \hat{\gamma}_{n,1}(1)^2 + o_P(1). \quad (\text{B.40})$$

Substituting $\hat{\gamma}_{n,1}(1) = \iota_0^2(\sqrt{\log n} + b + Z)/\sqrt{n} + o_P(n^{-1/2})$, we obtain $n\hat{\gamma}_{n,1}(1)^2/\iota_0^4 = (\sqrt{\log n} + b + Z)^2 + o_P(1)$. Therefore,

$$\text{BIC}_n(1) - \text{BIC}_n(0) = -2(b + Z)\sqrt{\log n} - (b + Z)^2 + o_P(1),$$

which is asymptotically determined by the sign of $b + Z$. Hence, $\hat{q}_n \wedge 1 = 1$ if and only if $Z > -b$ asymptotically, and

$$\mathbb{P}_b^{(n)}(\hat{q}_n = 1) \rightarrow \Phi(b), \quad \mathbb{P}_b^{(n)}(\hat{q}_n = 0) \rightarrow \Phi(-b). \quad (\text{B.41})$$

Since $\hat{\gamma}_{n,1}(1)$ and the event $\{\hat{q}_n = 1\}$ are both determined by the same random variable Z , the target and the selection are directly coupled. We compute the two contributions to $G_n(x)$ separately.

When MA(0) is selected, $\hat{\gamma}_{n,1}(0) = 0$, so $n^{1/2}(0 - \gamma_1^{(n)}) = -\iota_0^2(\sqrt{\log n} + b) \rightarrow -\infty$. For any fixed $x \in \mathbb{R}$, this is eventually $\leq x$, and the event $\{\hat{q}_n = 0\}$ occurs when $Z \leq -b$. Hence $\mathbb{P}_b^{(n)}(\hat{q}_n = 0, n^{1/2}(\hat{\gamma}_{n,1}(0) - \gamma_1^{(n)}) \leq x) \rightarrow \Phi(-b)$.

When MA(1) is selected, $n^{1/2}(\hat{\gamma}_{n,1}(1) - \gamma_1^{(n)}) = \iota_0^2 Z + o_P(1)$, and selection requires $Z > -b$. Therefore,

$$\mathbb{P}_b^{(n)}(\hat{q}_n = 1, n^{1/2}(\hat{\gamma}_{n,1}(1) - \gamma_1^{(n)}) \leq x) \rightarrow (\Phi(x/\iota_0^2) - \Phi(-b))_+.$$

Combining the two contributions,

$$G_\infty(x, b) := \lim_{n \rightarrow \infty} G_n(x) = \Phi(-b) + (\Phi(x/\iota_0^2) - \Phi(-b))_+ = \max(\Phi(x/\iota_0^2), \Phi(-b)). \quad (\text{B.42})$$

We verify that $G_\infty(x, b)$ is nonconstant in b for each fixed $x \in \mathbb{R}$. If $x < 0$: for $b \in [0, |x|/\iota_0^2]$, $G_\infty(x, b) = \Phi(-b)$, which is strictly decreasing in b . If $x \geq 0$: for $b < 0$ with $|b|$ sufficiently large, $\Phi(-b) > \Phi(x/\iota_0^2)$, so $G_\infty(x, b) = \Phi(-b)$, which depends on b . In both cases, $G_\infty(x, \cdot)$ is nonconstant.

Step 2. (Contiguity) We prove that $\mathbb{P}_b^{(n)}$ is contiguous with respect to $\mathbb{P}_0^{(n)}$ for every $b \in \mathbb{R}$. In view of Le Cam's first lemma (Lemma 6.4 in [van der Vaart \(2000\)](#)), denoting $\mathcal{U}_n = \log(d\mathbb{P}_b^{(n)}/d\mathbb{P}_0^{(n)})$, it suffices to show that $\mathcal{U}_n$ converges in distribution under $\mathbb{P}_0^{(n)}$ to a random variable Z satisfying $\mathbb{E}[\exp(Z)] \leq 1$.

We introduce the shorthand $\gamma^{(n,b)} = (\gamma_0^{(n)}(b), \gamma_1^{(n)}(b))$ for the true noise autocovariances under $\mathbb{P}_b^{(n)}$. From (B.36),

$$\gamma^{(n,b)} - \gamma^{(n,0)} = \left(\frac{\iota_0^2(2b\sqrt{\log n} + b^2)}{n}, \frac{\iota_0^2 b}{\sqrt{n}} \right). \quad (\text{B.43})$$

The log-likelihood ratio satisfies $\mathcal{U}_n = L_n(\sigma_0^2, \gamma^{(n,b)}) - L_n(\sigma_0^2, \gamma^{(n,0)}) + o_P(1)$. Denote the Hessian matrix $\tilde{H}_n = \frac{\partial^2 L_n(\hat{\gamma}_n(1))}{\partial \gamma \partial \gamma^\top}$. Expanding both terms around $\hat{\gamma}_n(1)$ (the QMLE under $q = 1$, at which the score vanishes), and using the mean value theorem, (B.37), and (B.38), we obtain

$$\begin{aligned} \mathcal{U}_n &= \frac{1}{2}(\gamma^{(n,b)} - \hat{\gamma}_n(1))^\top \tilde{H}_n (\gamma^{(n,b)} - \hat{\gamma}_n(1)) \\ &\quad - \frac{1}{2}(\gamma^{(n,0)} - \hat{\gamma}_n(1))^\top \tilde{H}_n (\gamma^{(n,0)} - \hat{\gamma}_n(1)) + o_P(1). \end{aligned} \quad (\text{B.44})$$

Setting $\delta^{(n)} = \gamma^{(n,b)} - \gamma^{(n,0)}$ from (B.43) and $v^{(n)} = \gamma^{(n,0)} - \hat{\gamma}_n(1)$, we rewrite (B.44) as

$$\mathcal{U}_n = \delta^{(n)\top} \tilde{H}_n v^{(n)} + \frac{1}{2} \delta^{(n)\top} \tilde{H}_n \delta^{(n)} + o_P(1).$$

Making use of $\tilde{H}_n = -\frac{n}{2}W(\gamma^*) + o_P(n)$ from (B.38), $n^{1/2}v^{(n)} = -n^{1/2}(\hat{\gamma}_n(1) - \gamma^{(n,0)}) \xrightarrow{d} \mathcal{N}(0, \text{AVAR}_1)$ from (B.37) (under $\mathbb{P}_0^{(n)}$), and $n^{1/2}\delta^{(n)} \rightarrow (0, \iota_0^2 b)$, it follows that $\mathcal{U}_n \xrightarrow{d} Z \sim \mathcal{N}(-b^2/2, b^2)$ under $\mathbb{P}_0^{(n)}$. Since $\mathbb{E}[\exp(Z)] = \exp(-b^2/2 + b^2/2) = 1$, Le Cam's first lemma gives that $\mathbb{P}_b^{(n)}$ is contiguous with respect to $\mathbb{P}_0^{(n)}$.

Step 3. (Conclusion) We have established: (i) under $\mathbb{P}_b^{(n)}$, $G_n(x) \rightarrow G_\infty(x, b)$ for each $x \in \mathbb{R}$; (ii) $G_\infty(x, b)$ is nonconstant in b for each fixed x ; (iii) $\mathbb{P}_b^{(n)}$ is contiguous with respect to $\mathbb{P}_0^{(n)}$ for every $b \in \mathbb{R}$. By Lemma 3.1 of [Leeb and Pötscher \(2006\)](#),

$$\liminf_{n \rightarrow \infty} \inf_{\hat{G}_n(x)} \sup_{b \in \mathbb{R}} \mathbb{P}_b^{(n)} \left(|\hat{G}_n(x) - G_n(x)| > \frac{1}{K} \right) > 0,$$

which concludes the proof. ■

B.4 Proof of Theorem 3

Proof. Let $\mathcal{P}$ denote the collection of all DGP sequences $\{\mathbb{P}^{(n)}, \iota^{(n)}, \theta^{(n)}\}_{n \geq 1}$ that satisfy Assumptions 1–5, and let r_n denote the rate for $\|\hat{\gamma}^{(n)}(\hat{q}_n) - \gamma^{(n)}\|^2$ displayed in the statement of Theorem 3. The theorem is equivalent to the statement that, for every $\varepsilon > 0$, there exists $M_\varepsilon < \infty$ with

$$\limsup_{n \rightarrow \infty} \sup_{\{\mathbb{P}, \iota^{(n)}, \theta^{(n)}\} \in \mathcal{P}} \mathbb{P}^{(n)} \left(\|\hat{\gamma}^{(n)}(\hat{q}_n) - \gamma^{(n)}\|^2 > M_\varepsilon r_n \right) \leq \varepsilon, \quad (\text{B.45})$$

together with the analogous uniform O_P bound on $\|\hat{\rho}^{(n)}(\hat{q}_n) - \rho^{(n)}\|^2$. For any set Γ consisting of parameter sequences $\{\iota^{(n)}\}_{n \geq 1}$, let $\mathcal{P}(\Gamma)$ be the collection of all DGP sequences within $\mathcal{P}$ under which $\{\iota^{(n)}\} \in \Gamma$, and introduce the two sets of parameter sequences

$$\Gamma_1(a^2) := \{\{\iota^{(n)} : n \geq 1\} : n^{1/2}\iota^{(n)} \leq a^2\}, \quad \Gamma_2 := \{\{\iota^{(n)} : n \geq 1\} : n^{1/2}\iota^{(n)} \rightarrow \infty\}.$$

We now establish that, for every $a^2 \in [0, \infty)$ and every sequence $\{\mathbb{P}, \iota^{(n)}, \theta^{(n)}\} \in \mathcal{P}(\Gamma_1(a^2) \cup \Gamma_2)$,

$$\|\hat{\gamma}^{(n)}(\hat{q}_n) - \gamma^{(n)}\|^2 = O_P(r_n). \quad (\text{B.46})$$

In view of the proof of Lemma B4 in the online supplemental appendix of [Da and Xiu \(2021\)](#), we have that under either $n^{1/2}\iota^{(n)} \rightarrow \infty$ or $n^{1/2}\iota^{(n)} \leq K$, and for all $a_n \rightarrow \infty$ and all fixed $0 < k < K$,

$$\begin{aligned} & \bar{L}_n^*(\sigma^{(n)}(q_n^*(k))^2, \gamma^{(n)}(q_n^*(k))) - \bar{L}_n^*(\sigma^{(n)}(\hat{q}_n)^2, \gamma^{(n)}(\hat{q}_n)^2) \\ &= \frac{\log n}{2} (q_n^*(k) - \hat{q}_n) + o_P(q_n^*(k) + a_n) + O_P(|q_n^*(k) - \hat{q}_n|). \end{aligned} \quad (\text{B.47})$$

The definition of $q_n^*(k)$, combined with Lemma A10 and (B.47), indicates that there exists a fixed k such that $q_n^*(k) - \hat{q}_n \leq 1$ with probability approaching one. Further, in view of Lemma A10 and using the bound on $n\psi_n^4 \sum_{j=q_n^*(k)}^\infty |\kappa_j^{(n)}|^2$, we obtain that $n\psi_n^4 \sum_{j=\hat{q}_n+1}^\infty |\kappa_j^{(n)}|^2 = O_P((\hat{q}_n + 1) \log n)$ and that $\hat{q}_n \leq O_P(q_n^*(k) + 1)$. Hence, it follows from Lemmas A5 and A6 that for all $0 \leq j \leq \hat{q}_n$,

$$|\hat{\gamma}^{(n)}(\hat{q}_n)_j - \gamma_j^{(n)}|^2 = O_P\left(n^{-1}(\iota^{(n)})^4(\hat{q}_n + 1) \log n + n^{-3}(n^{1/2}\iota^{(n)} + 1)(\hat{q}_n + 1)^3 \log n\right).$$

Here we also use the proof of Lemma B3 of [Da and Xiu \(2021\)](#). (B.46) directly follows. Continuous mapping theorem leads to the bound on $\|\hat{\rho}^{(n)}(\hat{q}_n) - \rho^{(n)}\|$. It remains to extend (B.46) to the uniform statement (B.45). Suppose, for contradiction, that (B.45) fails. Then there exists $\varepsilon_0 > 0$ such that for every $M < \infty$ and every $N < \infty$ one can find a DGP sequence $\{\mathbb{P}, \iota^{(n)}, \theta^{(n)}\} \in \mathcal{P}$ and an index $n \geq N$ for which $\mathbb{P}^{(n)}(\|\hat{\gamma}^{(n)}(\hat{q}_n) - \gamma^{(n)}\|^2 > Mr_n) > \varepsilon_0$. By induction we pick, for each $m \geq 1$, a DGP sequence $\{\check{\mathbb{P}}_m, \check{\iota}_m^{(n)}, \check{\theta}_m^{(n)}\} \in \mathcal{P}$ and an index $n_m > n_{m-1}$ (with $n_0 := 0$) such that

$$\check{\mathbb{P}}_m^{(n_m)}\left(\|\hat{\gamma}^{(n_m)}(\hat{q}_{n_m}) - \check{\gamma}_m^{(n_m)}\|^2 > m\check{r}_{m,n_m}\right) > \varepsilon_0, \quad (\text{B.48})$$

where $\check{r}_{m,n_m}$ stands for the rate r_n evaluated along the m -th DGP sequence at $n = n_m$. Consider the real sequence $\{n_m^{1/2}\check{\iota}_m^{(n_m)}\}_{m \geq 1}$. Either $\limsup_{m \rightarrow \infty} n_m^{1/2}\check{\iota}_m^{(n_m)} < \infty$, in which case there exist $a^2 \in (0, \infty)$ and an infinite subset $\mathcal{M} \subseteq \mathbb{N}$ with $n_m^{1/2}\check{\iota}_m^{(n_m)} \leq a^2$ for all $m \in \mathcal{M}$; or $\limsup_{m \rightarrow \infty} n_m^{1/2}\check{\iota}_m^{(n_m)} = \infty$, in which case there exists an infinite subset $\mathcal{M} \subseteq \mathbb{N}$ along which $n_m^{1/2}\check{\iota}_m^{(n_m)} \rightarrow \infty$. Write the elements of $\mathcal{M}$ in increasing order as $m_1 < m_2 < \dots$, and set $v_j := n_{m_j}$. Then $v_j \rightarrow \infty$ and (B.48) restricted to $\{m_j\}$ yields, for every $j \geq 1$,

$$\check{\mathbb{P}}_{m_j}^{(v_j)}\left(\|\hat{\gamma}^{(v_j)}(\hat{q}_{v_j}) - \check{\gamma}_{m_j}^{(v_j)}\|^2 > m_j\check{r}_{m_j,v_j}\right) > \varepsilon_0. \quad (\text{B.49})$$

Given this subsequence $\{v_j\}$, we construct an auxiliary parameter sequence $\{(\iota_*^{(n)}, \theta_*^{(n)})\}_{n \geq 1}$ as follows. Set $v_0 := 0$ and, for each $j \geq 1$ and every $n \in [v_j, v_{j+1})$, define

$$\iota_*^{(n)} := \check{\iota}_{m_j}^{(v_j)}(v_j/n)^{1/2}, \quad \theta_*^{(n)} := \check{\theta}_{m_j}^{(v_j)};$$

for $n < v_1$, set $(\iota_*^{(n)}, \theta_*^{(n)}) = (\check{\iota}_{m_1}^{(v_1)}(v_1/n)^{1/2}, \check{\theta}_{m_1}^{(v_1)})$. By construction, $\iota_*^{(v_j)} = \check{\iota}_{m_j}^{(v_j)}$ and $\theta_*^{(v_j)} = \check{\theta}_{m_j}^{(v_j)}$, so that the law at index $n = v_j$ agrees with the corresponding law of the m_j -th DGP sequence, i.e., $\mathbb{P}_*^{(v_j)} = \check{\mathbb{P}}_{m_j}^{(v_j)}$ and $r_{*,v_j} = \check{r}_{m_j,v_j}$. Moreover, for every $n \in [v_j, v_{j+1})$ we have $n^{1/2}\iota_*^{(n)} = v_j^{1/2}\check{\iota}_{m_j}^{(v_j)}$, which is uniformly bounded by a^2 in the first case above and tends to infinity with j in the second case; hence $\{\iota_*^{(n)}\} \in \Gamma_1(a^2) \cup \Gamma_2$ and the resulting DGP sequence $\{\mathbb{P}_*^{(n)}, \iota_*^{(n)}, \theta_*^{(n)}\}_{n \geq 1}$ lies in $\mathcal{P}(\Gamma_1(a^2) \cup \Gamma_2)$. Applying (B.46) to this auxiliary sequence produces a finite constant $M^* > 0$ such that

$$\limsup_{n \rightarrow \infty} \mathbb{P}_*^{(n)}\left(\|\hat{\gamma}_*^{(n)}(\hat{q}_{*,n}) - \gamma_*^{(n)}\|^2 > M^*r_{*,n}\right) \leq \varepsilon_0/2. \quad (\text{B.50})$$

Restricting (B.50) to the subsequence $\{v_j\}$ and using $\mathbb{P}_*^{(v_j)} = \check{\mathbb{P}}_{m_j}^{(v_j)}$, $r_{*,v_j} = \check{r}_{m_j,v_j}$, we obtain

$$\limsup_{j \rightarrow \infty} \check{\mathbb{P}}_{m_j}^{(v_j)}\left(\|\hat{\gamma}^{(v_j)}(\hat{q}_{v_j}) - \check{\gamma}_{m_j}^{(v_j)}\|^2 > M^*\check{r}_{m_j,v_j}\right) \leq \varepsilon_0/2.$$

On the other hand, $m_j \rightarrow \infty$, so for every sufficiently large j we have $m_j > M^*$. Since $\check{r}_{m_j, v_j} \geq 0$, the inclusion $\{\|\hat{\gamma}^{(v_j)} - \check{\gamma}_{m_j}^{(v_j)}\|^2 > m_j \check{r}_{m_j, v_j}\} \subseteq \{\|\hat{\gamma}^{(v_j)} - \check{\gamma}_{m_j}^{(v_j)}\|^2 > M^* \check{r}_{m_j, v_j}\}$ holds, so (B.49) yields

$$\mathbb{P}_{m_j}^{(v_j)}\left(\|\hat{\gamma}^{(v_j)}(\hat{q}_{v_j}) - \check{\gamma}_{m_j}^{(v_j)}\|^2 > M^* \check{r}_{m_j, v_j}\right) > \varepsilon_0$$

for all j sufficiently large. Taking $\limsup_{j \rightarrow \infty}$ gives $\varepsilon_0 \leq \varepsilon_0/2$, a contradiction. Hence (B.45) holds, and the O_P bound on $\|\hat{\gamma}^{(n)}(\hat{q}_n) - \gamma^{(n)}\|^2$ is uniform over $\mathcal{P}$. The uniform bound on $\|\hat{\rho}^{(n)}(\hat{q}_n) - \rho^{(n)}\|^2$ follows from the continuous mapping theorem. ■

B.5 Proof of Theorem 4

Proof. We observe that for all $(\sigma^2, \gamma) \in \Pi_n^{(\sigma^2, \gamma)}(q)$,

$$\mathbb{I}_n = \frac{\partial \log \Sigma_n(\sigma^2, \gamma)}{\partial \sigma^2} \sigma^2 + \sum_{j=0}^q \frac{\partial \log \Sigma_n(\sigma^2, \gamma)}{\partial \gamma_j} \gamma_j.$$

Thus, the theorem directly follows from the fact that for all finite q ,

$$\text{tr} \left(\frac{\partial \log \Sigma_n(\hat{\sigma}^2(q), \hat{\gamma}(q))}{\partial (\sigma^2, \gamma)_j} \right) = -\text{tr} \left(\frac{\partial \Sigma_n^{-1}(\hat{\sigma}^2(q), \hat{\gamma}(q))}{\partial (\sigma^2, \gamma)_j} Y_n Y_n^\top \right), \quad 1 \leq j \leq q+2,$$

which are the first-order conditions. ■

B.6 Proof of Theorem 5

Proof. Step 1. (Technical preparation) We define $(q+2) \times (q+2)$ matrix $\bar{W}_n(\sigma^2, \gamma)$ and $n_T \times n_T$ matrices $\mathcal{R}(k)$, with $1 \leq k \leq q+2$, as

$$\begin{aligned} \bar{W}_n(\sigma^2, \gamma) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{\partial \log f(\lambda; \sigma^2, \gamma, \Delta_n)}{\partial (\sigma^2, \gamma)} \right)^\top \frac{\partial \log f(\lambda; \sigma^2, \gamma, \Delta_n)}{\partial (\sigma^2, \gamma)} d\lambda, \\ \mathcal{R}(k) &= \delta_{k,1} \frac{\partial \Sigma_n(\sigma^2, \gamma)^{-1}}{\partial \sigma^2} + \mathbb{1}_{\{k \geq 2\}} \frac{\partial \Sigma_n(\sigma^2, \gamma)^{-1}}{\partial \gamma_{k-2}}. \end{aligned}$$

Throughout the proof we omit the argument of $\bar{W}_n(\sigma^2, \gamma)$ and $\widetilde{W}_n(\sigma^2, \gamma)$. We can calculate, using Lemmas A1, A3, and A4,

$$(\bar{W}_n^{-1})_{1,1} = 4\sigma^3 \zeta \Delta_n^{-1/2} + O(1), \quad (\bar{W}_n^{-1})_{1,k+1} = -\frac{\sigma^2}{\zeta^2} \sum_{r=1}^{q+1} (2 - \delta_{r,1}) W(\gamma)_{r,k}^{-1} + O(\Delta_n^{1/2}), \quad (\text{B.51})$$

$$(\bar{W}_n^{-1})_{j+1,k+1} = W(\gamma)_{j,k}^{-1} + O(\Delta_n^{1/2}), \quad (n_T \widetilde{W}_n^{-1})_{j,k} - (\bar{W}_n^{-1})_{j,k} = O(\Delta_n + \delta_{j,1} \delta_{k,1}). \quad (\text{B.52})$$

On the other hand, also using $n^{1/2+\alpha} \leq i \leq n - n^{1/2+\alpha}$, we have, for $r \geq 0$,

$$\mathcal{R}(1)_{i,i} = -(4\zeta\sigma^3\Delta_n^{1/2})^{-1} + O(1), \quad \mathcal{R}(k)_{i,i} = -(2 - \delta_{k,2})(4\sigma\zeta^3\Delta_n^{1/2})^{-1} + O(1), \quad (\text{B.53})$$

$$\mathcal{R}(k)_{i,i+r} - \mathcal{R}(k)_{i,i} = \delta_{k,1} \frac{1}{4\zeta^2\sigma^2} \left(\frac{1 - z_n^r}{1 - z_n} - rz_n^r \right) + \mathbb{1}_{\{k \geq 2\}} \frac{2 - \delta_{k,2}}{4\zeta^4} \left(\frac{1 - z_n^r}{1 - z_n} + rz_n^r \right) + O(1), \quad (\text{B.54})$$

where z_n is defined in Lemma A3. If we further restrict $0 \leq r \leq K$, then it holds that

$$\mathcal{R}(1)_{i,i+r} - \mathcal{R}(1)_{i,i} = \frac{\Delta_n^{1/2} r^2}{8\sigma\zeta^3} + O(\Delta_n), \quad (\text{B.55})$$

$$\mathcal{R}(k)_{i,i+r} - \mathcal{R}(k)_{i,i} = -\frac{1}{4\pi} \sum_{s=0}^{r-1} (2 - \delta_{s,0})(r - s) \int_{-\pi}^{\pi} \frac{\partial f^{-1}(\lambda; \gamma)}{\partial \gamma_{k-1}} e^{i\lambda s} d\lambda + O(\Delta_n^{1/2}). \quad (\text{B.56})$$

Step 2 (Main proof) Using $(2\pi)^{-1} \int_{-\pi}^{\pi} (\partial f^{-1}(\lambda; \gamma) / \partial \gamma_{k-1}) e^{i\lambda s} d\lambda = (1 - \delta_{s,0}/2)W(\gamma)_{k,s+1}$, we derive from (B.56) and the second part of (B.51) that for $0 \leq r \leq K$,

$$\sum_{k=2}^{q+2} (\bar{W}_n^{-1})_{1,k} (\mathcal{R}(k)_{i,i+r} - \mathcal{R}(k)_{i,i}) = -\frac{\sigma^2 r^2}{2\zeta^2} + O(\Delta_n^{1/2}).$$

Combined with (B.55) and the first part of (B.51), we prove claim (i). Using (B.53) and the second part of (B.52), we have, for $2 \leq l \leq q+2$ and $0 \leq r \leq K$,

$$\sum_{k=2}^{q+2} (\bar{W}_n^{-1} - n_T \widetilde{W}_n^{-1})_{l,k} \mathcal{R}(k)_{i,i+r} = O(\Delta_n^{1/2}). \quad (\text{B.57})$$

In view of Lemmas A1 and A3, it holds by definition that

$$\mathcal{R}(k)_{i,i+r} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\delta_{k,1} \frac{\partial}{\partial \sigma^2} + \mathbb{1}_{\{k \geq 2\}} \frac{\partial}{\partial \gamma_{k-2}} \right) f^{-1}(\lambda; \sigma^2, \gamma, \Delta_n) \cos r\lambda d\lambda + O(\Delta_n^{1/2}).$$

Hence, by observing that $\cos r\lambda = \Delta_n^{-1} \partial f(\lambda; \sigma^2, \gamma, \Delta_n) / \partial \sigma^2 - \frac{1}{2} \sum_{k=0}^r (r-k) \partial f(\lambda; \sigma^2, \gamma, \Delta_n) / \partial \gamma_k$ for $0 \leq r \leq q+1$, we have, for $2 \leq l \leq q+2$ and $0 \leq r \leq K$,

$$\sum_{k=2}^{q+2} (\bar{W}_n^{-1})_{l,k} \mathcal{R}(k)_{i,i+r} = \mathbb{1}_{\{l \leq r+1\}} \frac{r+2-l}{2} + O(\Delta_n^{1/2}). \quad (\text{B.58})$$

Combining (B.57) and (B.58) proves claim (ii). Claim (iii) comes directly from the expressions of $(\bar{W}_n^{-1})_{j,k}$ provided by (B.51) and (B.52) and the expression of $\mathcal{R}(k)_{i,i+r} - \mathcal{R}(k)_{i,i}$ provided by (B.55). For the first part of claim (iii) we additionally use (B.53) to obtain $\mathcal{W}(\sigma^2, \gamma; 1)_{i,i}$, whereas for its second part we use claim (ii). ■

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